An Archival ALMA Search for Unresolved Spectral-Carrier Technosignatures toward Stars within 40 pc at 114–873 GHz

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ABSTRACT

Above 30 GHz, narrowband transmitter searches have reached one spectral line toward some tens of stars. We uniformly search 405 fine-channel windows toward 61 nearby systems in the ALMA archive, spanning 47.7 GHz of union bandwidth in disjoint intervals, plus 1258 coarser ones without drift discrimination. Windows are filtered for an unresolved carrier drifting linearly in the stellar frame. Threshold crossings are tested against catalogued transitions at the star's velocity, then sought again in later blocks. Nine injections in ten are recovered at 1.5×10^{15} W for the median system, for a carrier narrower than the instrumental channel drifting below our default ceiling of 12.0 nHz, 6 per cent of the generally recommended ceiling. The injections carry no atmospheric decorrelation, so these limits are optimistic by it. No confirmed carrier is found. Of 56 crossings, 18 are attributed and 38 are not; the four coarse-window ones come from one block that fails a quality criterion. None of the unattributed crossings recurred, including the two that lead their own control fields, while 13 attributed to circumstellar or foreground CO do recur. The repeats exclude a carrier whose stellar-frame frequency returns to within $\pm 0.21\,\mathrm{km\,s^{-1}}$ of the first epoch's and radiates through both observations, not a persistent transmitter of any kind. For 7 of them the repeats are too shallow for even that test at the trigger amplitude. The remainder is consistent with chance, on a step from the control positions to the star measured from only 8 reserved events. A calibration residual could mimic a carrier in 9 windows and is measured to act in none. Line emission, not interference, produces every astrophysical false positive and the strongest crossings. What limits the search is inherited coverage, not depth: 23 systems have no confirmation epoch.

Subject headings: extraterrestrial intelligence, radio lines: stars, submillimetre: stars, surveys, astrobiology

1. INTRODUCTION

The public ALMA archive holds thousands of hours of millimetre and submillimetre observations taken for other reasons, mostly studies of dust discs and planet formation. Their visibilities allow a spectrum at the position of the star at the centre of each field, at a resolution and depth set by the original observer's correlator setup. This paper searches those spectra for a narrowband transmitter at the star's own position. No such signal was found.

Searches for *technosignatures* — detectable imprints of technology, intended or not (Sheikh 2020; Vidal et al. 2026) — have examined many thousands of stars over six decades, yet the last published accounting of that effort covers only 6×10^{-18} of the space of transmitter position, power, frequency, bandwidth, polarisation and repetition rate (Wright et al. 2018), and essentially all of it lies below ∼30 GHz. Two targeted searches have reached nearby stars above that frequency. Steffes & DeBoer (1994) observed 40 solar-type stars within 23 pc at 203 GHz, and Mauersberger et al. (1996) included τ Cet and ϵ Eri among 17 targets at the same frequency. Each covered a single line, over no more than 35 MHz, with no search over drift rate and no test of persistence; Mason et al. (2025) has since searched more distant stars in Band 3. This survey covers 47.7 GHz in 26 disjoint intervals, searches a grid of drift rates rather than a fixed frequency, tests every crossing against the 512 control positions of its own field, and uses repeat observations to test persistence.

The band has physical advantages and physical costs. Interstellar dispersion and scintillation are negligible at these frequencies, so a carrier arrives essentially undistorted, and a given aperture delivers far more gain per watt supplied. Against that, receiver system temperatures are higher, the atmosphere is opaque in places at both ends of the link, and the same high gain narrows the beam: the 12 m benchmark of §6.3 radiates into a beam only 34 arcsec across, so the Earth must lie inside a beam that was deliberately aimed. What else makes the band hard belongs to the archive rather than to the sky, and §2 sets that out.

The signal sought is an *unresolved spectral carrier* — emission confined to a few channels, unresolved at the stellar position, and present at more than one epoch. Recurrence rather than brightness is what separates an interesting event from a detection (Sheikh et al. 2021), and the archive holds a second, separated epoch for some of these systems and not for others, so what could be searched and what could be confirmed are different extents. The frequency is allowed to drift, as it must for a transmitter carried on a rotating, orbiting body (§4.1). "Narrowband" describes the assumed transmitter and not the resolving power of the archived channels. The targets were chosen by other programmes for other purposes, so every statistical statement here is about an archive-selected sample (§3.1).

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2. BACKGROUND

Radio searches for technosignatures began at centimetre wavelengths and have largely stayed there. The first proposals and targeted experiments (Cocconi & Morrison 1959; Drake 1961), the programmes reviewed by Tarter (2001) and Project Phoenix (Backus & Project Phoenix Team 2002) all worked below a few gigahertz. The modern generation divides into targeted and commensal surveys at centimetre and metre wavelengths (Zhang et al. 2020; Czech et al. 2021; Li et al. 2026; Johnson et al. 2023; Choza et al. 2024), several of them riding back-ends built for other science, such as the MWA's correlator (Morrison et al. 2023), with wide-field low-frequency coverage from that array (Tremblay & Tingay 2024) and single-dish C- and K-band work at 6 and 18 GHz with the Sardinia Radio Telescope (Manunza et al. 2025). Machine-learning interference rejection is now central to that effort (Ma et al. 2023; Jacobson-Bell et al. 2025; Pardo et al. 2025). The classical threshold search used here lacks that capability and instead uses a spatial test on the star against control positions in the same field, which a single dish cannot make. A narrowband search also reaches only one class of signal: broadband planetary-scale leakage (Garrett 2026) is outside it by construction.

Above about 30 GHz the record is nearly empty, and that is what makes this frequency range new. ALMA's 10 receiver bands cover 35 to 950 GHz, and one of them, Band 3, has been searched for technosignatures. The lowest, 35–50 GHz, does hold observations of 4 stars of this sample, but all from one programme, all in a continuum mode whose 31.25 MHz channels are 2048 times coarser than the narrowest channel searched here, and all proprietary until 2027: the coverage exists and cannot carry a carrier search. The band above it holds none at all. For continuity with the centimetre literature, the Arecibo planetary radar had an EIRP of 2.0×10^{13} W at 2.4 GHz (Sheikh et al. 2025); an aperture of that diameter is not realisable in the millimetre band, so the illustrative benchmark of §6.3 is instead a dish of 12 m. The limits of this paper are stated as EIRP and need no transmitter model.

Mason et al. (2025) is the only prior millimetre search to cover more than a single line, and the only archival one; the earlier nearby-star searches of Steffes & DeBoer (1994) and Mauersberger et al. (1996) covered one line each. Two Band 3 spectral windows, 117 MHz between them against the 47.7 GHz union of this survey, were searched toward 28 Gaia DR3 stars lying in the fields of 4 ALMA calibrators, observed for other science and so usable for free. Their nearest target is at 1.01 kpc, where a 5σ flux threshold corresponds to $\mathrm{EIRP}_{\min} > 6.9 \times 10^{17}$ W. That target lies between 25 and 314 times further away than the stars of the carrier search here, so at a fixed received flux an EIRP ratio between the two surveys would be 3–5 orders of magnitude of d^2 and almost nothing else. The quantity that does not depend on distance is the minimum detectable *received* flux, $F_{\min} = S_{\min}\,\delta\nu$, and it has to be compared at the channel width each experiment recorded: $S_{\min}$ falls as $\delta\nu^{-1/2}$ while $F_{\min}$ rises as $\delta\nu^{1/2}$, so putting a coarse window on a fine channel arithmetically credits it with a depth it never reached. Mason et al. reach 3.8×10^{-23}–5.8×10^{-23} W m^{-2} across their four fields, on a 30.52 kHz channel at a bare 5σ flux threshold. On that same threshold convention and on native channels throughout, the windows searched here reach 4.8×10^{-24} W m^{-2} at best — a factor 10 deeper, in a channel 15.3 kHz wide — and 3.0×10^{-22} W m^{-2} at the median, a factor 6 shallower in a channel 488 kHz wide. Per channel this survey is therefore the shallower of the two except in its finest windows, and its advantage lies elsewhere: 61 stellar systems within 40 pc rather than distant stars in four calibrator fields, and a frequency range reaching about nine times higher. Mason et al. also identified the obstacles that shape the present method: drift rates scale with frequency, smearing erodes sensitivity unless drift is searched for, and the molecular line forest raises the confusion background. Every window reported here was re-reduced and re-searched from the raw archive with the pipeline of §4.

TABLE 1
The comparison programmes of Figure 1, each on its own detection threshold and on the common 5σ basis. $\mathrm{EIRP}_{\min}$ is the deepest limit each paper reports, and the factor is 5/(S/N), the minimum detectable flux being linear in the threshold in every one of their own expressions for it.

programme	S/N	factor	$\Delta\nu_{\mathrm{ch}}$	$\mathrm{EIRP}_{\min}$ (W)	at 5σ (W)
Enriquez + 17	25	0.20	2.8 Hz	10^{13}	2×10^{12}
Margot + 23	10	0.50	3.0 Hz	1.35×10^{13}	6.75×10^{12}
Price + 20	10	0.50	2.8 Hz	2×10^{12}	10^{12}
Mason + 25	5	1.00	30.5 kHz	6.91×10^{17}	6.91×10^{17}

Figure 1(a) places this survey among published searches, in minimum detectable EIRP against distance. For a carrier narrower than every programme's channel that limit is a total radiated power rather than a spectral power density, so the axis is commensurable. The detection thresholds behind the published limits are not: of the 4 programmes drawn, 3 quote their limits at a threshold other than 5σ, and the minimum detectable flux is linear in that threshold in each of their own expressions for it. Table 1 gives each programme's threshold, the factor that puts it on the common 5σ basis and the channel width it used, and the figure is drawn after that rescaling, with each published value kept beside it. The survey's own filled symbols are bare 5σ thresholds on the same convention; the open symbols above them are EIRP_{90}, the power recovered nine times in ten through the single-epoch trigger stage, and every limit in this paper is quoted on that. Any EIRP advantage toward the nearest systems is a d^2 effect and not an instrumental one. Panel (b) is the other axis, the native channel width of every window searched here beside the channel of each comparison programme. What distinguishes this survey is its frequency range and the proximity of its targets, not its depth in any one channel.

A non-detection becomes a quantitative statement only through a stated search volume. Wright et al. (2018) formalised that volume, and Margot et al. (2023) its conversion into an upper bound on the fraction of stars hosting a transmitter; Sheikh et al. (2025) asked where Earth's own technosignatures would be detectable. Such a bound is valid only if the sample represents the population it is meant to describe. This sample was assembled by other observers for other purposes, so it does not, and §3 states what it is instead.

The statistical treatment this search needs is unusually

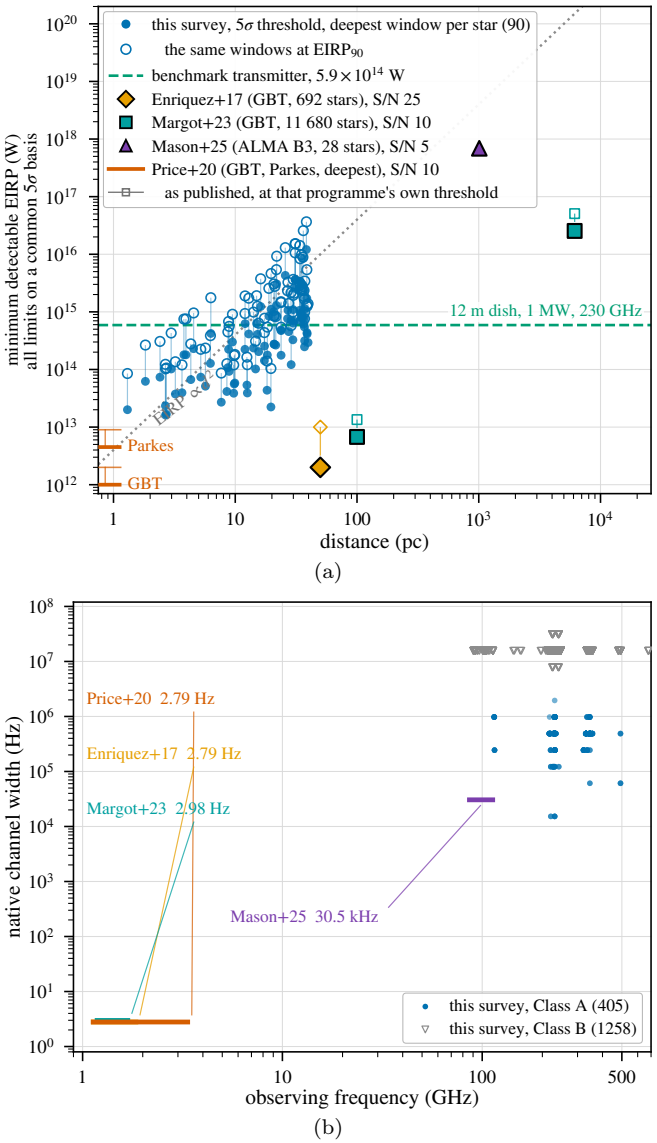


FIG. 1.— Where this survey sits among published radio technosignature searches. Every limit plotted is a total-power limit for a signal narrower than the native channel of the experiment that set it. Those channels differ by 7 orders of magnitude, from 2.79 Hz to 31.2 MHz, so the programmes are sensitive to different classes of spectral feature and a signal detectable by one of them need not be detectable by another at any power. **(a)** Minimum detectable EIRP against distance, all on a common 5σ basis. Blue, this survey, the deepest window per star: *filled*, that window's bare 5σ threshold, which carries no correction for the instrument's spectral response and so stands on the same footing as every other threshold in the panel; *open*, the same window at EIRP_{90}, the single-epoch trigger completeness of §5.1, which does carry that response, joined to its own filled symbol. Orange, Enriquez et al. (2017); aqua, Margot et al. (2023); violet, Mason et al. (2025); left-edge ticks, Price et al. (2020). Each of these is drawn at its published limit rescaled to 5σ by the factor in Table 1, with the published value itself as the hollow marker joined above it. Grey dotted, EIRP $\propto d^2$ at constant received flux; green dashed, the illustrative 12 m benchmark transmitter of §6.3. No symbol in the panel carries a confirmation criterion. **(b)** Native channel width against observing frequency for every searched window (filled, Class A; open, Class B), with the comparison programmes as horizontal segments at the widths of Table 1.

detailed, and the appendices set it out in full rather than in summary.

3. THE SAMPLE

This search covers 1663 spectral windows toward 90 stars in 83 systems within 40 pc, drawn from 407 public ALMA execution blocks and 4.1 TB of raw data through one frozen pipeline, 281 h of telescope time and 1098 h summed over windows. That set is the *primary census*: every sensitivity, every expectation and every limit here is computed over it, and so are 50 of the 56 threshold crossings this paper reports. Of the rest, 4 are coarse-channel events in the one block that fails the data-quality criterion of §4.2 and 2 are fine-channel events in archival-tail blocks; both sets are kept in the ledger and enter no expectation. Four further sets of blocks appear without being part of the census: that archival extension, a hold-out reserved before any block on its own side of the rule was searched, an out-of-sample control beyond 40 pc, and a rank-calibration sample drawn from the blocks already counted. Table A1 names them in rows that sum, and Table A2 resolves every count of windows, crossings and systems into the set it is counted over.

Star means one stellar component, *system* a bound group counted once however many of its components were observed, and *target* one ALMA pointing.

The census is two experiments: 405 fine-channel windows toward 66 stars in 61 systems resolve a drifting carrier — *Class A*, and every sensitivity and exclusion quoted here refers to them unless stated otherwise — while the remaining 1258 coarse-channel windows support a secondary search for unresolved spectral excess with no drift discrimination, *Class B*.

The inclusion criteria were applied by query. A star enters if its Gaia DR3 parallax is at least 25 mas, measured to better than 20 per cent, and if its proper-motion-corrected position at the observation epoch falls inside the quoted field of view of at least one public execution block. Distances are direct inversions, $d = 1/\varpi$, and the star need not sit at the pointing centre. One further rule applies to the window and not to the star: a window is kept only where the primary-beam response at the star is at least 0.22, the offset at which the Gaussian the pipeline applies and the response of a blocked circular aperture part by 10 per cent. It removes nothing: the furthest window in the census lies at $0.82\,\lambda/D$ against a limit at $0.83\,\lambda/D$, so the limit does not stand clear of the distribution, and the windows it is nearest are the 8 off-axis ones on ϵ Eridani, which the survey searches. The precision clause never binds: the worst fractional parallax error among the stars searched is 2.5 per cent. The query returns 168 work-list entries; the quoted field of view is optimistic, and only 115 of those stars are in fact contained by a public field, of which 90 yielded a searchable window. One target has no catalogue name — a serendipitous field star that appears throughout as ALMA J153702653−33192492, Gaia DR3 6015967739554000768, an M5 dwarf at 35.0 pc with no measured systemic velocity.

Disc, planet and colour enter as descriptive metadata only. One exclusion, though, is a property of the catalogue and not of the sky: a Gaia DR3 parallax needs a Gaia DR3 source, and the brightest stars have none. Every one of the 22 stars in this volume brighter than second

magnitude is absent from DR3 astrometry, so 257 of the 4227 Hipparcos stars whose own parallaxes meet the criteria above never reached the work list. Of these, seven lie inside a public field of view — Sirius A, α Centauri A, Vega, Fomalhaut, α Centauri B, β Leonis and MCC 124 — and hold 38 fine-channel and 374 coarse-channel windows in 83 execution blocks and 1.0 TB of raw data, none of it searched here. Sirius shows what that costs. The three blocks that put Sirius B in this census are pointed at Sirius A, whose catalogue position lies within 0.4 arcsec of the delivered phase centre, while the companion was read 7.4–8.0 arcsec off axis in the same 12 windows. Each block's continuum map holds the primary at 27–46 times its own noise with spectral index 2.1, displaced 3.1–3.7 arcsec from that catalogue position by its own orbit, and nothing at the companion: the brightest star in the sky was measured three times by this survey and reported at neither position. Sirius A, α Centauri A and α Centauri B carry no fine-channel window at all, so they would add a constraint on unresolved excess rather than a carrier limit; the four that do are Vega, Fomalhaut, β Leonis and MCC 124, three of them A stars with bright debris discs. A search of this volume should admit these stars on Hipparcos astrometry where Gaia has none.

Two removals act on windows rather than on stars, and account for the whole difference between the 1721 rows the extraction wrote and the 1663 retained: 45 repeated rows of one window and 13 failing a quality cut that is physical rather than chosen; the primary-beam limit above removes none. Since $\sigma\sqrt{t_{\rm on}\Delta\nu_{\rm ch}}$ is set by system temperature and collecting area, a window more than 100 times below the sample median in it is not measuring the sky; those removed lie 4.7×10^{3}–3.0×10^{4} times below and the nearest retained 24 times, so the level is not a tuned one. No integration-time cut was applied; the shortest window retained carries 21 s on source.

3.1. *The archival selection*

This sample was not designed: it is whatever previous ALMA programmes happened to observe among the stars Gaia DR3 catalogues, and 90 per cent of the stars searched were targeted under a debris-disc or planet-formation proposal. Table 2 measures the bias against a volume-limited reference census of which the searched stars are a subset, classifying both columns by one rule so that every ratio below is like against like. Both columns are Gaia-defined, so neither contains the bright stars above and the ratios are internally consistent without them.

The dominant selection effect is distance. Stars within 5 pc are over-represented by a factor 53, and the enhancement falls monotonically with distance to a factor of two *under*-representation beyond 30 pc. Colour comes second: A stars are over-represented by about 17 and F and G stars by factors of three, because those are the stars with bright debris discs. For the A stars that factor is a floor, because the catalogue requirement is itself colour-selective: it removes 39 of the 196 stars in the volume as blue as an A star, 3.7 times its rate among all the others. M dwarfs are depleted — 41 of the 90 searched stars are M dwarfs, a factor 0.7 of their share of the reference census, and only 28 of them carry the fine-channel coverage that can resolve a drifting carrier at all. Coverage is also concentrated: 285 of the 1663 windows come from 3 systems. No statement in this paper is therefore an occurrence rate for the solar neighbourhood, and the selection acts hardest on exactly the targets a designed search would pick: of the 16 sample systems within 15 pc with a confirmed planet, 6 carry no Class A coverage and so no drift-resolving limit.

TABLE 2
Selection function of the searched sample against a volume-limited reference census. Both columns are the same Gaia DR3 selection — $\varpi > 0$, $\sigma_\varpi/\varpi \leq 0.1$ and $1/\varpi \leq 40$ pc, 17 566 objects — of which the 90 searched stars are rows, so the two are selected alike by construction; and both are classified by one rule, in which BP−RP colour sets the class and the absolute G magnitude separates the degenerates that colour alone would place among the A stars. The rule reaches 17 278 census objects and 88 of the 90 searched stars, the rest carrying no Gaia BP or RP photometry. "Ratio" is the searched column fraction over the census column fraction; > 1 is over-representation. The strongest selection effect here is distance, not colour.

Property	Census	Searched	Ratio
Distance			
$d \leq 5$ pc	48 (0.3%)	13 (14.4%)	52.9
5–10 pc	240 (1.4%)	13 (14.4%)	10.6
10–20 pc	2069 (11.8%)	18 (20.0%)	1.7
20–30 pc	5391 (30.7%)	19 (21.1%)	0.7
30–40 pc	9818 (55.9%)	27 (30.0%)	0.5
Colour class[a]			
A	61 (0.4%)	5 (6.2%)	17.2
F	473 (2.8%)	8 (10.0%)	3.6
G	978 (5.8%)	15 (18.8%)	3.2
K	2221 (13.2%)	11 (13.8%)	1.0
M	13 082 (77.8%)	41 (51.2%)	0.7
Other axes			
Confirmed planet hosts[b]	26 (15.5%)	24 (26.7%)	1.7
White dwarfs[a]	463 (2.7%)	8 (8.9%)	3.3
Searched stars / systems	–	90 / 83	–

[a] Colour bins, not MK types. The boundaries are the colours at which the census's own temperature–colour relation crosses 7500, 6000, 5300 and 3900 K: BP−RP < 0.36 for A, 0.36–0.72 for F, 0.72–0.95 for G, 0.95–1.77 for K and > 1.77 for M. Over the 8594 census objects that also carry a Gaia temperature the two classifications agree for 98 per cent, and every disagreement is between neighbouring bins. A star is counted as degenerate when it lies more than 4 mag below the main sequence measured from the census itself: the two sequences are separated by 5 to 8 mag here, so moving the cut by a magnitude either way changes the count by under 3 per cent. The five classes are normalised over the 80 non-degenerate searched stars and the 16 815 non-degenerate census objects.

[b] Both columns are the same positional join, of the 168-entry ALMA-covered work list and of its searched subset, to the NASA Exoplanet Archive `pscomppars` table accessed 2026 October 4. The 24 searched hosts carry 51 confirmed planets, 22 of the 24 surviving the deuterium-burning mass limit; the Gaia census carries no planet information, so the census column in this row is the work list and not the 17 566-object Gaia census.

3.2. *Frequency coverage and archival scope*

The Class A experiment covers $\Delta\nu_A = 47.7$ GHz of unique sky frequency, and that coverage is sparse: 26 disjoint islands between 114.0 and 873.1 GHz rather than a band, so those endpoints bound where observations exist and not what was covered. With the coarse-channel windows the archival coverage is 118.1 GHz in 46 intervals, of which 113.1 GHz survives the molecular-line mask of §4. Fig. 2 shows the distribution: a transmitter between the islands was never observable here, whatever its power.

The archive is not exhausted either. 177 further blocks

lie inside the same query but were unprocessed when the census was frozen; 144 have since been calibrated and searched, 115 of them reported here as repeat epochs, and the other 29 enter no number anywhere, for the reasons Table A1 gives row by row. The largest single gap is the 18 blocks whose raw visibilities the archive holds but never pipeline-calibrated, 237 GB in all; no effort of ours would close that.

4. DATA AND METHODOLOGY

The search looks for one kind of signal: an unresolved carrier at the star's own position, narrow compared with a single correlator channel, and drifting in sky frequency because the transmitter accelerates along the line of sight. Over a long track its power is spread across many channels, so a search that treats each channel on its own will miss it. The analysis de-drifts the data before it measures them.

4.1. *The de-drifted search statistic*

Each window's dynamic spectrum is stacked along every trial drift rate $\dot{\nu}$. The search statistic is the largest stacked signal-to-noise ratio anywhere on the resulting channel×drift grid,

$$T(\mathbf{x}) = \max_{\dot{\nu},\,\nu} \frac{\sum_t I(\mathbf{x}, t, \nu + \dot{\nu}t) \,/\sigma^2(t,\nu)}{\left[\sum_t 1/\sigma^2(t,\nu)\right]^{1/2}}, \qquad (1)$$

where $I(\mathbf{x}, t, \nu)$ is the spectrum at sky position $\mathbf{x}$ in integration t and channel ν, and $\sigma(t,\nu)$ is the noise scale there. Equation 1 is the matched filter for a constant-drift carrier: it adds the signal coherently along the track and divides by the noise of the stacked spectrum, so T is an absolute signal-to-noise ratio and not a local contrast. The noise scale is the median absolute deviation across 512 control positions at each integration and each channel, scaled to its Gaussian-equivalent standard deviation: local in time and frequency, global in position, and shared by the star and every control. The statistic is evaluated at one position at a time, identically at the star and at each control, which is what makes the two comparable. $T_\star$ denotes its value at the star.

Spectra are extracted at the star's apparent position, never from images. For each integration and each native channel the pipeline forms the weighted vector average of the phase-shifted visibilities, in total intensity: the mean $I = (XX + YY)/2$ of the two parallel-hand correlations. That average is the dirty-map value at the extraction position — an amplitude, not a deconvolved flux density, so in a field with extended emission it is not a point-source measurement. The data are used as the archive delivered them, through ALMA's own calibration and quality assurance, with their flagging carried through, every delivered channel searched, and no regridding, deconvolution or edge trimming. A per-integration median along the *frequency* axis removes the continuum; no median is taken along time, so a one- or two-channel feature cannot be removed by it. One subtlety matters for every power quoted below: the delivered channels are smoothed, so a channel's effective noise bandwidth is not its separation and flux density times channel width is not yet a radiated power. A response correction computed from the correlator's own channel shape supplies the difference (Appendix A).

One convention holds everywhere in the paper. The search runs on the topocentric sky frequencies ALMA delivered, and every drift rate quoted is the apparent one at the telescope, which is what the data contain and what a reader can recompute from the archived channel axes. Two steps need the star's own frame and both take it: a crossing is compared with the line catalogue, and a repeat epoch is read at the predicted cell, after each block's frequency axis has been corrected for the Earth's motion at that epoch and then for the star's systemic velocity.

Two power quantities carry the main text, both defined in Table 3: the trigger level $P_{\rm trig} = 4\pi d^2 S\,\Delta\nu$ at distance d, for a carrier that makes $T_\star$ reach $T_{\rm trig} = 5$ in one native channel of width $\Delta\nu$, and the sensitivity EIRP_{90}. The second is measured, not assumed: synthetic carriers of known amplitude were injected into calibrated visibilities and recovered through the unmodified pipeline. Three campaigns did that, each with its own denominator. The completeness itself is measured on 49 of the 405 fine-channel windows, injected directly, and on 5 936 carriers over 35 windows of the coarse class. What the spatial rank costs is measured separately, on 8 860 carriers over 52 of 56 attempted block–window configurations. The primary-beam correction is tested on a third set of 40 scored units, 10 of them also among the directly injected windows. Between them they span both ALMA arrays, four bands and the sample's full range of channel widths. Every limit quoted in this paper is the EIRP_{90} of its own window, given to two significant figures beside its native channel width; the measured completeness is §5.1 and its error budget is Table 4. Two comparisons bear on the amplitude scale and they are not the same kind of check. An independent reduction of the same AU Mic visibilities is reproduced to 0.994 ± 0.014, which tests the extraction and not ALMA's flux transfer, an error in which would appear in both reductions alike; the external comparison is with the published quiescent flux density of ϵ Eridani, recovered to 1.10 ± 0.31 (Burton et al. 2022) and so a 28 per cent measurement. The absolute scale is therefore carried at ALMA's own band-dependent accuracy specification, and the 14 Class A windows in Bands 8 and 10, at 492–493 and 849–873 GHz, accordingly carry ±21 per cent where the sample as a whole carries ±11.

The drift range searched is bounded by an acceleration, not by a frequency. Drift rate and line-of-sight acceleration are one quantity in two sets of units,

$$\frac{\dot{\nu}}{\nu} = \frac{a_{\rm los}}{c} \qquad \text{and so} \qquad a_{\rm los} = \frac{c\,\dot{\nu}}{\nu}, \qquad (2)$$

so a ceiling expressed as $\dot{\nu}/\nu$ is the same physical limit in every band. The drift and acceleration ceilings adopted for each window were computed from Equation 2 window by window, and reproduce it to 1×10^{-4} over all 405 Class A windows. The default ceiling is 12.0–13.3 $\mathrm{Hz\,s^{-1}\,GHz^{-1}}$, which is 12.0–13.3 nHz in the fractional units the literature uses, and is the acceleration of a transmitter orbiting a solar-type star at a few hundredths of an au. It is the Mason et al. (2025) fiducial with a threefold margin, widened for a confirmed exoplanet host whenever the innermost known planet demands more; at unit margin the search would have been blind to two of the windows that produced a crossing.

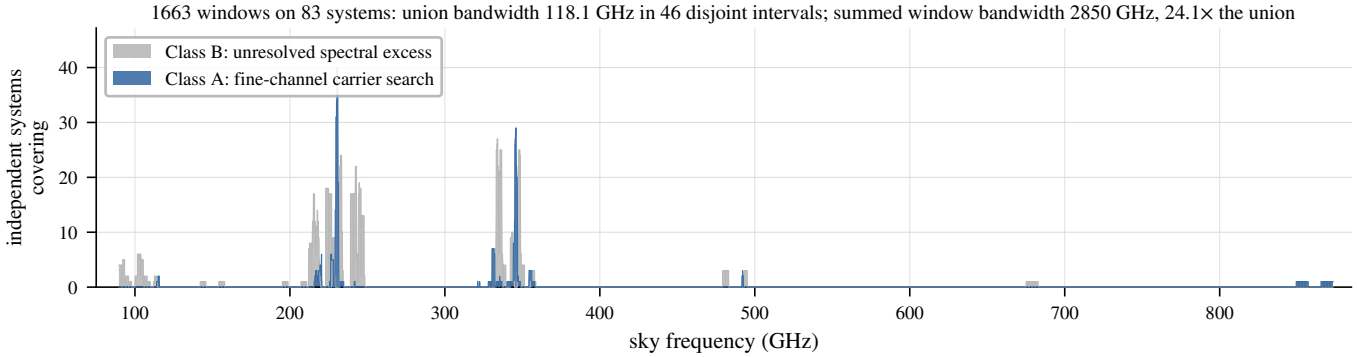


FIG. 2.— Where this survey searched. *Lower panel:* each searched window is a horizontal segment on its host system's row, rows ordered by distance; heavy blue marks the fine-channel carrier search (Class A) and light grey the coarse-channel spectral-excess search (Class B). *Upper panel:* the number of systems covering each gigahertz of sky frequency. The summed window bandwidth is 24.1× the union, so the union measures frequency coverage and the summed bandwidth search effort.

It is 6 per cent of the 200 nHz that Sheikh et al. (2019) recommend as a ceiling for a general search, their figure being set by the closest observed exoplanet orbit rather than by an orbit of a few hundredths of an au, and the restriction here is deliberate. The grid is uniform in $\dot{\nu}$, and fine enough that a carrier midway between two trials loses no more than 0.26 of a native channel over the longest track. Three limits follow: a deliberately chirped carrier is not distinguished from an accelerating platform; a transmitter need not be bound to a known planet, so drifts faster than the ceiling are unconstrained; and orbital and rotational acceleration add, so a transmitter on a rapidly rotating planet can exceed the ceiling on its spin alone — the fastest rotation measured for any planetary-mass object, that of β Pictoris b, which orbits one of the stars searched here, corresponds to 19.4 nHz (Sheikh et al. 2019). For the most demanding known case, transiting TRAPPIST-1 b, 29 per cent of the orbital phase lies above the default ceiling and 1 per cent above the widened grid used on that star's own windows (Fig. 3).

Only part of the archive can carry a drift search at all, and the two classes reported here are that division (Table 3). Channelisation in the sample is bimodal with an empty gap between the modes, so the split is unambiguous, and it coincides with the physical criterion that matters: whether the maximum searched drift crosses of order one channel or more over the on-source track, $\eta_{\rm drift} = |\dot{\nu}_{\rm max}|\tau_{\rm track}/\Delta\nu_{\rm ch} \gtrsim 1$. Each window's own $\eta_{\rm drift}$, trial-drift count and acceleration ceiling follow from its channel width, its track length and Equation 2, so the sample may be reclassified on that criterion instead; the two do not partition the survey identically. The two classes are physically different experiments, results are given per class, and they are combined only where that is appropriate.

4.2. *From a crossing to a confirmed signal*

Every window passes through the same three tests in the same order, and three words name what comes out of them: a *crossing*, an *unattributed crossing* and, had one survived, a *confirmed signal* (Fig. 4). Each is set out below as the quantity it measures, the number that quantity is compared with, when that number was fixed, and what passing it does and does not establish. There is no fourth test, and none of the three is a significance test.

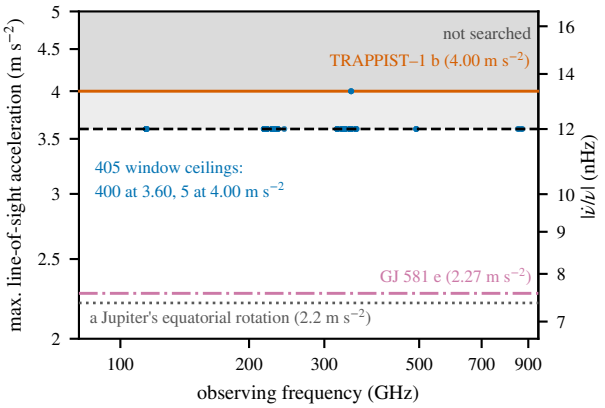


FIG. 3.— What the drift search reaches, as a physical selection function. The drift ceiling of each of the 405 Class A windows (points), converted to line-of-sight acceleration by Equation 2, against observing frequency; the shaded region above is unsearched. The right-hand axis is the same quantity as a fractional drift rate, $|\dot{\nu}/\nu| = a/c$, which is the unit the literature states drift reach in. The boundary drawn is the default ceiling, 3.60 m s^{-2}, equivalently 12.0 nHz, which is also the median; it runs to 4.00 m s^{-2} on the 5 TRAPPIST–1 windows. Horizontal lines are reference accelerations: the two most demanding known planets of any target, TRAPPIST–1 b (4.00 m s^{-2}) and GJ 581 e (2.27 m s^{-2}), and the equatorial rotation of a Jupiter, 2.2 m s^{-2} or 7.4 nHz. Earth's own rotation and orbit, 0.034 and 0.0059 m s^{-2}, lie below the foot of the axis. **Transmitters whose combined orbital and rotational acceleration exceeds the ceiling are not constrained by this search, and the ceiling is a budget they share**: a transmitter on the equator of a rotating giant spends 61 per cent of it on the planet's spin alone. Nor are unbound transmitters, or any platform accelerating for its own reasons.

(i) *Trigger → a crossing.* The quantity is $T_\star$, the dedrifted statistic of Eq. 1 read at the stellar position; the criterion is $T_\star \geq T_{\rm trig} = 5$ in at least one channel×drift cell; and both the statistic and the level were fixed before the archival search began. $T_{\rm trig}$ **is a trigger threshold and not a detection significance:** T is not shown to follow any standard distribution under the null, so the level carries no tail probability, and a window holds between 112 and 6.87×10^6 of those cells, so one level buys very different protection in a coarse single-pointing window and in a finely channelised one on a long track. Every cell that reaches the level is recorded, not only the largest in each window, because a carrier is masked as effectively by a noise excursion as by a line. A crossing establishes only that it goes on

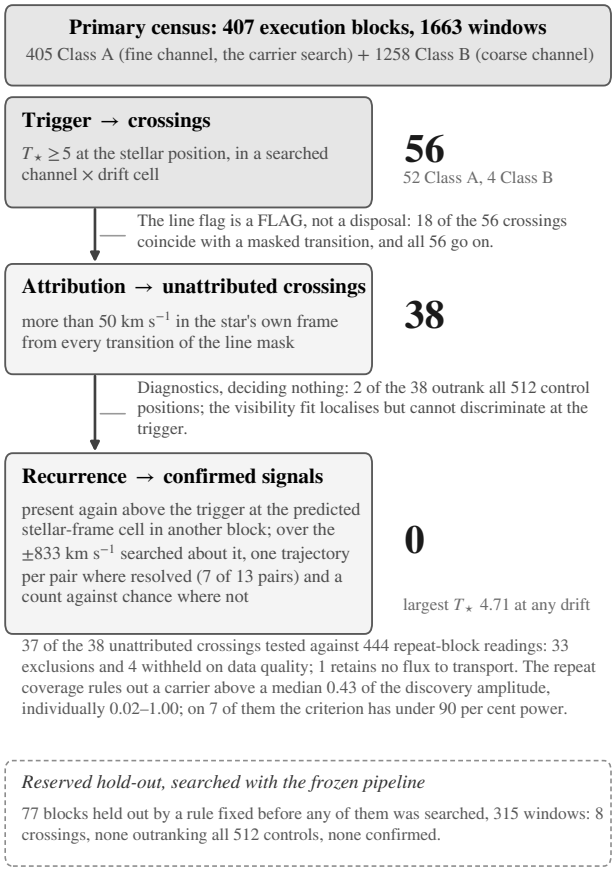


FIG. 4.— The detection logic, with the count at each level. Three levels and no more: a *crossing* is an event requiring investigation, an *unattributed crossing* is a target for follow-up, and a *confirmed signal* would be a detection. Coincidence with a catalogued transition is a flag and not a disposal, so all 56 crossings continue past it; the spatial rank and the visibility fit are diagnostics attached to the middle level and decide nothing. The reserved hold-out is drawn as a separate branch because a transmitter in it would matter whatever its calibration role. Every count is read from the ledger, which lists every crossing the search produced, including those in the archival-tail blocks and the quality-failing block that no expectation is computed over, so the figure, Table D1 and the text cannot disagree. The counts drawn are the primary crossings, one to a window; the 21 secondary crossings of §5.3 enter the same three levels and three reach the second, none the third.

to the next two tests.

(ii) *Line attribution → an unattributed crossing.* The quantity is the crossing's velocity offset from the nearest transition of a molecular-line catalogue, evaluated in the star's own rest frame; the criterion is $\pm 50\,\mathrm{km\,s^{-1}}$. The statistic and the trigger were fixed before the archival search began; the line catalogue is a single database query made afterwards, on 2026 October 6, so the dispositions rest on the physical evidence set out beside each one and not on the order in which the list was assembled. The half-width follows the Keplerian line-of-sight velocities of gas surviving beyond a few au, with margin for the systemic velocity. A crossing inside that window is *attributed* where the species is one that such systems are known to carry: a coincidence with a catalogued transition makes an astrophysical origin far more likely than a transmitter, the more so where a second transition of the same species or known circumstellar emission is present. A further three crossings coincide with a species no debris disc has been shown to carry; their offsets are reported and they are counted as unattributed (Appendix C). A coincidence is in any case a statement about a catalogue rather than a property of a frequency, so this is an attribution and not an exclusion, and **every crossing goes on to the recurrence test whichever side of the window it falls**, wherever a later block covers the frequency. An unattributed crossing is a target for follow-up and not a detection. The window's price is that the survey constrains nothing inside it (Appendix C).

(iii) *Independent recurrence → a confirmed signal.* The quantity is $T_\star$ at the cell the first epoch predicts, read in another execution block of the same star at a single drift rate after the frame correction of §4.1; the criterion is that it reach the trigger again; and the test was specified before the repeat epochs were searched. It is a weak criterion, and the archive's uneven temporal sampling is why: only part of the sample has epochs separated well enough to apply it at all.

Two diagnostics hang off the second test, and the word is exact: each says which crossing deserves attention first, and neither decides anything. The first ranks $T_\star$ against the same statistic at 512 control positions in the same window, drawn once from a fixed seed into an annulus placed by each block's own synthesised and primary beams so that no control sits on the star; its lowest value, $1/(512+1)$, means the star beat every control. It is a ranking filter and not a significance test, because the stellar and control positions are not demonstrably exchangeable: out of sample, on 315 windows the frozen pipeline had never seen, the stellar rank has median 0.405 where one half is expected if the two were interchangeable, and a parameter-free correction for the annulus geometry does not restore an exchangeable statistic (Appendix B). It also responds to compactness rather than to origin, as β Pictoris shows: genuine carbon monoxide ranks below its own control ring there, because the resolved disc lifts the ring along with the star. The second diagnostic is a visibility-domain fit, which asks whether a crossing's emission is at the stellar coordinates rather than elsewhere in the field; it cannot discriminate at the trigger either, because a noise maximum at the star has flux at the star (§5.5). **Neither gates a disposition, so neither is charged to the sensitivity**, and the limits here are correspondingly deeper than a chain charging for the rank would give. Throughout, p denotes a genuine significance level, and no probability computed against the rank's own null is called one.

One execution block is not a null, and a uniform rule says so. The comparison with the control positions needs each block's ensemble to behave like noise, and a block fails when the median control statistic over its own windows, scaled by the median of that quantity over its class, exceeds twice that scale. Of the 407 primary-census blocks, 1 fails — A002_Xc7fa6f_X28a8 toward BD+05 1668, at 4.1 where the next worst is 1.12. Its 4 windows are all coarse-channel, so the Class A results are identical with it and without it — the same 405 windows and 52 crossings either way — but it holds 4 of the

56 crossings, the survey's entire Class B crossing population. **The quality-passing data therefore hold 52 threshold crossings, all of them Class A**, and that is the number every expectation here is set against. The other 4 are reported in full in the ledger as the record of a block whose own control ensemble does not satisfy the noise model the rest are compared with, and not as crossings of the same standing (Appendix D).

Because a fixed threshold is not a fixed false-alarm rate, the number of crossings expected by chance is computed window by window, from the exceedance rate measured at each window's own control positions; the reserved blocks then test the one assumption that makes, which is that a star behaves like a control position in its own field (§5.3.1).

Three corrections precede the statistic: the extraction field is chosen so that the star lies inside the delivered primary-beam field and not merely inside the quoted field of view; the stellar position is propagated to the observation epoch for proper motion, though not for the annual parallax, whose omission is corrected window by window in §5.1; and on-source time is accounted per integration from the delivered exposure records rather than from the scan intentions. That the chain finds what it is meant to find is shown twice over, by the injected carriers above and by β Pictoris CO — a real feature at a known stellar position, recovered and localised in two transitions and 9 execution blocks (§5.4).

5. RESULTS

The primary experiment is the Class A search for a drifting, unresolved spectral carrier: 405 fine-channel windows toward 61 systems. The Class B search for an unresolved spectral excess is secondary and carries no drift discrimination: 1258 coarse windows. Together they are the 1663 windows of the primary census. **No event satisfied the confirmation criterion set out in §4.2.** What follows is first how deep the search reaches, and then what it found and how each finding was disposed of.

5.1. *Sensitivity: the power at which a carrier would have been found*

The sensitivity of this survey is a recovery completeness of one epoch, and not a threshold. Artificial carriers were injected at a random sub-channel phase and a random drift from each window's own grid and recovered through the unchanged pipeline; the power at which nine in ten come back is EIRP_{90}, the only sensitivity quoted here, measured through the single-epoch trigger and the localisation at the stellar position inside one execution block. **It is therefore the power at which a carrier would have produced a crossing and not the power at which one would have been confirmed**, because confirmation needs a second and separated epoch, which the archive supplies for some systems and not others whatever their depth (§5.6.1). The 512-position spatial rank decides nothing and is not charged against it either: charging it would cost a further factor 1.31 and in 3 bright-ring windows could not be met at any injected power (Fig. 5).

Measured that way the completeness is one number for the class. Nine carriers in ten are recovered at $3.03\,P_{\mathrm{trig}}$; the sampling error of that pooled value is ±2 per cent,

TABLE 3
The quantities and the three words this paper uses, defined once. Every term below is used in this sense throughout and in no other. The distinction the first three rows draw is the one the limits of §5.1 turn on: a crossing is a single-epoch event, and a confirmed signal needs a second and separated epoch that no sensitivity supplies.

Term	Definition
Crossing	A channel × drift cell at the stellar position reaching $T_\star \geq 5$ in one execution block; neighbouring channels above the trigger belong to one crossing. The largest in a window is its *primary* crossing and any other a *secondary* one, and because the largest cell is the window's own statistic a secondary crossing exists only where a primary one does. An event requiring investigation; no claimed significance.
Unattributed crossing	A crossing lying more than $50\,\mathrm{km\,s^{-1}}$ in the star's frame from every transition of the line catalogue. That is a reproducible frequency coincidence; which coincidences are astrophysically plausible is a separate judgement, stated with each one. A target for follow-up; not a detection.
Confirmed signal	An *unattributed* crossing recovered at the predicted stellar-frame frequency and drift in an independently separated block. The criterion is applied to every crossing, and a crossing coincident with a catalogued transition that satisfies it is a recurring molecular line: 13 of them are, which is this stage's positive control (§5.6). The reading is taken at one channel at one drift, so it holds only for a carrier whose stellar-frame frequency moved by between ±0.08 and $\pm0.63\,\mathrm{km\,s^{-1}}$ at most between the epochs compared; over the $\pm833\,\mathrm{km\,s^{-1}}$ also searched about that cell, a pair of crossings is tested against a single bounded trajectory where the drift grid resolves the interval between them (7 of 13 pairs, Table 5). It can be applied only where the archive observed the same frequency again, so for the 23 Class A systems where it did not, no sensitivity makes a confirmed signal possible.
P_{trig}	The power of a carrier that makes $T_\star$ reach the trigger in one native channel. An internal pipeline level: not a sensitivity, not a significance.
EIRP_{90}	The power at which nine injected carriers in ten produce a crossing at the star: a measured single-epoch trigger completeness, not a detection probability. Measured for the 49 injected Class A windows; for the other 356 an estimate from the class-average calibration.
Class A	The primary experiment: 405 fine-channel windows, median 488.3 kHz, toward 61 systems over 47.7 GHz of union bandwidth, in which the searched drift crosses of order a channel or more.
Class B	The secondary experiment: 1258 coarse-channel windows, mode 15.625 MHz, constraining a persistent unresolved excess. No drift discrimination, and never a carrier search.
The sample	Neither volume-limited nor representative of the solar neighbourhood: the stars ALMA happened to observe for other reasons (§3).

which is a statement about the class average and about no individual window. It is a total power set against a nominal threshold: $P_{\rm trig}$ is the bare 5σ of the window itself, and the loss a channel-confined carrier suffers to the correlator's spectral response is carried by the injected tones, so it sits inside the measured factor — $\times 2.29$ of it — and is not applied to the result as well (Appendix A). Every one of the 49 directly injected Class A windows resolves a ninety per cent point of its own, and the value does not depend on the brightness of its control ring: the 5 windows whose ring carries resolved emission come back at $3.24\,P_{\rm trig}$ against 2.94 for the rest (Fig. 5). Class B carries its own factor $4.17\,P_{\rm trig}$, measured the same way from 5 936 carriers injected into 35 coarse windows of 14 execution blocks; it is what sets the limits for Barnard's Star, Wolf 359 and Proxima.

The resulting limits. The median Class A window reaches 2.8×10^{15} W of equivalent isotropic radiated power and the median system, on its best window, 1.5×10^{15} W, with a spread across systems of 8.7×10^{13}–3.7×10^{16} W (Fig. 6a). The round 10^{15} W level is reached for 25 of them. The calibration of the factor is good to $\times 0.89$–$\times 1.11$ (Table 4); that is the interval on the median and on the counts, and not on any one window's limit. One bias is not inside it, runs one way only, and is *not* included in the values quoted here: an injected carrier is added to already calibrated visibilities and so suffers no atmospheric decorrelation, whereas a real carrier does. Measured block by block from the phase rms the observatory records on each block's phase calibrator (Cortes et al. 2026, §§10.4.11, 11.3.1), through the standard phase-scatter relation (Richards et al. 2022, Eq. 3), it leaves the median limit optimistic by 1 per cent and the median window above 400 GHz by 8 per cent, so the limits at the top of the frequency range are the least secure quoted here. **The 128 blocks whose delivery exposes no such record are not a random subset of the 407**: their longest baseline has a median of 279 m against 46 for the 279 that do, and it is the long baselines that lose coherence. Matching each of their windows to the measured blocks within a factor of two in baseline length puts the correction at 1.9 per cent in the median and 21 at the ninetieth percentile, so their limits are conditional on a calibration performance the archive does not report. *That is why this term is stated and not folded in*: the three factors that are applied are available for every window of the census, while this one is available for 279 blocks of 407 and the rest are measurably the harder ones, so folding it in where it is known would leave a set of limits that is neither corrected nor uncorrected.

Three further factors act window by window, and all three are divided out of the limits above rather than quoted beside them (Appendix A). The first is the primary-beam response at the star's own offset from the pointing centre, which is what turns a measured noise into a flux at the star: taken at the full width the observatory documents, it costs nothing in the median window and reaches 23.2 per cent in the 12 windows furthest off axis. The search's phase centre also omits the annual parallax, which displaces the star by at most 0.67 of a synthesised beam and makes the median Class A limit shallower by 0.4 per cent, while ϵ Eri in `A002_X99c183_X4f1e` has no astrometric keys and is left uncorrected. That correction is concentrated rather than general, which is why the median limit moves far more than the median window's own factor (Table 4): 219 of the 405 windows need none at all, while 14 lose more than a tenth and lie around the middle of the distribution of limits. The third the injections do not reproduce at all, each carrier being laid down at one frequency per integration while a real one sweeps within it. **It costs nothing in the median window and up to $\times 1.53$ in the 13 finest-channel ones, which are also the deepest**: the best Class A window moves from 6.8×10^{13} to 8.7×10^{13} W, and HIP 10679 drops out of the 26 systems that would otherwise reach 10^{15} W.

Carriers were injected directly into 49 of the 405 Class A windows, 12 per cent, and the result transferred to the remaining 356, which is the dominant caveat on all of the above. The injected windows are the Class A windows of execution blocks drawn in a stratified random order fixed in advance and without reference to which windows hold a crossing; they hold 7 crossings against the 6.3 the population rate predicts. *The unit is the block and not the window*, because a block's calibrated visibilities are what an injection campaign has to rebuild, so band and array were fixed by the draw and the conditions that vary within a block were compared with the population afterwards. **That comparison covers 6 properties** — channel width, integration time, noise, control-ring brightness, the channels a carrier at the drift ceiling crosses in one track, and the intra-integration retention — and on every one the injected windows span the central eight-tenths of the population, five agreeing in the median to 6.3 per cent. The sixth, the noise level, does not: the injected windows are $\times 1.7$ noisier, which follows from drawing whole blocks, and it is also the property the measurement is built to be insensitive to, each ladder being in units of its own window's threshold. That is measured rather than assumed: splitting the injected set at the median of each property in turn moves the ninety per cent point by at most $0.27\,P_{\rm trig}$. The draw controlled the array, but the achieved mix still departs from the population by 22 per cent of the sample; re-weighting the pooled curve by the survey's own composition gives $2.98\,P_{\rm trig}$ against the adopted 3.03.

A window that was not injected therefore carries an estimated EIRP_{90}, and across the injected windows the ninety per cent point ranges from $\times 0.80$ to $\times 1.58$ of the adopted factor: that dispersion, not the calibration interval above, is what one uninjected window's own limit is uncertain by. Because a system's limit is its best window, a minimum over several such estimates could favour windows whose own factor falls below the class average; drawing each window's factor from the 49 measured ones instead, over 4 000 trials, moves the per-system median deeper by 0.5 per cent, holds the count at 10^{15} W to 24–25 in nine trials in ten and spreads one system's own limit by -17 to $+9$ per cent, so the estimate is mildly conservative. But **the best window was itself injected into for only 6 of the 61 systems**: for the other 55 the headline limit is an estimate.

A carrier that is not always radiating is penalised in proportion. The de-drifted stack is coherent across a whole execution block, so over 3 888 trials at duty cycles down to 10 per cent the recovered amplitude followed the duty cycle to 1 per cent: a transmitter radiating a tenth

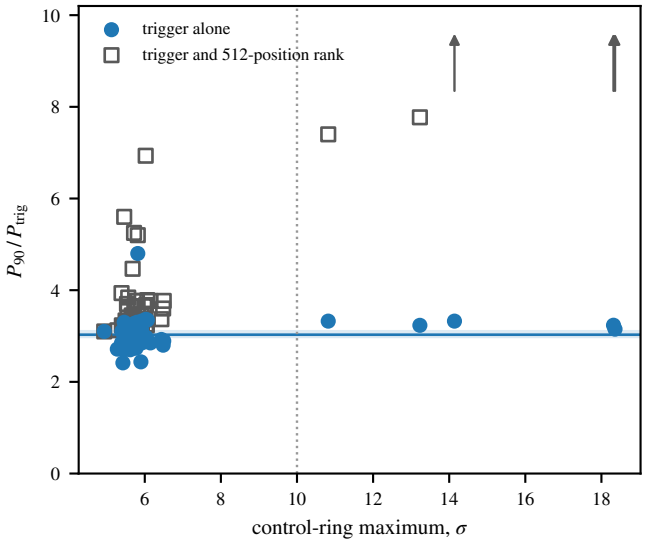


FIG. 5.— Why one completeness factor serves the whole class. Each of the 49 of the 405 Class A windows injected directly is plotted at the power that recovers nine carriers in ten, against the maximum of its own 512-position control ring. Filled circles are the criterion adopted here — the trigger and localisation at the star — with the pooled value and its uncertainty as the line and band; they show no trend with the ring. Open squares add the demand that the carrier outrank every control position, and arrows mark the windows that then reach ninety per cent at no injected power. Dotted, the 10σ ring brightness above which a ring carries resolved emission rather than noise.

of the time needs $10\times$ the power quoted above. Those trials carry no drift, so the joint drifting and intermittent case is not measured here.

5.2. *The nearest stars: Barnard's Star and Wolf 359*

Barnard's Star is the second-nearest stellar system and Wolf 359 the third if brown dwarfs are excluded, and to our knowledge these are the first millimetre or sub-millimetre technosignature limits for either; earlier radio searches of the nearest stars have worked at centimetre wavelengths (Enriquez et al. 2017; Price et al. 2020; Margot et al. 2023). Both are non-detections. *Both limits come from the secondary Class B experiment and not from the carrier search*: the archive holds only coarse Band 6 windows toward these stars, so what is constrained is an unresolved spectral excess and not a drifting carrier. On the EIRP_{90} convention of §5.1 Barnard's Star reaches 2.7×10^{14}–4.0×10^{14} W over 12 windows and Wolf 359 3.1×10^{14}–4.5×10^{14} W over 12.

5.3. *Threshold crossings and how each was disposed of*

The result is a chain of three tests, and it is read in one direction. Of the windows the search covers — the 1663 of the primary census and the archival-tail blocks beside them — 56 hold a cell reaching $T \geq 5$ at the stellar position. 4 lie in the one execution block that fails the quality criterion of §5.3.2, fixed before it was applied, so the population that characterises the search is the 52 in quality-passing data and those 4 are reported beside it. The molecular-line mask, $\pm50\,\mathrm{km\,s^{-1}}$ wide in each star's own rest frame, places 21 of the 52 within that distance of a catalogued transition; three of those coincidences are with a species no debris disc has been shown to carry and are reported rather than acted on (Appendix C), leaving 18 attributed and 34 unattributed. The attributions are kinematic and not an accident of bandwidth: in the delivered frame, 11 of the 51 crossings with a measured peak frequency fall within $\pm50\,\mathrm{km\,s^{-1}}$ of a transition these setups were tuned to, against 4.8 expected were each peak free to land anywhere on its own window's grid (Appendix E). Of the unattributed, 2 outrank all 512 control positions of their own window, and none was found again in any later epoch covering its frequency. **No crossing survives the chain, and this survey reports no detection.** Figure 4 counts what survives each step and Tables D1 and D2 list every crossing.

Anything bright in a window hides a weaker carrier behind it, so the weaker cells were sought too. The de-drifted statistic is retained in every channel, so behind a window's tabulated crossing — its largest cell — weaker above-trigger cells can be read as well, neighbouring channels counting as one. Re-read that way, the 399 of the 405 Class A windows that retain such a profile — the other 6 retain none — hold 49 maxima at or above the trigger, one for each of the census's 50 Class A crossings but HD 14055's, whose own window retains no spectrum, and twenty-one weaker cells besides, in 9 windows and as many as 8 in one. 18 are the resolved carbon monoxide of a disc spread over channels; the three unattributed ones, toward HD 14055, 61 Vir and β Pic, each have 3–16 covering blocks and none reaches the trigger in any of them, the deepest repeat being $T_\star = 4.4$. Applied to the window maxima the rule returns 50 of the 56 tabulated crossings to 0.002 of a channel; the 6 it misses are the 4 of the failing block, whose repaired profile was not retained, and crossings whose own window retains no usable profile. Cells are counted in the controls too, and outside the mask the star holds 1.05–1.15 times as many as a control position at every level from $T = 3$ to the trigger, with no step where a carrier search would put one; the expectation that follows is §5.3.1's.

5.3.1. *Most of the crossings are expected by chance*

The trigger is a threshold on a statistic and not a false-alarm rate: the number of searched channel × drift cells per window spans 4.8 decades, so one level buys very unequal protection. How many crossings chance should produce is therefore measured window by window, over the 405 Class A windows of the census and with one condition on both sides of the comparison: a cell at the stellar position that reaches the trigger and that the line mask does not attribute. A window's own 512 control positions measure how often a position in that field reaches the trigger, and the channel each of those exceedances lands on measures how often one would be attributed: 88 of 302 fall inside an attribution window, which takes 29 per cent of the expectation, where a maximum free to land anywhere on its own grid would take 20 per cent — the fields that hold line emission being where a control position most often triggers. The control positions differ from the star in one respect that bears on this: they carry no continuum, so a calibration error that varied with frequency would raise the star and not them. Measured on the continuum these stars are found to have, the windows where such an error could reach the trigger hold no crossing and bound it below 6.6 per cent (Appendix B); and the mild stellar excess of §5.3 does not follow that continuum as such an error would, standing at 1.01 in the

TABLE 4

WHAT SETS $EIRP_{90}$, AND IN WHICH DIRECTION. THE UNCERTAINTY ON THE CALIBRATION IS AN UNCERTAINTY ON THE CLASS-AVERAGE FACTOR, AND SO ON ANY STATISTIC FORMED OVER MANY WINDOWS; ITS FLUX TERM IS BAND-DEPENDENT, SO THE SUBMILLIMETRE WINDOWS CARRY THE LARGER COMBINED FIGURE. THE SPREAD BETWEEN WINDOWS IS AN OBSERVED DISPERSION AND IS THE STATEMENT ABOUT ONE WINDOW'S OWN LIMIT; THE SIGNED TERMS ARE BIASES, STATED WITH THEIR DIRECTION RATHER THAN FOLDED INTO AN ERROR BAR, AND THE DECORRELATION TERM IS MEASURED IN 279 OF THE 407 BLOCKS AND NOT IMPUTED TO THE REST.

	value	what it does to the limit
Class-average completeness	×3.03	nine carriers in ten recovered at this multiple of a window's own trigger power. *Measured* in the 49 Class A windows injected directly and *estimated*, at this factor, in the other 356
Uncertainty on that calibration (per cent), combined in quadrature		
The factor itself	±2	sampling error of the pooled value over the 49 injected windows
Absolute flux scale	±10.2	the observatory's own band-dependent specification (ALMA Proposer's Guide (Privon et al. 2026), with the Technical Handbook's quality-assurance criterion (Cortes et al. 2026) at Band 8, where the two differ): ±5 in Bands 1–5; ±10 in Bands 6 and 7; ±20 in Bands 8–10, weighted over this sample's bands
Distance and visibility scale	±4.5	the Gaia parallaxes, which enter as d^2, and the block-to-block visibility scale
Combined	$-11/+11$	×0.89–×1.11 on the factor, and so on the median limit and on the system counts; ±21 for the 14 Class A windows in Bands 8 and 10, at 492–493 and 849–873 GHz
Spread between windows. An observed dispersion, not an error bar.		
Window to window	×0.80–×1.58	the 49 injected windows resolve 2.41–4.80 $P_{\rm trig}$ individually. This, and not the interval above, is what an estimated limit is uncertain by; it is wider, and it applies to 55 of the 61 systems
Applied. Folded into every limit quoted, and so not combined with the above.		
Intra-integration smearing	×1.00–×1.53	a carrier sweeps through part of a channel within one integration and the injected ones do not: $\eta_{\rm smear}$ at the drift that sets a ninety per cent point, from each window's own integration length where it is recorded (1054 windows) and from the longest delivered anywhere where it is not (609)
Annual parallax	×1.00–×2.06	the phase centre omits it, so each limit is divided by the amplitude a compact source retains at its own displacement, a median 0.13 per cent
Signed terms. A bias is not an error bar and is not added to one.		
Channel response	×2.29	the share of a channel-confined carrier the peak channel keeps. Carried by the injections, so already inside the completeness above, which is this factor times a margin of ×1.325 over the nominal trigger
its residual	+8	the injected profile falls short of the exact response: *conservative*
Decorrelation	$+0/+1$	measured block by block from the phase rms the observatory records on each block's phase calibrator, which an injected carrier does not suffer and a real one does: +8 above 400 GHz, and more than +20 in 5 per cent of windows

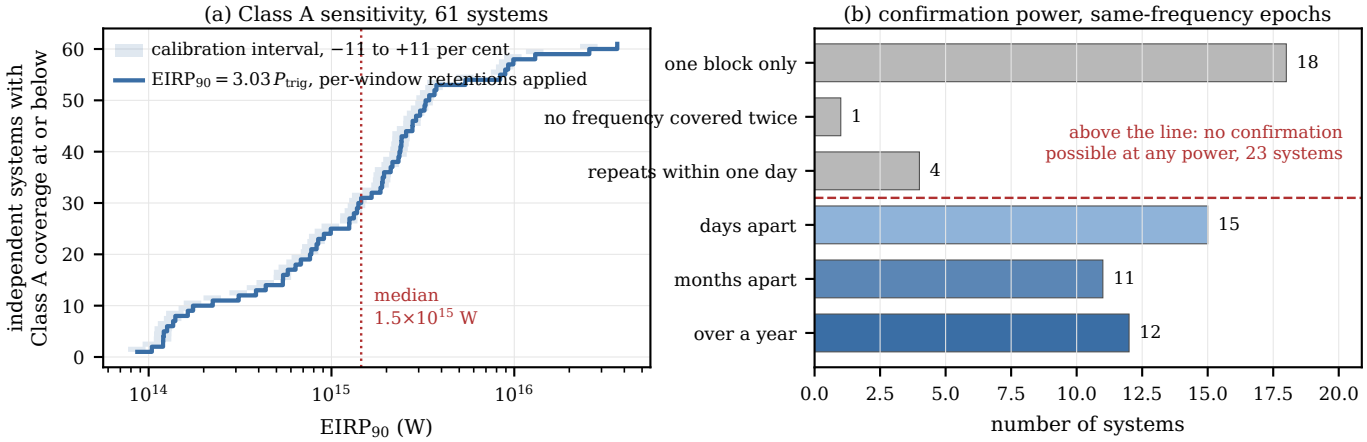


FIG. 6.— **The two completenesses of the carrier search, over the same 61 systems.** *(a)* Single-epoch trigger completeness: the cumulative number of systems whose best window reaches a given $EIRP_{90}$ (Table 3), evaluated as $3.03\,P_{\rm trig}$ with each window's own primary-beam, annual-parallax and smearing retentions applied; the shaded band is the calibration interval of Table 4. *(b)* Confirmation completeness, which no amount of power buys: systems grouped by the longest separation between two execution blocks covering a common frequency, more than 1 day counting as independent. **Left of the dashed line, 23 systems have no such pair, and for them confirmation was not possible at any sensitivity in panel (a)**: what panel (a) gives them is a crossing limit, not a detection limit.

brightest third of the 350 windows whose star has a photospheric estimate against 1.06 in the faintest. Chance predicts 25.8 unattributed Class A crossings, with 11–43 allowed by resampling the execution blocks the windows come from and 9–47 once the scatter in a count is allowed for as well, and 32 are found. Counting cells rather than windows on both sides moves the two together: the same 3 192 control positions predict 28.1 cells, 2.2 of them beside a window's largest, against 34 and 3 found and 13–48 allowed, because a control position holds a second above-trigger cell about as often as a star does (§5.3). A further two unattributed crossings, both toward HD 14055, lie in archival-tail blocks the catalogue does not tabulate window by window, and enter neither side.

The expectation is a sum of rates measured at control positions, and the 8 events of the reserved blocks measure how far a star departs from them: carrying that difference through moves it to 27.6 and the count agrees with chance either way, with an added population of as many as 23 crossings sitting inside the range unremarked.

What tests each crossing individually is the recurrence test of §5.6.

5.3.2. *One execution block accounts for four of the crossings*

4 crossings belong to BD+05 1668 (GJ 273, Luyten's Star), all in one ACA execution block, `A002_Xc7fa6f_X28a8`, whose whole field is above threshold: 512 of 512 control positions exceed $T = 5$, where the median of a window's own control ensemble is 2.75 across the survey. Its single-integration noise is as predicted (0.1609 Jy against 0.1611 Jy) but does not integrate down — a jackknife on the channel mean gives 0.030 Jy against 0.0048 Jy thermal, 6.2 times too large, while neighbouring channels stay uncorrelated (0.007). What the two numbers describe is a residual that persists from one integration to the next and differs from channel to channel. Equation 1 divides by the noise of the stacked spectrum, which falls as the root of the integration count, so a residual that does not average down accumulates along a drift track faster than the denominator grows, and it does so at every position in the field; and having no part common to the channels of one integration, it is not what the continuum median of §4.1 removes. The 24 sibling executions of the same field, array and setup reach only $T_\star = 4.26$ and produce no crossing. *What is inflated is the field and not the star:* the strongest of the four reaches $T_\star = 73.4$ where its own controls peak at 74.7, and 1–421 of the 512 stand above the star. The criterion fails this block and no other (Appendix D.2), so its 4 crossings are reported apart from the 52; Appendix E carries what the block might instead be. Milder cases of the same condition are not unique to it: read window by window the same diagnostic flags 5 windows in 2 blocks, and the 1 flagged crossing outside this block — HD 48370 — is withheld for it too.

5.3.3. *Stellar flares are excluded by a broadband prediction*

The sample contains known flare stars. But a millimetre flare raises every channel of a window alike, so one able to produce a single-channel crossing at the trigger must produce a $5\sqrt{N}\sigma$ continuum event in the same integrations. Over the 48 crossings with a recoverable continuum measurement that prediction runs 49–4 717σ, against an observed continuum statistic of median -0.29 with $|Z| < 3$ in 45 of them. One block settles it: AU Mic `A002_Xa42f75_X319` holds both a crossing ($T_\star = 5.27$) and the millimetre flare of MacGregor et al. (2020), and *that flare contributes 0.17σ to the single-channel statistic* — raising the continuum at $Z = 10.3$ while the crossing's own de-drifted track reaches $Z_{\max} = 2.1$.

5.4. *β Pictoris: an astrophysical positive control, and the obstacle it identifies*

A search that finds nothing must show that it could have found something, and the survey's strongest features show exactly that. The frozen pipeline recovers the β Pictoris circumstellar carbon monoxide blind, without being told the disc is there. It recovers it in two rotational transitions and in 10 execution blocks spanning 8.4 yr. Those blocks belong to 6 observing sessions, several of them executions of one session on one tuning, and each transition is recovered in more than one session: the closest two sessions to hold the same transition are 25 d apart, so what repeats here repeats between epochs and not inside one. In 9 of the blocks the star also outranks every control: 5 Band 3 windows at $T_\star$ up to 30.76, and 4 Band 6 windows at 11.68. The identification is kinematic and does not rest on those statistics. Carried through the frame chain — to the barycentre at each block's own epoch, then to the star's rest frame — all 10 crossings lie within 0.5 km s^{-1} of the stellar systemic velocity, in the two lowest CO transitions of a star with mapped circumstellar CO (Dent et al. 2014; Matrà et al. 2017). The Band 3 peaks agree across blocks to 1.00 channel once each block's own tuning is removed, so any apparent drift is far below the grid searched. The alternative reading requires a transmitter placed by coincidence at matched offsets from two CO transitions of one star.

It is also the control that shows the recurrence test of §5.6 working in the positive direction: every flagged β Pictoris carbon-monoxide feature is recovered in another execution block, and each transition in a block of another session as well, the screened Band 6 window returning $T_\star = 8.95$ in the second block of its own recurrence set. A non-recurrence elsewhere means nothing unless the same machinery recovers a repeat where one exists.

One of those blocks is a counterexample, and it is the more useful half of the control. In it several of the 512 control positions exceed the stellar statistic, so the spatial rank disfavours a window holding carbon monoxide known to be real and centred on the star; the disc is resolved on the scale of the primary beam in that configuration, which is the geometry the statistic is built to reject. The rank is therefore a test of compactness and not of authenticity, and on a real extended source it errs conservatively — a stronger statement than a clean positive control, because it bounds the diagnostic's reach as well as demonstrating it (Fig. B1).

HD 48370 is the complementary case and is equally instructive. Its crossing reaches $T_\star = 26.83$, but the control ring rises with the star, to 18.36, as emission filling the primary beam does. The velocity matches the prominent CO(2→1) that Cataldi et al. (2023) report toward the source and flag as possible foreground cloud emission. The block's ^{13}CO(2→1) window is bright at the ring and faint at the star. What identifies the extended emission is therefore the *level* of the control ring and not the star's rank against it — the star here outranks every control — which is why the window-level flag of Appendix D.2 is defined on the ring's median.

Taken together, these two results identify the dominant obstacle for millimetre technosignature searches, and it is not sensitivity. The strongest apparent signals in this survey are circumstellar and foreground molecular emission, and 18 of the 56 crossings land on a catalogued transition. The mask that removes them costs 6.4 per cent of the Class A union bandwidth over 12 species, and it is nevertheless the single most productive test in the chain. The warning for future work is about line lists rather than about depth, and Appendix C puts a number on it.

5.5. *The two crossings that lead their own control fields*

Frequencies in this section and the next are in the star's own rest frame, which is the frame the recurrence test registers in; the delivered ones are in Table 6 and in the

ledger.

Two crossings reach a stellar statistic above every one of the 512 control positions in their own window: 61 Vir at 346.3232 GHz and CP−72 2713 at 344.2965 GHz, both in Band 7. Both are tabulated in Table 6, the first is drawn in Fig. 7, and neither is a detection. The control rank is recorded for them because it says which crossing was examined first; it disposes of nothing (§4). CP−72 2713's crossing lies $-12\,\mathrm{km\,s^{-1}}$ from SO(8_8–7_7), a coincidence reported as a note rather than as a disposition (Appendix C). Both events go on to the recurrence test. Each was also re-fitted in the visibility domain, as 51 of the 56 crossings were, by phase rotating to the star and following the recovered drift; 41 of those fits are consistent with a point source at the star, and injected sources in the bin these two occupy return $\mathrm{Re}/\sigma = 6.58 \pm 1.97$, so the fit excludes none. Its value is that a pedestal fed through a sidelobe from elsewhere in the field, which a dirty map at the star cannot tell from a source at the phase centre, is not a point source there. Of the 512 control positions the rank screen uses, 8 retain a dynamic spectrum and 12 are reachable in the visibilities.

Neither crossing is present when the same stellar-frame frequency is observed again, and Table 6 gives both readings of that silence. The nearest covering block alone, 1.97 and 2.05 hours from each discovery, assumes least and rules out anything above 0.95 and 0.71 of the discovery amplitude; the combination over all 3 and 5 covering blocks deepens that to 0.56 and 0.66, but asks the same return of baselines of 3 and 2 days rather than of hours. Each reading covers a half-channel, ± 0.211 and $\pm 0.213\,\mathrm{km\,s^{-1}}$, which each carrier's own fitted drift would cross in 77 and 119 seconds: what is asked is a return to that frequency, not a stability at it (§5.6). At the faintest amplitude consistent with each crossing having been found at all, the criterion would have recovered a persisting carrier with probability 1.00 and 0.66, so for CP−72 2713 the silence is not an exclusion and is not read as one. Appendix B follows its crossing through every step of the chain.

5.6. *Recurrence, applied to every unattributed crossing*

One trajectory model carries every statement in this section: a carrier at a fixed rest frequency, whose line-of-sight velocity relative to its star is the same at the two epochs compared to within half a channel, and which drifts through each observation at the rate fitted there. Everything follows from it, read in the star's own frame. The predicted channel is set by the Earth's motion at the repeat epoch and the star's systemic velocity and by nothing else — that frame moves the cell by up to 116 channels, and for η Crv by −80.2 — while the fitted drift enters only as the de-drift within each observation. The statistic is read at that one channel, maximised over nothing. A *confirmed signal* is an *unattributed* crossing that satisfies the criterion (Table 3); a line-coincident crossing that satisfies it is a recurring molecular line, and is reported as one.

None of the unattributed crossings recurred, while 13 crossings attributed to circumstellar or foreground carbon monoxide do recur, as expected. Those 13 are the positive control of this stage and they test it on both of the ways real emission arises here: 10 are the resolved circumstellar disc of β Pic and 3 are foreground gas toward HD 285968, whose sightline carries catalogued cloud emission at the crossing's own velocity in the local standard of rest (Appendix C). No exclusion is formed for them, because what one would exclude has just been observed. Close to the trigger the control is thinner than that count suggests: every circumstellar recovery has $T_\star \geq 8.0$, and the 2 attributed crossings that are not recovered are the two faintest tested, both at $T_\star = 5.41$, so there the stage is controlled by HD 285968 alone — by foreground gas rather than by a point source. 37 of the 38 unattributed crossings were tested, over 444 readings in 151 execution blocks, and at the predicted cell the largest statistic over all of them is +2.00 at the carried drift and 4.71 over every drift trial, against a trigger of 5. The exception is HD 14055 `X1001fdc_X29ae`: 30 blocks cover its frequency, but its own window retains no spectrum, so there is no discovery amplitude to transport to them and no reading is quoted for it.

Every exclusion is conditional on that model, and the condition is narrow. The reading is taken at one cell at one drift, so it constrains only a carrier whose stellar-frame frequency returned to within half that cell: 488 kHz, or ± 0.08 to $\pm 0.63\,\mathrm{km\,s^{-1}}$ across the 33 crossings that carry an exclusion, median ± 0.21. It is not a requirement of frequency stability, and the crossings are not stable: their fitted drifts are line-of-sight accelerations of 0.5 to $3.6\,\mathrm{m\,s^{-2}}$, the carrier of Fig. 7 crosses 22 channels of its window inside the discovery observation, and at those rates one leaves the half-channel the reading covers in 77 and 119 seconds. The single cell therefore asks only that the carrier be back where it was — stable over the gap, or periodic and sampled at the same phase — and says nothing about one its own motion has carried elsewhere. Everything beyond that class is addressed by searching the whole covering window at every drift of the grid at each repeat epoch: $\pm 833\,\mathrm{km\,s^{-1}}$ in the median window, which spans the entire velocity excursion an orbit at the grid's own $3.6\,\mathrm{m\,s^{-2}}$ ceiling can produce, $176\,\mathrm{km\,s^{-1}}$. A catalogued window covers 32 of these 33 crossings, and the search spans that excursion in 30 of them; the other 1 is covered by no catalogued window at all.

The registration has a limit in time, and it belongs to the model rather than to the arithmetic. A fitted drift is known to half a grid step, $67\,\mathrm{Hz\,s^{-1}}$ in the median window; carried to the nearest covering block, that half step displaces a carrier at constant acceleration by 5 channels in the median case, by more than half a channel for 23 of the 33, and by 648 over the longest baseline. Half a step reaches half a channel after 2.0 hours, so a single-channel reading is registered that finely only over the first hours of a gap. The same cell was therefore read at every trial of the window's own grid as well, which bounds an error in the carried drift whatever its size: over all 444 readings the largest value anywhere is 4.71 against a trigger of 5, and under that wider reading the amplitude ruled out falls from a median 0.43 to 0.72 of the discovery amplitude with 7 of the 33 ruling out nothing — exactly the crossings the recovery probability below already reports as untested rather than excluded. No disposition rests on the carried drift.

The amplitude each exclusion reaches is the fraction of the discovery amplitude still excluded at the trigger, $(T_{\mathrm{rep}} + 5)/T_{\mathrm{pers}}$. Of the 38 unattributed crossings, 33

carry a quoted exclusion; 4 take their discovery amplitude from the block that fails the quality criterion of §5.3.2 and 1 retains no discovery spectrum, so none can be formed for those. Over the 33 the repeat coverage rules out a carrier at a median 0.43 of the discovery amplitude and the deepest at 0.017 of it, but the individual limits run to 1.00, and for 3 of them — β Pic at 241.77 and 241.86 GHz and HD 110411 at 219.44 GHz — nothing below 0.84 is ruled out. Combining every covering block buys most of that: the nearest block alone rules out a median 0.98 of the discovery amplitude, and 15 of the 33 rule out nothing. The chance that a carrier would have satisfied the criterion is evaluated at the trigger amplitude $5S/T_\star$ rather than at the discovery amplitude, which is selected as the largest of many cells at a statistic of about five and is biased high (Appendix B): it exceeds 90 per cent for 26 of the 33 and for the remaining 7 it is β Pic at 241.86 GHz (0.38), ALMA J153702653−33192492 at 347.19 GHz (0.50), β Pic at 241.77 GHz (0.61), CP−72 2713 at 344.30 GHz (0.66), CD−57 1054 at 346.53 GHz (0.82), AU Mic at 230.67 GHz (0.85) and HD 110411 at 219.44 GHz (0.88). Those 7 are untested rather than excluded. These are conditional detection probabilities under the persistent-carrier model adopted here, not probabilities that a candidate was noise: a crossing that does not recur has not been shown to be anything else. They are also computed from the model of the reading rather than measured on carriers injected and recovered near the trigger, for which the archive offers no control: what is available is line emission at a known frequency, which does not drift (§5.4), and the measured behaviour of the reading itself at the predicted cell (Appendix B), so the recovery of a *drifting* carrier close to the threshold is modelled here and not demonstrated.

Part of that lost power is the threshold and not the archive. A statistic of five was set for a window of 10^2 to 10^6 cells; at one pre-specified cell it is a far stricter demand, with a probability of 3×10^{-7} of being reached by noise and 1×10^{-4} false confirmations expected across all 444 readings. A threshold matched to that many readings would be 2.8, and the largest reading taken at the carried drift is 2.48. At 2.8 the criterion would have reached 90 per cent recovery probability at the trigger amplitude for all 33 crossings that carry an exclusion rather than 26 of them, and still nothing would have been confirmed.

The 28 repeat windows that do hold something above the trigger are themselves crossings, so a carrier present again would appear in its repeat block as one; their number agrees with the 28 chance predicts in trials of that size, but each pair must still be asked whether one transmitter could have produced both members. Inside that interval there are 16 pairs in which at least one member is unattributed; Table 5 asks the question of the 13 of them, across 5 stars, whose two crossings each carry an independently fitted drift and a block epoch measured from their own data. The other 3 cannot be asked: 2 involve the crossing whose discovery products do not survive and 1 a line-attributed crossing for which no drift was fitted. Both fitted drifts are carried into the star's frame first, by removing the observatory's own line-of-sight acceleration: at most 40 Hz s^{-1} here, but diurnal, so extrapolating it across days would add an excursion that is not there.

Two criteria follow, and they partition the pairs differently. Where the drift grid resolves the interval — 7 of the 13 pairs — the elapsed time and the two drifts fix the offset a carrier at constant line-of-sight acceleration would show, and all 7 fail that comparison, by 3.6 standard deviations in the mildest case. The weaker demand is any trajectory whose drift stays inside the searched grid and whose curvature is bounded by the same acceleration ceiling, and it is the one each verdict rests on: it excludes 3 pairs — 1 needing a mean drift the search never covered and 2 needing the line-of-sight acceleration to reverse faster than orbital motion about the host star can reverse it — and leaves 10 not separable, among them 4 of the 7 that the constant-acceleration comparison resolves. An acceleration ceiling alone bounds no reversal time, and the assumption that supplies one is that the acceleration be gravitational, which ties the velocity amplitude to it: at 3.6 m s^{-2} the tightest bound orbit about a star of at most 2 $M_\odot$ has a radius of 0.057 au, a quarter period of 21 hours and a line-of-sight velocity amplitude of 176 km s^{-1}, while these 2 pairs demand 136 and 73 km s^{-1}. A transmitter accelerated by something other than that orbit — carried on the surface of a rotating body, say — reverses faster at the same acceleration, and this criterion does not reach it. β Pictoris is judged by the same criterion. Its two crossings sit at 241.7739 and 241.8594 GHz in blocks 24.4 hours apart, 106 km s^{-1} apart, and both fitted drifts are negative, −1611 and −679 Hz s^{-1}, so a single carrier would have had to rise in frequency between the epochs; that needs a mean line-of-sight acceleration of 1.21 m s^{-2}, inside the 3.6 m s^{-2} ceiling, so the sign settles nothing on its own. The reversal does: the pair requires one in 16.5 hours, 1.3 times faster than that orbit permits, or a velocity amplitude of 136 km s^{-1} against its 176 — the narrowest margin of the 3 exclusions and the only one inside a factor of two. The same test accepts the one pair that is a single emitter, the CO(2–1) line of HD 285968 recovered at one stellar-frame frequency 5.0 days apart, 0.6 standard deviations from the prediction; it is 1 of the 23 pairs whose two members are both already attributed to a molecular line, the only one of them for which a drift and an epoch were both recorded. For the 10 pairs it cannot separate, 2 days to nearly two years apart, nothing measurable tells one carrier from two, and only the agreement of their number with chance bears on them.

The 2 attributed crossings of the 15 tested that are not recovered both remain consistent with their attribution. HR 1010's crossing at 230.5287 GHz is −12.0 km s^{-1} from CO(2–1) and is bounded at $T_{\rm rep} < 4.6$ against the $T_{\rm pers} = 8.4$ real carbon monoxide at the discovery flux would have given, a disagreement of only 2.1σ once that window's noise is measured over the 47 minutes it integrates, 11.5 mJy, where its sibling blocks also lie. ALMA J153702653−33192492 is a serendipitous field star with no measured systemic velocity, so its frame is the barycentric one; its crossing at 230.5392 GHz is +1.6 km s^{-1} from the same transition, bounded at $T_{\rm rep} < 4.2$ against $T_{\rm pers} = 6.2$, or 1.4σ. Its attribution rests on a velocity rather than on this test: the sightline carries catalogued foreground emission at +4.9 km s^{-1} in the local standard of rest, 0.5 km s^{-1} from the crossing's own velocity in that frame (Appendix C). Its repeat blocks

show where that gas lies, not whether it can be recovered. All 6 share one 7 m configuration and beams of 4.8 to 5.3 arcsec, so an extended cloud is recovered alike in each; in every one the ring reaches 10.9 to 13.8 against 4.2 to 4.8 at the phase centre, with 193 to 496 of the 512 controls above it. The gas is bright and it is not at the star. Toward HD 285968 it is: over its 4 covering blocks the ring reaches only 6.4 to 8.1 against 4.8 to 6.0 at the star. Whether a foreground line recurs at the phase centre is set by where the cloud is bright, which the control ring measures in the same map.

5.6.1. *What can be searched, and what can be confirmed*

A sensitivity limit and a confirmation limit are different statements. This experiment searched 61 systems and 405 Class A windows over 47.7 GHz of union bandwidth, and could have confirmed a signal in 38 of those systems — those observed again more than a day later at a frequency the first block covered — and in 12 of them more than a year later (Fig. 8). Table A2 prints it beside the larger count that ignores frequency, and retuning is the difference.

An independent execution block is also not an independent epoch. The baselines run from 0.46 hours to 1.8 yr, and for 21 of the 37 crossings that carry a discovery amplitude at all — the 33 with a quoted exclusion together with the 4 withheld — the nearest covering block falls within a day. Over months and years these tests bound no duty cycle, and for the other 23 systems they bound nothing at all.

5.6.2. *The reserved hold-out and the control beyond 40 pc*

Two sets of execution blocks enter no census statistic and calibrate instead (Table A1); both were searched with the frozen pipeline, both hold data on real targets and both are reported here as searches. The hold-out — 77 blocks, 315 windows, 36 stars and 214 h on source, held out by a rule fixed before any of them was searched and sharing no execution block with the census (Appendix B.1) — contains 8 threshold crossings, the strongest at $T_\star = 8.64$, and not one of them outranks its own control positions: the fewest controls above the star in any of them is 5 of 512. 7 were tested for recurrence on the same code path as the census, and exactly one is present again: HD 285968 at $+9\,\mathrm{km\,s^{-1}}$ from CO(2–1), whose 4 later blocks combine to $T_{\rm rep} = +6.71$ at the carried drift, the value Table 7 prints, while the strongest single block reaches 5.87 at its own best drift. It is a molecular line, so its recovery is the test working in the positive direction here as well; for every other tested crossing the repeat coverage rules out a carrier at 0.47 of the discovery amplitude or less (Table 7).

The control beyond 40 pc is 47 blocks and 198 windows toward 22 background sources, 88 h on source, with 8 crossings whose strongest is $T_\star = 12.7$ and none of which outranks its own controls. No systemic velocity is catalogued for any of them, so the set supports no limit and shows only that the pipeline behaves the same way on fields it was not tuned for. The third reserved set calibrates the spatial rank and is not external at all (Appendix B).

5.7. *A stellar-frame multi-epoch stacked search*

TABLE 5
Can two crossings be one carrier? Every pair of crossings toward one star, in two execution blocks, lying within $\pm 833\,\mathrm{km\,s^{-1}}$ of each other in the star's frame — the half-width of the median spectral window a repeat could be found in — and carrying between them two independently fitted drifts and two block epochs measured from their own data: 13 pairs with an unattributed member, of the 16 inside that interval. Below the rule is the control, the 1 of 23 pairs of already attributed crossings for which a drift and an epoch were both recorded; a test that cannot return one carrier for it is not a test. Stars are identified by position, so the two archival-tail crossings of HD 14055 pair with its catalogued ones. $\Delta\nu$ is the stellar-frame offset and $\dot\nu_1$, $\dot\nu_2$ the drifts fitted independently in the two blocks and carried into that frame. A transmitter at constant line-of-sight acceleration would put the pair at $\Delta\nu = \frac{1}{2}(\dot\nu_1 + \dot\nu_2)\Delta t$, and *miss* is how far it is from that in units of the drift grid's resolution carried over Δt; a dash means that resolution exceeds the window searched, so the comparison has no power. The last column is the verdict under the weaker criterion of a bounded trajectory inside the same grid and acceleration ceiling: *no* where one carrier is excluded, *open* where the pair is not separable, *yes* for the control row. The drifts are in the star's frame while the grid was stepped in the observatory's, and a fitted drift is refined inside one grid step, so a printed value can lie beyond the outermost trial of its own window: the largest here is $4184\,\mathrm{Hz\,s^{-1}}$ as fitted, 0.72 of a $59\,\mathrm{Hz\,s^{-1}}$ step past that trial, and gains a further $32\,\mathrm{Hz\,s^{-1}}$ on conversion into the star's frame. The grid itself stops at the 12.0 nHz ceiling of §4; that window's outermost trial is 11.99 nHz.

Star	blocks	Δt (d)	$\dot\nu_1$ ($\mathrm{Hz\,s^{-1}}$)	$\dot\nu_2$	$\Delta\nu$ (MHz)	miss (σ)	one carrier?
61 Vir	X822/X82f	3.00	+1092	+3221	+755.4	15	open
J1537−3319	X10359/X398	123	−2000	+2390	+223.5	3.6	open
AU Mic	X1838/X319	310	+1440	+2645	−159.0	—	open
HD 14055	X2b2ee/X2c3bb	0.09	−2390	+4216	+825.2	2447	no
	X24be0/Xd46	1.92	+1134	−3823	+63.0	41	open
	X2c3bb/X3835	2.00	+4216	+1800	−514.6	144	no
	X2b2ee/X3835	2.09	−2390	+1800	+310.6	48	open
	X3835/Xd526	607	+1800	−2563	−555.5	—	open
	X2b2ee/Xd526	609	−2390	−2563	−244.9	—	open
	X3835/X25c5e	649	+1800	+2611	+423.3	—	open
	X2c3bb/X25c5e	651	+4216	+2611	−91.3	—	open
	X2b2ee/X25c5e	651	−2390	+2611	+733.8	—	open
β Pic	X3a90/Xa9df	1.02	−1611	−679	+85.5	132	no
HD 285968	X108f/X5b4e	4.99	−29	−29	−0.002	0.6	yes

Combining the epochs of a star in that star's own rest frame deepens the limit. 56 of the 83 surveyed systems hold more than one searched execution block, and 59 of the surveyed stars can be stacked, 5 of them from blocks beyond the census. Over 320 stacked groups, 2219 epoch-windows and 1431 h, the flux limit improves by a median factor 1.52 over each star's best single epoch and 4.45 at most. 10 further stars stack too, all of them control stars beyond 40 pc, and are excluded from those figures; over all 69 stacked stars, whose stacks are mostly shallow two-epoch ones, the median gain is 1.46. Every stacked limit is measured by injection into the stack itself and never read off a stacked spectrum as a threshold.

A fixed stellar-frame frequency is not a fixed observed frequency, so the epochs are registered before they are added. The barycentric term alone swings by a median $8.2\,\mathrm{km\,s^{-1}}$ between the epochs of one star and by as much as 53.9 — many channels at the 488 kHz Class A width — so epochs added as observed would dilute rather than reinforce. Each epoch is transformed to the stellar rest frame, divided by its own primary-beam response and combined by inverse variance over the channels of

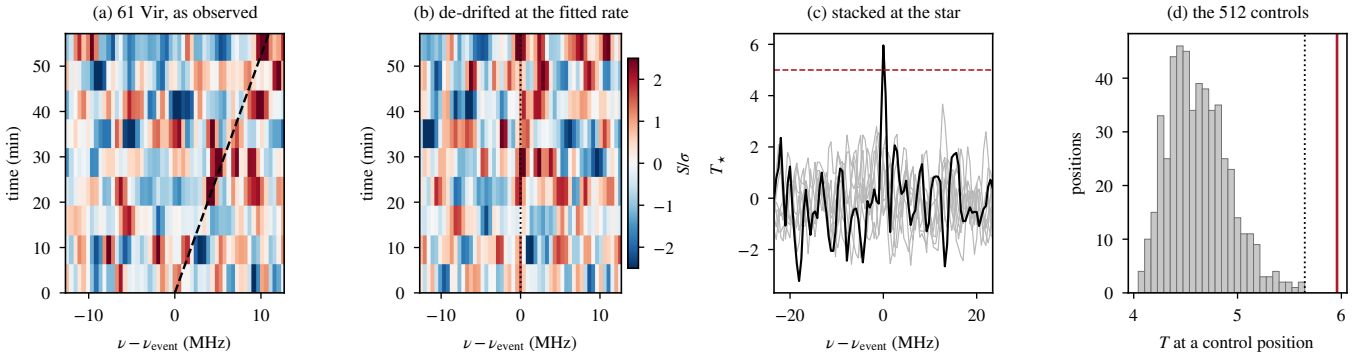


FIG. 7.— **The strongest unattributed crossing of the quality-passing data, and the control field it leads.** Frequencies are in the star's rest frame. *(a)* The dynamic spectrum at 61 Vir's position around the crossing at 346.3232 GHz, observed 2017 May 20, with the fitted drift of $+3182\,\mathrm{Hz\,s^{-1}}$ as the dashed track; the 225 integrations are binned into 9 and plotted in units of each bin's noise. At 119 ± 20 mJy against 303 mJy per integration the carrier is 0.39 standard deviations in any one of them, and exists only in the coherent stack of all 38 minutes on source. *(b)* The same rows shifted by that drift, which carries the carrier 22 channels of 488.3 kHz across the observation. *(c)* The de-drifted stack at the star (black) and at the 8 control positions with retained dynamic spectra (grey), in units of its own noise; dashed, the trigger. *(d)* The same statistic at each of the 512 rank-screen control positions: the star (solid) against the highest control (dotted). The recurrence test, for this crossing and for the second one, CP−72 2713 at 344.2965 GHz, is in Table 6, which also gives the implied isotropic-equivalent powers, 5.1×10^{14} W at 8.5 pc here and 8.9×10^{14} W at 36.7 pc there.

common stellar-frame coverage, keyed on the star's catalogued identity rather than the directory the search ran in. Spectra with pathological archived visibility weights are excluded first (27 of 2838, which would otherwise dominate the sum). The transform is verified on a real line: HD 285968's molecular line collapses from a 19.26-channel topocentric spread to 0.798 channels over 4 epochs spanning 44 d.

The criterion was fixed before any stellar position was examined: a stacked $Z \geq 6.0$ above every stacked control maximum, no epoch supplying more than 0.60 of the weighted sum, and at least 0.50 of the signal surviving removal of the strongest epoch. Controls are stacked identically and every clause evaluated at them, measuring the whole test's false-alarm rate at 0.21 per cent against a 1 per cent ceiling fixed in advance.

The gain is $\sqrt{N_{\rm eff}}$ and not $\sqrt{N}$, where $N_{\rm eff} = \sigma_{\rm min}^2 \sum_e \sigma_e^{-2}$ is the equivalent number of equal-depth epochs and the median $N_{\rm eff}/N$ is 0.65. What a stack realises is $\sqrt{N_{\rm eff}}$ divided by the measured noise of its own stacked spectrum, and that noise follows the inverse-variance prediction to 0.992 in the median, with a per-stack scatter of 7 per cent and nine stacks in ten between 0.81 and 1.18. Against $\sqrt{N}$ the realised gain is 0.78, the shortfall being unequal epoch depth, so an epoch count alone misleads. HD 161868 stacks 13 epochs for a gain of only 1.15: its $N_{\rm eff}$ is 1.7, so $\sqrt{N_{\rm eff}}$ is 1.29, and its stacked spectrum came out 1.12 times as noisy as predicted.

The deepest stacked limit is Proxima Centauri's, and it is a Class B limit. Proxima is covered by 66 blocks over 85 h spanning 6.3 yr, all of them coarse 15.625 MHz windows, so the stack constrains an unresolved spectral excess and discriminates no drift. 25 of those blocks belong to the primary census, 33 were searched afterwards from the same public archive and 8 come from the reserved hold-out, all observed over the census blocks' own years. Stacking the census blocks alone, the set every other limit here uses, gives $\mathrm{EIRP}_{90} = 2.2 \times 10^{13}$ W; adding the rest of the archive deepens that by 1.26 to 1.73×10^{13} W, against 5.51×10^{13} W for the best single epoch on the same injection-calibrated convention. Its $N_{\rm eff}$ is only 7.9, so $\sqrt{N_{\rm eff}} = 2.80$ of the measured 3.19 is what combining the epochs buys, the rest being this spectrum's own noise. The limit holds for an emitter steady in the stellar frame to within one 15.625 MHz channel, $20.9\,\mathrm{km\,s^{-1}}$ peak to peak, with no inter-epoch drift above $7.8 \times 10^{-2}\,\mathrm{Hz\,s^{-1}}$. At 1.30 pc that depth is mostly geometry; the gain is what stacking bought.

2 groups pass the full criterion and both are known lines. They are β Pictoris' CO $J = 1$–0 at $+3.09\,\mathrm{km\,s^{-1}}$ over 8 epochs ($Z = 45.3$) and its $J = 2$–1 at $+3.55\,\mathrm{km\,s^{-1}}$ ($Z = 12.1$), agreeing to $0.46\,\mathrm{km\,s^{-1}}$ or 0.73 of a CO(1–0) channel: the astrophysical positive control passing in the stack. The survey's own $\pm50\,\mathrm{km\,s^{-1}}$ mask leaves none of them unattributed. HD 285968 reaches $Z = 12.89$ on the line used above as the registration standard and is rejected by its own controls at 14.19: the emission is extended, as a cloud is and a transmitter is not.

Every stacked limit above holds only for a carrier that stays inside one channel. At CO(2–1) one 488 kHz channel spans $0.63\,\mathrm{km\,s^{-1}}$, the whole excursion a stacked carrier is allowed, peak to peak. A transmitter on a short-period giant planet, its line-of-sight velocity swinging by $\pm30\,\mathrm{km\,s^{-1}}$, crosses 94 of those channels peak to peak and de-registers completely, against 3.0 of a 15.625 MHz window: the coarse stacks buy reflex tolerance with bandwidth. Drift divides the same way, a stack in 488 kHz channels being confined to $1.3 \times 10^{-1}\,\mathrm{Hz\,s^{-1}}$ over the median inter-epoch span of 44 d. No crossing of §5.6 drifts that slowly: the slowest measured is $519\,\mathrm{Hz\,s^{-1}}$, which leaves one channel in 16 minutes, and in 33 of 38 cases the stack covering a crossing contains its discovery block. Toward the 20 stars carrying them the stack is a deeper limit on a non-drifting carrier, not a test of these events. Each stacked limit therefore constrains only emitters stable in the stellar frame to within its own channel; the single-epoch limits of §6.3 apply generally.

6. DISCUSSION

6.1. *What is different about a millimetre technosignature search*

The most transferable result of this experiment is not the non-detection but a measurement of what limits a

TABLE 6
The two events, the crossings whose stellar statistic exceeds all 512 control positions of their own window; the first of them is Fig. 7. Repeat epochs measured are the blocks at which the predicted stellar-frame cell was read, the remainder of them coming from the archival extension. $T_{\rm pers}$ is what a carrier persisting at the discovery amplitude S would have given there and $T_{\rm rep}$ is what is measured, combining every repeat epoch by inverse variance at the carried drift: one cell and one trial, so it may be negative, while the two indented readings beneath it are maxima and so positive on average whatever is there — the second of them the same cell at every trial of the window's own drift grid, which bounds an error in the carried drift. The nearest block alone follows, because the combination asks the model of §5.6 of every block across the whole baseline while the single reading asks it only of the two observations. The exclusion is the fraction of S still ruled out at the trigger, $(T_{\rm rep}+5)/T_{\rm pers}$; the displacement permitted is half a channel in the star's frame. The last row but one is the chance $\Phi(5T_{\rm pers}/T_\star-5)$ that a carrier at the trigger amplitude $5S/T_\star$ would have satisfied the criterion, and in brackets the same at S, which is selected as the largest of many cells and biased high.

	61 Vir	CP−72 2713
Frequency, stellar frame (GHz)	346.323236	344.296455
as delivered, topocentric	346.313811	344.269746
Drift rate ($\mathrm{Hz\,s^{-1}}$)	+3182	−2056
$T_\star$	5.958	5.809
Highest control T	5.65	5.68
Fitted amplitude (mJy)	118.8 ± 19.9	11.3 ± 2.0
Peak-channel EIRP (W)	5.1×10^{14}	8.9×10^{14}
Repeat epochs measured	3	5
of them in the primary census	3	1
census repeats with no retained spectrum	1	0
Nearest covering epoch, separation (h)	−1.97	+2.05
covering epochs more than a day away	2	4
Longest baseline (d)	3	2
T expected if persistent, $T_{\rm pers}$	9.70	6.30
T measured, matched drift, $T_{\rm rep}$	+0.39	−0.82
largest single epoch	+1.65	+0.96
maximised over every drift	2.71	3.60
Nearest block alone, $T_{\rm pers}$	5.28	6.21
$T_{\rm rep}$	−0.01	−0.57
Amplitude still excluded, $(T_{\rm rep}+5)/T_{\rm pers}$	0.56 S	0.66 S
from the nearest block alone	0.95 S	0.71 S
Displacement permitted ($\mathrm{km\,s^{-1}}$)	±0.211	±0.213
Recovery chance at $5S/T_\star$ (at S)	1.00 (1.00)	0.66 (0.90)
Disposition	not confirmed	not confirmed

carrier search at these wavelengths. It is not depth: the power a transmitter must radiate to be found here is a median 1.5×10^{15} W per system, and more power buys nothing at a frequency the archive never tuned to or never returned to. What limits the search is that inherited coverage. Molecular-line confusion is a different matter: it costs bandwidth, and it decides how a crossing is disposed of once one is found.

Line emission, and not interference, is the astrophysical false positive here. Catalogued molecular transitions, almost all of them carbon monoxide, account for 18 of the 52 crossings in quality-passing data, and no crossing in this survey is attributed to interference (Appendix E). Those coincidences are also the strongest events, but they are not the commonest: the 34 unattributed crossings are no more numerous than the control positions of their own fields predict (§5.3.1), so thermal noise at a trigger of 5 accounts for most crossings by number while line emission accounts for the ones that stand out. 10 of the 12 windows that reached the spatial stage are disc or foreground carbon monoxide, all but one the resolved disc of β Pictoris and the exception a cloud along the sightline to HD 48370. The remedy that works is a velocity-space mask evaluated in each star's own rest frame, and it is not free: it removes 6.4 per cent of the Class A union bandwidth.

Spatially extended emission is the second discriminant, and it has no analogue at centimetre wavelengths. A line filling the primary beam raises the star and the control annulus together instead of cancelling, so the spatial statistic loses power exactly where circumstellar gas is brightest. β Pictoris shows the benign face: the frozen pipeline recovers its circumstellar carbon monoxide blind, in two transitions, which is the positive control the experiment needed. HD 48370 shows the other. Its emission — foreground cloud, not a disc (Appendix C) — lies near a band edge, and that one window contributes 59 per cent of every near-edge threshold exceedance in the survey: pooled, the near-edge rate is ×2.464 the interior rate; without that window ×1.020, and on a held-out half ×1.181 ($p=0.24$). What looked like a property of band edges is one disc, so no edge correction is applied to any limit.

The channelisation is inherited. 1258 of the 1663 windows have channels so coarse that every drift in the grid stays inside one channel over a whole track; they measure an amplitude but cannot distinguish a drifting carrier from a stationary one, which is why the carrier search rests on 405 of them. The widths on offer here run from 15.3 kHz to 31.25 MHz, about 3.3 orders of magnitude; set beside the 2.8 Hz channels of centimetre-band carrier searches the range becomes 7 orders (Fig. 1(b)), and nothing in the search could choose them.

Repeat epochs turn a crossing into a testable claim, and an archive supplies them only incidentally. Let the *confirmation coverage* $C_{\rm conf}(\nu)$ be the number of systems toward which two independent execution blocks more than a day apart both cover the sky frequency ν, so that a carrier found once there could have been sought again. It must be stated against frequency rather than as a fraction of the sample, because repeat visits re-tune and a second epoch is of no use at a frequency it does not cover. Its support is 24.7 GHz of union bandwidth, 52 per cent of the 47.7 GHz searched, falling to 8.5 GHz beyond a year, and at its best near 231 GHz it reaches 27 systems (Fig. 8). The searched extent is about twice the extent over which a persistent carrier could have been confirmed, and for close to half the sample the archive constrains an intermittent emitter hardly at all, whatever its power.

6.2. *The search space this archive opens*

Earlier searches above about 30 GHz reached nearby stars at one frequency only, and the one earlier millimetre search of more than a single line covered 117 MHz of Band 3 toward far more distant stars (§2); Band 3 is searched here, in 112 windows down to 90 GHz including β Pictoris' CO $J=1$–0 control. Of the 10 MHz–115 GHz haystack axis of Wright et al. (2018) this experiment covers 19.9 GHz, only 1.0 GHz of it carrier search because the Class A union begins at 114 GHz; the rest lies above

TABLE 7

The threshold crossings of the reserved hold-out, searched with the same frozen pipeline and disposed of by the same tests as the census. $\Delta v_\star$ is the offset from the nearest masked transition in the star's own frame and $n_{\rm rep}$ the number of covering repeat blocks; $T_{\rm pers}$ and $T_{\rm rep}$ are as in Table D1. The one crossing that is present again is a carbon monoxide line.

Star	block	$\nu_{\rm topo}$ (GHz)	$T_\star$	line	$\Delta v_\star$	$n_{\rm rep}$	$T_{\rm pers}$	$T_{\rm rep}$	disposition
HD 285968	Xdaef5a_Xdd9b	230.5081	8.642	CO(2–1)	+8.7	4	11.6	+6.71	attributed; present again
HD 14055	X107376c_X6126	330.7123	5.513	^{13}CO(3–2)	+112.6	30	312.0	+0.56	unattributed; does not recur
TRAPPIST−1	Xd0e450_X259b	346.2454	5.326	SO(9_8–8_7)	−306.6	5	11.1	−0.32	unattributed; does not recur
HD 172555	X107fa96_X1300f	344.6066	5.144	SO(8_8–7_7)	+251.3	—	—	—	unattributed; no repeat coverage
HD 285968	Xdb4b9a_Xa7c	229.9129	5.106	CO(2–1)	−768.0	4	11.2	+0.21	unattributed; does not recur
HR 1010	Xc6b674_X23f2	346.2301	5.044	SO(9_8–8_7)	−238.7	7	25.9	−0.80	unattributed; does not recur
HD 10647	Xff99e1_X9cdf	345.7059	5.017	SO(2_3–2_1)	+37.1	13	142.1	−0.93	unattributed; does not recur
HD 170773	Xff294f_X1fb	345.5597	5.001	SO(2_3–2_1)	−113.8	4	59.3	+1.45	unattributed; does not recur

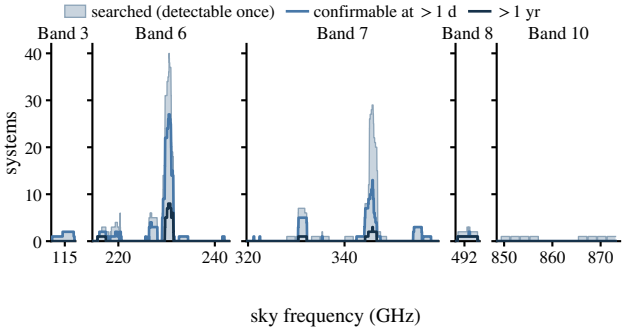


Fig. 8.— What could be searched against what could be confirmed. For the Class A experiment, against sky frequency: the number of systems toward which a carrier could have been detected once (grey), the number for which a second independent execution block more than a day later also covers that frequency (blue), and the same beyond a year (dark). One panel per populated ALMA band, widths in proportion to the frequency each contributes, as in Fig. 9.

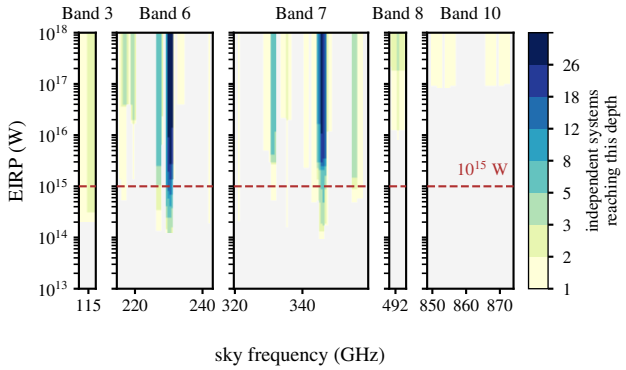


Fig. 9.— Where the experiment could have found something. Shading gives the number of independent systems in which an unresolved carrier of the plotted EIRP at the plotted sky frequency would have been recovered nine times in ten *in a single epoch*; unshaded means none. That is a recovery completeness and not a probability of confirmation: confirmation requires a second independent epoch covering the same cell, which Figure 8 maps and which no amount of transmitted power can supply. One panel per populated ALMA band, with panel widths in proportion to the frequency each contributes: the 47.7 GHz Class A union is 26 disjoint islands across 114.0–873.1 GHz, which a continuous abscissa would draw as hairlines. The dashed line is 1×10^{15} W.

that axis.

Union bandwidth overstates what that buys: different frequencies were searched toward different numbers of stars and to very different limits. Figure 9 gives instead the number of independent systems that would have yielded a recovery at ninety per cent completeness, against sky frequency and transmitter power. At an EIRP of 1×10^{15} W the covered frequency falls to 19.1 GHz, 40 per cent of the union, reaching 25 systems and 17 at the best-covered frequency near 231 GHz; with no limit on power that frequency reaches 40 of the 61 systems, and half the sample only above 3.9×10^{15} W.

The conventional measure for comparing programmes is the continuous-waveform transmitter figure of merit of Enriquez et al. (2017), CWTFM $= \zeta_{\rm AO}\,{\rm EIRP}_{\rm min}/(N_\star \nu_{\rm rel})$ with $\nu_{\rm rel} = \Delta\nu/\nu_{\rm mid}$, smaller being better. Those programmes quote ${\rm EIRP}_{\rm min}$ at a bare detection threshold, and not at the same one: Enriquez et al. (2017) quote at S/N = 25 and Margot et al. (2023) at 10, where this search triggers at five (Table 1). The measure is linear in ${\rm EIRP}_{\rm min}$, so putting every programme on the common footing of five moves the published values — 0.85 becomes 0.17 and 0.14 becomes 0.072 — and it runs against this survey, making the comparison programmes deeper than their own papers report. Margot et al. (2023) publish no CWTFM; that value is the same expression on the three quantities they do give, 11 680 stars, 1.15–1.73 GHz and EIRP $> 1.35 \times 10^{13}$ W. This experiment takes no such factor. The denominator used here modifies theirs: in place of a single survey-wide fractional bandwidth it is the star-bandwidth product $\sum_i \Delta\nu_i/\nu_i$, one term per system, in which $\Delta\nu_i$ is the union of that system's own searched windows and ν_i the midpoint of the span those windows cover. That is not the frequency of its best window, which is what Table F1 lists beside the coverage: formed with those frequencies instead the same sum is 0.70, and the convention adopted here is the smaller of the two. The two are the same quantity for a survey in which every star is observed over the same band, which is what the programmes compared here are, while for an archive a single fractional bandwidth credits every system with frequencies searched toward another. Measured term by term it is 0.64 over the 61 systems against 5.9 on the uniform convention, a factor 9.2, because the median system was covered over only 1.84 GHz of the 47.7 GHz union. On that denominator the Class A experiment is 9.5×10^5 on the sample's largest threshold power and 3.7×10^4 on the median system, and 2.9×10^6 and 1.1×10^5 on the ${\rm EIRP}_{90}$ adopted everywhere else here — a factor 1.3×10^5 above the published Enriquez value, widening to 6.7×10^5 once that value is put on the common threshold. That gap is 2.8 decades of haystack and 2.4 decades of depth, so neither explains it alone: this is a far smaller haystack than a wideband centimetre survey of thousands of stars and a shallower one, and the measure compares surveys rather than estimating occurrence rates.

6.3. *What this experiment excludes, and what it does not*

The signal these limits describe. Every limit here applies to one kind of signal: an unresolved carrier at the stellar position, confined to a single 15.3–1953 kHz channel, lying outside the molecular-line mask, radiating while one of the archival blocks is on sky, and changing frequency linearly at no more than its own window's ceiling, 12.0–13.3 nHz. Such a carrier would have produced a crossing nine times in ten above the EIRP_{90} of §5.1, and none survived toward any of the 61 systems the carrier search covers. Four conditions bound it as sharply as power does: the coverage is sparse and inherited (Fig. 9); the drift grid reaches an acceleration that contains every known planet of the sample but one, while a transmitter on a rapidly rotating planet spends most of that reach on its spin alone (Fig. 3); emission broader than one channel is removed with the spectral baseline; and a carrier on for part of a block comes back at that fraction of its amplitude, while one absent throughout is not constrained at all.

Recurrence applies over a far smaller domain. A crossing is not a detection (Table 3), and where the archive observed only once no power makes one. Recurrence could be tested only for the systems the archive revisited at a frequency the first epoch covered (§5.6.1); for the rest the confirmation completeness is zero at any power. Where the test does apply it has power only against a carrier as strong as the trigger demands of the repeat, so a shallow repeat excludes nothing (§5.6).

The result is a non-detection and not an exclusion of persistent transmitters. An exclusion is read at the one cell the first epoch predicts, so the transmitter it tests is one that comes back to that cell, and four restrictions on that model are ours. A transmitter operating intermittently need not have been radiating during the second block. A second block need not cover the frequency at all, and over about half the searched bandwidth none does (§6.1). The interval that cell permits is half a channel in the star's own frame, and the drifts fitted to these crossings themselves, 0.5–3.6 $\mathrm{m\,s^{-2}}$, carry a carrier across it in 77–119 s: a transmitter accelerating like that is in the predicted cell again only if its motion is periodic and the second epoch catches it at the matching phase. Anything else is left to the wide search over $\pm 833\,\mathrm{km\,s^{-1}}$ at every drift (§5.6). A transmitter may also change frequency deliberately, and nothing here addresses that. Outside the adopted criterion, then, nothing is excluded: neither emission inside the $\pm 50\,\mathrm{km\,s^{-1}}$ molecular-line mask at any strength, nor any emission from the 17 566 stars within 40 pc that ALMA has not observed, which is almost all of them.

An illustrative engineering benchmark. A 12 m dish radiating 1 MW at 230 GHz with an aperture efficiency of 0.7 has a directive gain $G = \eta(\pi D\nu/c)^2 = 5.9 \times 10^8$ and so an EIRP of 5.9×10^{14} W, a depth this survey reaches for 17 of the 61 Class A systems. The same gain makes its beam $\Omega = 4\pi/G = 2.1 \times 10^{-8}$ sr, 34 arcsec across and illuminating 1.7×10^{-9} of the sky. EIRP is not transmitter power — the same megawatt radiated isotropically would be 10^6 W — so these limits apply to beamed transmitters whose beam contains the Solar System.

Two blind spots. The search forms Stokes I as (XX + YY)/2, which retains the full intensity of a circularly polarised carrier; what is lost is not sensitivity but the discrimination polarisation would offer against unpolarised astrophysical emission, because ALMA's feeds are linear and the circularity lies in the cross-hands, which this analysis does not extract. And the frequency coverage is the archive's and not a choice; none of the "magic frequency" arguments that have guided centimetre-band searches falls in an ALMA band.

6.4. *What a dedicated ALMA technosignature experiment should do differently*

A purpose-built experiment on the same telescope would be far stronger, and the changes it needs follow from the limitations above. *Channelise finely everywhere:* the 1258 coarse-channel windows cannot carry a carrier search at all. *Choose the targets and the frequencies:* an archive weighted toward dusty discs is the worst case for molecular-line confusion, the M dwarfs that dominate the nearby habitable-zone planet census are the least covered, and windows placed deliberately away from the bright transitions would recover most of the 3.05 GHz of the 47.7 GHz Class A union that the mask costs — all three are decisions a proposal can simply reverse. *Repeat the epochs on purpose, at the same tuning:* two or three visits separated by days and by years turn a crossing into a testable claim, and the separation is a scheduling constraint rather than extra sensitivity; Fig. 8 is what its absence costs. *Retain the visibilities:* rotating the phase centre to the star leaves a genuine point source entirely in the real part of the visibility. That asks whether the data contain a source *where the star is*, which is a stronger question than any rank against neighbouring positions in an image.

7. CONCLUSIONS

1. The two earlier searches to reach nearby stars above 30 GHz each covered a single line, over no more than 35 MHz, and neither used its repeats as a test of persistence (§2). What is new here is the bandwidth, at least 1363 times theirs, the search over drift rate, and the test for recurrence; and an archive assembled for other science is at present the only way to reach that combination. The one ALMA band below those searched that holds any data on these stars is proprietary until 2027 and channelised far too coarsely to carry a carrier search.
2. The primary experiment is 405 fine-channel windows toward 61 stellar systems, covering 47.7 GHz of union bandwidth in 26 disjoint intervals, searched for an unresolved carrier whose frequency drifts. A supplementary coarse-channel search for an unresolved spectral excess adds 1258 windows toward 82 systems and 90.5 GHz of union bandwidth, with no drift discrimination. Both come from one uniform search of 407 public execution blocks, 4.1 TB of raw data, and between them they span 90–873 GHz.
3. EIRP_{90} is a measured completeness of one epoch and not a threshold: the power at which nine of ten carriers injected into the real calibrated visibilities come back through the unchanged trigger and localisation at the star, inside a single execution block, for a carrier narrower than that window's own channel; one spread over many channels meets a differ-

ent limit entirely. The median system's best window reaches 1.5×10^{15} W, with a spread across systems of 8.7×10^{13}–3.7×10^{16} W, and 25 systems reach 10^{15} W; carriers were injected into 49 of the 405 windows and the rest carry the class-average factor, whose measured dispersion moves that median by 0.5 per cent, so a single window's value is an estimate. One bias runs one way only and is outside that calibration: an injected carrier suffers no atmospheric decorrelation where a real one would, so every figure here is optimistic by 1 per cent in the median window and by 8 per cent at the top of the frequency range. No limit quoted here carries that correction, and for the 128 blocks whose phase stability cannot be measured it is conditional on their own calibration performance.

4. The primary census yields 50 Class A threshold crossings and four Class B, and the archival extension a further two Class A: 56 in all, each reported with its disposition. The four Class B crossings are the whole of that search's crossing population and all come from the single execution block that fails the data-quality criterion of Appendix D.2, so the quality-passing data hold 52 crossings, all Class A. Of the 56, 21 fall within $\pm 50\,\mathrm{km\,s^{-1}}$ of a catalogued transition at the star's own velocity, a coincidence that is reproducible and involves no judgement. Of those, three involve a species not yet demonstrated in any debris disc and are reported rather than acted on, leaving 18 attributed and 38 not. The mask costs 6.4 per cent of the Class A union bandwidth. Recording every cell above the trigger, rather than each window's strongest, adds 21 more: three survive attribution and none recurs, so no disposition changes. Chance, measured from each window's own control positions and the channels its control maxima land on, predicts 25.8 unattributed Class A crossings over the census against the 32 found there, or 27.6 with the measured star-to-control transfer applied, and resampling the execution blocks allows 11–43, widening to 9–47 when the scatter of a single count is included; the archival extension's two crossings enter neither side. The count is compatible with that empirical control expectation, which is a weaker statement than a demonstration that the crossings are noise: the width of the range is its limit, an added population of as many as 23 crossings would sit inside it unremarked, and the step from a control position to the star that carries the expectation to 27.6 is itself measured from the 8 events of the reserved blocks, which allow anything from 0.48 to 1.79 of the control rate. The one instrumental mechanism that could raise the statistic at the star and not at its controls is a calibration residual varying with frequency, and against the continuum the pipeline records at the extraction position — dust inside the beam included — nine windows toward two stars would admit one at 3.8–13 per cent, inside the documented range. In every one of them the largest statistic at any channel and drift is 3.1 and none holds a crossing, which puts such a residual below 6.6 per cent: the mechanism is available to those windows and is measured not to be acting in them.

5. Every unattributed crossing was tested for recurrence in the star's own frame, over the whole of each further window covering the event frequency and at every drift searched, because an accelerating carrier need not return to one channel. None was found again: the largest statistic at any predicted cell is 4.71 against $T_{\mathrm{trig}} = 5$. The same criterion recovers 13 of the 15 line-attributed crossings tested, which is the positive control of the stage. Of the 33 crossings that carry an exclusion, the repeat coverage rules out a carrier at a median 0.43 of the discovery amplitude with individual limits from 0.017 to 1.00, so the median is not a bound on the set; combining every covering block assumes a carrier that returns to one stellar-frame channel at the start of each, and where the repeat is shallow the criterion has little power and its silence is not an exclusion. Judged at the trigger amplitude rather than at the discovery amplitude, 7 of these crossings are untested rather than excluded, among them one of the two that lead their own control fields. The criterion is responsible for part of that and not the archive alone: a trigger sized for a whole window is a far heavier demand at a single pre-specified cell, and matched to that one reading it would stand at 2.8, where all 33 would carry useful power rather than 26 of them. What these tests exclude is a carrier whose stellar-frame frequency returns to within ± 0.08–$\pm 0.63\,\mathrm{km\,s^{-1}}$ of the first epoch's and radiates through both observations — one stable over the interval between them, or periodic at matched phase, and not one free to drift as the fitted rates do (§5.6). The conclusion this supports is that no unresolved spectral carrier satisfying the adopted recurrence criterion was confirmed, and not that persistent transmitters are excluded above a power threshold.

6. What this survey could search and what it could confirm are different extents. Of the 61 systems the carrier search covered, 38 were observed again more than 1 day later at a frequency the first block covered; for the other 23 confirmation was beyond reach at any sensitivity, and about those systems the null reported here says nothing at all.

7. Combining each surveyed star's epochs in its own rest frame deepens the flux limit by a median factor 1.52. For Proxima Centauri the stacked limit, measured by injection into the stack, reaches 2.2×10^{13} W over the primary census alone and 1.73×10^{13} W once the rest of the archive is added, in 15.625 MHz channels — a Class B limit on an unresolved excess, carrying no drift discrimination — for an emitter steady in the stellar frame to within one channel across 6.3 yr.

8. These results constrain unresolved carriers only within the domain actually searched (§6.3). The drift reach is part of that domain: the grid stops at 12.0 nHz by default, 6 per cent of the ceiling recommended for a general carrier search (Sheikh et al. 2019), so an unbound transmitter, one on a rapidly rotating planet, or one whose combined orbital and rotational acceleration exceeds that ceiling is not constrained here at all. What transfers beyond the domain is the obstacle, that channelisation chosen by other observers, and not collecting area, decides which kind of search is possible at all.

AUTHOR CONTRIBUTIONS

G.J.W. conceived the survey, defined the target sample and the selection criteria, and wrote the manuscript. R.D. built and operated the archival retrieval, calibration and search pipeline, implemented the spatial control statistic and the injection–recovery campaigns, and produced the per-window catalogue and its regeneration scripts. Both authors designed the vetting procedure, chose the masked species, carried out the statistical validation reported in §4.1 and Appendix D, and agreed the dispositions of Table D1.

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COMPETING INTERESTS

The authors declare no competing interests. VBRL Holdings is a privately held company through which R.D. conducts independent research, named here solely as his affiliation; it had no role in the design of the survey, the selection of targets, the analysis, the decision to publish or the content of this paper, and provided no funding for the work.

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This paper makes use of the following ALMA data: the 67 projects listed in Table A3, each cited as `ADS/JAO.ALMA#` followed by its project code. The 407 primary-census execution blocks searched span 2013 October 6 to 2025 July 11, and we thank the principal investigators and teams who proposed and obtained them. ALMA is a partnership of ESO (representing its member states), NSF (USA) and NINS (Japan), together with NRC (Canada), NSTC and ASIAA (Taiwan), and KASI (Republic of Korea), in cooperation with the Republic of Chile. The Joint ALMA Observatory is operated by ESO, AUI/NRAO and NAOJ. This work has made use of data from the European Space Agency (ESA) mission *Gaia* (Gaia Collaboration et al. 2023), processed by the *Gaia* Data Processing and Analysis Consortium (DPAC). This research has made use of the SIMBAD database (Wenger et al. 2000) (CDS, Strasbourg, France); of the NASA Exoplanet Archive (Akeson et al. 2013), which is operated by the California Institute of Technology under contract with the National Aeronautics and Space Administration under the Exoplanet Exploration Program; and of the Splatalogue spectral-line database (Remijan et al. 2007).

Use of generative artificial intelligence. Large language models were used in this work. They wrote most of the software: the archival retrieval and calibration scripts, the search and vetting pipeline, the statistical tests, and the generators that produce every number, table and figure in this paper. They also drafted text, which the authors revised. The authors set the scientific question, chose every criterion, and are responsible for the content of this paper.

Testing and validation of that analysis. No result here is accepted on a model's assertion, and four things are done instead. No number in this paper is typed: each is written by the program that computes it from the data and re-derived whenever the paper is built, so a value in the text cannot differ from the analysis behind it. Before the manuscript is settled, every generated product is deleted and rebuilt from the data and compared with its predecessor byte for byte, which a program depending on the state left by an earlier run would not survive. A separate audit re-derives the identities that must hold among the published values — counts that must partition, quantities that must sum, numbers reached by two routes that must agree — and stops the build where one does not. And every assertion in the analysis was first made to fail on purpose, on deliberately wrong input, before it was relied on, since an assertion that has never fired is no evidence that anything is right. Every reference cited here has been checked against the published literature.

DATA AVAILABILITY

Every observation this paper rests on is public, and is identified here precisely enough to be fetched. The 407 primary-census execution blocks and the blocks of every other set named in Table A1 are held in the ALMA Science Archive under the 67 project codes of Table A3, and are retrieved there by the execution-block identifier, which Tables D1 and D2 give for every one of the 56 threshold crossings. ALMA's own citation convention for archival data is the project code rather than a persistent identifier, so the archive's identifiers are given in place of one, which is the recommended practice where a data set carries none of its own (Chen et al. 2022). The ancillary catalogues are public in the same sense: Gaia DR3 for the astrometry and the colours, the NASA Exoplanet Archive for the planet counts, and the CDMS and JPL line catalogues through Splatalogue for the mask, queried once on 2026-10-06.

The derived results are in the paper. Table A1 partitions the blocks, windows, stars, systems and hours of every set the search defines; Table A2 carries every headline count, the five fates of the unprocessed blocks and the crossing partition; Appendix F and Table F1 give the sensitivity, the frequency, the channel width, the covered bandwidth and the confirmability of every system of the carrier search; Tables D1 and D2 list all 56 crossings with their frequencies in both frames, the stellar and control statistics, the nearest masked transition and the recurrence test; Table 6 gives the two leading events in full; and Table 7 does the same for the reserved hold-out.

What a reader can and cannot repeat is worth saying plainly. The observations can be obtained in full from the archive and the method is specified in enough detail to be reimplemented, so the search can be carried out again independently, which is the stronger test in any case. The calibrated visibilities, the per-window products and the code that produced them are not distributed with this paper, so this particular execution of the search cannot be re-run by a reader. Two choices that could create or remove an unattributed crossing were fixed before the data they govern were searched: the calibration hold-out rule of Appendix B.1 and the block list its first guard depends on were fixed 3 days before the first reserved block was searched, and the disposition criteria were fixed on 2026 September 11, before the line list was queried. The line list is a dated query of a public catalogue and can be repeated exactly as it stands; the ordering of the other

two is a statement by the authors, and a reader has no way to check it.

SOFTWARE

The pipeline depends on CASA 6.7.6 (casatools/casatasks 6.7.6-14; CASA Team et al. 2022), Python 3 with astropy (Astropy Collaboration et al. 2022), numpy (Harris et al. 2020) and scipy (Virtanen et al. 2020), and the ALMA Science Archive and SIMBAD TAP interfaces through astroquery (Ginsburg et al. 2019); all are free and open-source, at the versions cited.

APPENDIX

A. SPECTRAL RESPONSE, FREQUENCY CONVENTIONS AND THE PRIMARY-BEAM RESPONSE

This appendix sets out the three things every threshold quoted in this paper is checked from: the frame of each frequency, the instrumental response, and the primary-beam factor that turns a measured noise into a flux at the star.

A.1. *Frequency frames, linewidth and smearing*

Every frequency axis here — searched spectra, tabulated ranges and crossing frequencies alike — is *topocentric* as delivered by the ALMA correlator, and the search applies no velocity-frame shift. Velocities enter at the vetting stage, where the line mask carries every crossing frequency to the barycentre at its own block's epoch and then to the stellar frame before comparing it with a transition (Appendix C). The residual error is then the catalogue radial-velocity uncertainty alone, a few $\mathrm{km\,s^{-1}}$ within 40 pc; second-order relativistic terms at the $50\,\mathrm{km\,s^{-1}}$ scale are $\lesssim 5$ kHz at 350 GHz, below the finest native channel searched. The stacking is drift-aligned and inverse-variance weighted: coherent in frequency trajectory, not in electric-field phase.

The native channel width is also the linewidth convention for every EIRP threshold. The statistic is the single-channel flux density after drift correction, so $\mathrm{EIRP_{trig}}$ is the minimum *total* power of a channel-confined carrier, flat in the assumed linewidth below the channel and never a spectral power density; flatness is exact only for a centred tone, and a boundary-straddling one divides its power between two channels, which is the loss the injections carry and one reason EIRP values are quoted to two significant figures. The Hz-resolution surveys' limits are total powers in the same sense, each the best case of its own channelisation, which makes the axes of Fig. 1(a) commensurable; and no reprocessing recovers resolution the correlator never recorded. Channel widths come from the measurement sets' own CHAN_WIDTH; MS v2 defines no "EFFECTIVE_BANDWIDTH" column, so the quoted channelisations are separations and are not claimed to be noise bandwidths.

ALMA applies online Hanning smoothing by default (Cortes et al. 2026), so the spectral response is ∼2 channels wide and neighbouring channels are correlated. For the 0.25/0.5/0.25 kernel the coefficients are $\rho_1 = 2/3 = 0.667$ and $\rho_2 = 0.167$, with $\sigma_{\mathrm{smoothed}}/\sigma_{\mathrm{raw}} = 0.612$: the correlation is not confined to adjacent channels, the lag-two term being a sixth. The peak channel holds 0.50 of an unresolved carrier's flux at a channel centre and 0.42 on a boundary, a median 0.48 over sub-channel phase.

That loss enters the limits exactly once, and it is the thresholds rather than the limits that are free of it. A window's trigger flux is the bare 5σ of its own spectrum, so setting the *total* power of a channel-confined carrier against it would be optimistic by ×2.00–×2.36. The injected carriers are not set against it in that way: each is deposited with the correlator's own channel response, at a sub-channel phase drawn afresh, and recovered through the same trigger, which reads the peak channel. The measured completeness therefore already contains the loss, and factorises into it exactly: $3.03\,P_{\mathrm{trig}}$ is ×2.29 for the channel response and ×1.325 for the margin a peak channel needs above the nominal trigger for nine carriers in ten to return. So P_{trig}, the sensitivity S_{min} below and the filled symbols of Fig. 1(a) are nominal throughout, which is what makes them commensurable with other programmes' detection thresholds, while $\mathrm{EIRP_{90}}$ carries the correction and is not corrected again. The deposition falls 8 per cent short of the ideal response averaged over phase, so the completeness is conservative by that much; and where in a channel a carrier was placed makes no difference to what is recovered, because at a typical searched drift rate it crosses about 11 channels during one observation and samples the whole response.

Two measurements fix the smoothing state. The retained per-integration residuals of the second-stage local null have a lag-one spectral autocorrelation of 0.65 against the Hanning prediction 0.667, so those spectra are smoothed and un-averaged; and the archive's reported effective resolution is exactly twice the channel separation in 1516 of the 1663 windows, the other 147 reporting a smaller ratio, the signature of online channel averaging, where the peak-channel loss is smaller. *The correlation model is truncated at lag two*: the baseline step, a per-integration median over up to 65 channels, could imprint longer-range correlation that the retained spectra carry no residuals to measure, which would make the independent-trials count of Appendix D an overestimate.

Transmitters wider than one channel. The baseline filter (§4.1) is flat below roughly half its effective width W and falls to zero above it, so W is the linewidth ceiling: 65–74 channels over the fine class and 30–59 over the coarse one, that is ≈32 MHz at 488 kHz and ≈0.9 GHz at 15.62 MHz. Below it a top-hat transmitter spanning N_{ch} channels divides its flux between them at unchanged noise, so its threshold is N_{ch} times the single-channel one.

Intra-integration smearing, which the injections do not model. Each injected tone is deposited at one frequency per integration, so the campaign carries the channel response and the drift between integrations but not the sweep within one. A carrier moving Δ channels during an integration keeps $\eta_{\mathrm{smear}} = \mathrm{sinc}(\pi\Delta/2)$ of its peak amplitude, and every limit in this paper is divided by it. The Δ that sets a ninety per cent point is the one at the ninetieth percentile of each window's own drift draw, 89 per cent of the searched ceiling over the campaign's 10 332 tones. The integration length is recorded for 1054 of the 1663 windows, 455 of them in the delivered metadata and 599 more in the exposure records the search kept; for the other 609 the factor is *bounded* at 11.6 s, the longest integration delivered anywhere, where none

falls below 0.997. Nothing moves in the median window, and 13 windows on 6 stars rise by more than one per cent, up to ×1.53: HIP 10679, HD 172555, β Pic, η Crv, CP−72 2713 and HD 48370. Re-scoring the 49 injected windows' own ladders with every tone de-rated by its own $\eta_{\rm smear}$ gives a penalty never larger than the one applied and smaller by at most 8 per cent, so the correction errs the safe way.

A.2. *The primary-beam response*

The sensitivity quoted for a window is $S_{\rm min} = 5\sigma/A$, where A is the primary-beam response at the star's offset from the phase centre: the noise σ is measured in apparent flux, the limit is quoted in true flux, and the two differ by A. That offset is not a nuisance. Archival fields are pointed at a catalogue position, often decades old, while these are high-proper-motion stars, so the star can sit well off axis: over the 1663 windows the offset has a median of 0.089″, a 99th percentile of 7.98″ and a maximum of 17.99″, and A runs from 0.999982 in the median down to 0.2308, below 0.99 on 99 windows and below 0.9 on 46. Every one of the low-response windows is a physically explicable J2000 pointing on a fast-moving star rather than an error.

The response applied is a Gaussian of the full width the observatory documents, $1.13\,\lambda/D$ (Cortes et al. 2026), evaluated at each window's *own* dish diameter. The dish matters: the 1058 rows taken on the 1.7× smaller compact-array antennas have a correspondingly wider beam, and a Gaussian forced to $D = 12$ m reproduces the per-window geometry only to 7.7×10^{-2} against 8.7×10^{-5} for the per-dish form. That width is a physical choice rather than a convention: the blocked-aperture response, which carries the shadow of the secondary and is what CASA applies (CASA Team et al. 2022), has a half-power width of $1.15\,\lambda/D$ on a 12 m antenna and $1.14\,\lambda/D$ on a 7 m one, so the Gaussian used here is that beam to 2 per cent and departs from it by at most 6.2 per cent anywhere in the sample.

A window is used where the two forms agree. At low response a small error in the beam model becomes a large error in inferred flux, and the two forms part company as the offset grows, so the limit on retention is the offset at which they differ by 10 per cent: $0.83\,\lambda/D$ from the pointing centre, where the response is 0.22. The furthest window in the primary census lies at $0.82\,\lambda/D$, where the two agree to 6.2 per cent, so the criterion excludes nothing from it; it is stated because it is what admits the one star this survey covers far off axis.

That star is ϵ Eridani, at 3.22 pc the nearest in the sample with Band 6 coverage. 20 of its 44 windows, in 11 execution blocks, place it at the pointing centre and need no correction at all; the other 24 lie 0.75–$0.82\,\lambda/D$ off it, where the response is 0.23–0.29. Those offsets are inside the main lobe — the blocked-aperture response does not reach its first null until $1.36\,\lambda/D$ — and there the two beam forms agree to 7 per cent, which is the extra uncertainty those limits carry. No window reaches the trigger, the largest statistic being 4.7. Over 6.6 window-hours the 11 fine windows reach 1.2×10^{14} W at 488–977 kHz and the 33 coarse ones 4.7×10^{14}–4.7×10^{15} W. 12 of these windows fall inside the primary census and carry its statistics, counted with it in Table A1; the other 32 belong to the archival extension, and those carry no record of their own phase centre or intra-integration retention. Degraded by the largest either term takes anywhere in the census, ×1.57, the deepest of the 32 still reaches 1.8×10^{14} W, deeper than 51 of the 61 systems the carrier search covers. The flux-scale comparison of §4 is a separate continuum measurement on the same star, a block mean over the 4 compact-array executions that place it at the pointing centre, 0.22 h in Band 6, where no beam correction enters either.

The correction is applied exactly once. The offset is computed for every measurement set from its own `FIELD::PHASE_DIR` and the propagated stellar position, so no window carries a zero-by-assumption offset, and it enters in one place only: $S_{\rm min} = 5\sigma/A$ closes on all 1663 catalogued rows to 8.7×10^{-5} in relative terms. On 10 rows whose products are not retained A is back-solved from $5\sigma/S_{\rm min}$ and flagged as such, so there the closure is arithmetic; the test is the other 1653 rows, where A comes from the search's own stored geometry and closes to 8.7×10^{-5}. Nothing else needs correcting: the statistic of Eq. 1, the control ranks and the attribution are *ratios* formed inside one window, in which A divides out identically.

The star's apparent direction includes the annual parallax, and the amplitude a compact source retains when that is left out of the assumed phase centre is measured rather than modelled. A small fraction of a synthesised beam attenuates a compact source and does nothing else, while a displacement approaching a beam puts the phase wrong on the longest baselines, so the size of the displacement decides whether a scalar correction is legitimate at all. Re-extracting the 8 windows of 2 blocks furthest out at the full apparent position — proper motion, parallax and radial velocity where catalogued — leaves each window's noise unchanged to 0.20 per cent and alters no disposition. It cannot test the amplitude factor, which describes a source those windows do not hold: the statistic merely scatters from 0.78 to 1.15 about 0.98 rather than approaching the 0.49 a source would give. A carrier injected at the apparent position and recovered from the *same* visibilities at both does return the normalised dirty beam, on that block's own baselines and with no model in it: 0.92 against a predicted 0.89 at 0.25 of a beam, and 0.58 against 0.49 at 0.67. It exceeds the model at the larger displacement because maximising over a plane of trajectories lets a partly decohered carrier still find a cell, so a limit corrected by the model is shallower than it need be.

The injection campaign is blind to A by construction, and the bias that follows is exactly zero. The injector deposits an unattenuated amplitude at the star's own direction cosines and refers the ladder to the *apparent* 5σ on all 40 scored units, so an injected apparent amplitude recovered nine times in ten implies a true flux larger by $1/A$ while $P_{\rm trig}$ is the power of a true $5\sigma/A$ and A cancels identically; 4 of the units have apparent and corrected $S_{\rm min}$ differing by more than a part in a thousand, so the cancellation is tested rather than vacuous. Being self-consistent in apparent flux, the campaign cannot falsify the beam model; and it reaches out to an offset of 4.98″, at which the response is 0.9102, whereas 83 of the 1663 released windows, on 6 stars, lie further out than any injected one, down to the sample minimum $A = 0.2308$,

so a limit quoted for one of those rests on an extrapolated curve; their 6 crossings are ratios and unaffected. That set is defined by offset and is not the 46 windows whose response falls below 0.9, which is a different criterion and a smaller set.

A few rows carry a catalogued noise that no retained product reproduces, and they are declared rather than reconciled: on 4 windows of HD 285968 in block `A002_Xd98580_X3bf6` the catalogued σ is 0.88–1.32 per cent *higher* than the surviving product's, so the limits there are weaker and not stronger. With the 10 windows whose products are not retained the declared provenance gap is 14 windows.

B. THE SPATIAL RANK, AND WHAT IT CAN SUPPORT

The spatial rank asks whether an excess sits at the star or anywhere else in the field. It is a diagnostic, no disposition rests on it, and this appendix is the measurement that settles why: The star-versus-control rank is an empirically useful way of deciding which crossing to examine first, but the stellar and control positions are not demonstrably exchangeable under the null. The rank therefore has no calibrated per-window false-alarm probability and is not used as evidence for detection.

The star lies at the phase centre, inside the control annulus, and the asymmetry is real. *It is not a primary-beam effect.* Spectra are extracted from phase-rotated visibilities in apparent flux and are never corrected for the primary-beam response (Appendix A), so the familiar rise of image noise with distance from the pointing centre cannot operate here; were it operating it would make the control probes noisier than the star and place the stellar rank *above* one half, the opposite of what is measured. What the control ensemble does show is a weak rise of its own level with radius, and the obvious repair is to remove it: each probe is standardised within its own radius bin, with no free parameter.

Applied out of sample, the correction does not repair the statistic. Frozen — probe radii, bin edges and standardisation alike — and applied unmodified to 778 windows that played no part in its construction, it moves the median stellar rank from 0.409 to 0.407 against the exchangeable one half, further from it, in both out-of-sample partitions separately; rescoring the whole survey with the gradient removed moves the crossings outranking all their controls from 12 windows to 11, and leaves the survey median rank slightly further from one half than before. The gradient is real but the residual is not radial, and neither normalisation yields a calibrated tail probability, so we keep the frozen rank and rest every disposition on physical evidence or on recurrence.

A calibration error that varied with frequency would explain the displacement, and it is the only mechanism on offer. An error identical in every channel is removed exactly by the per-integration median along frequency; a channel-dependent one is not, and two ordinary errors are channel-dependent: a residual in the bandpass, and a system temperature measured in the coarse correlator mode and interpolated onto the science channels, which near a telluric line leaves a narrow feature behind (Hunter et al. 2018). Both multiply the continuum, so both act at the star and not on empty sky, both stand still in topocentric frequency and both are coherent between integrations: they would stack at near-zero drift and put the stellar rank *below* one half, the sign observed.

Amplitude alone does not exclude them. What such an error multiplies is the whole continuum at the extraction position, a dirty-map amplitude that includes any dust inside the synthesised beam; what it must beat is the noise of the de-drifted, block-combined spectrum, since a coherent residual stacks as a carrier does. A fractional error ϵ deposits $\epsilon S_{\rm c}$ at the star, so a crossing at the trigger needs $\epsilon = 5\sigma/S_{\rm c}$. The survey's own continuum maps — 7 of them, each used only on windows of its own band — give 63 mJy at L 836-122, a 66σ detection and a factor 1.3×10^4 above that star's photosphere; on the measured continuum 9 windows of 2 stars need only 3.8–13 per cent, inside the 5–20 per cent a coarse system temperature is known to leave at an ozone line; and at AU Mic, where the same extractor measures 29.2 mJy near 106 GHz, 1.4 per cent would suffice.

Those windows measure the residual instead, and it is too small. In all 9 of them the largest statistic anywhere — every channel, every drift — is $T_\star = 3.1$ where the trigger is five, the largest control probe 5.4, and none holds a crossing. Read as a fraction of the continuum beside it, the coherent channel-dependent residual at the extraction position is below 6.6 per cent in all of them and below 1.9 per cent at L 836-122, whose continuum was measured in the one execution block its windows come from — beneath the 5 per cent floor of the documented range. Their median stellar rank, 0.61, lies on the far side of one half from the displacement the mechanism is invoked to explain. CD-57 1054 carries the whole pattern within one star: 6 of its windows would admit 9–13 per cent and none produces anything, while its one crossing lies in the window where 71 per cent would be needed. The crossings are not where the continuum is: 52 of the 56 are in fine-channel windows, where this route needs 36–4115 mJy within one beam of the star and only 6 ask less than the brightest continuum this sample has anywhere. The bandpass branch fails by amplitude everywhere, needing 4× the 1 per cent its worst band is specified to (Kameno 2014); and no crossing of either sample lies within 25 MHz of a catalogued atmospheric transition (Appendix E), against the 31.2 MHz resolution at which the system temperature is measured. One further question the shared noise scale raises is settled by the same arithmetic: over the 1607 windows carrying no crossing the stellar statistic divided by the median of the 512 control statistics — which *is* the ratio of the two noises — has median 1.000 (0.994–1.004).

B.1. *The reserved hold-out, and what it calibrates*

The statistic and the mask were settled on the searched sample and cannot also calibrate themselves, so a hold-out was reserved by a rule fixed and recorded before any reserved block was searched: a block is held out if and only if $\mathrm{sha256}(\mathrm{UID})_{[:8]} \bmod 5 = 0$, subject to two guards that use identifiers alone — a block on the pre-registered list stays in the survey, and no system loses every block — so that the assignment follows from the UIDs and that list alone. The statistic, the trigger and the recurrence criterion were all fixed before the rule itself, and the first reserved block was searched 3 days later, so no diagnostic built afterwards can promote a window. Of 484 blocks

TABLE A1

The survey ledger, part one: execution blocks and windows. Each row carries its own number, and a group sums to the row whose number heads it — rows 3.1.1 to 3.1.4 to row 3.1, rows 3.1 to 3.5 to row 3. The primary census is the survey: unless a statement names one of the other sets, it is computed over that row. Hours are window-hours: on-source time summed over spectral windows, so the classes partition the hours as they partition the windows. The telescope on-source time behind the primary census is 281 h, of which 153 h carries at least one Class A window and 277 h at least one Class B window — figures that overlap and must not be added. Confirmed planet counts here and in the target list come from the NASA Exoplanet Archive **pscomppars** table (Akeson et al. 2013), accessed 2026 October 4. Part two, Table A2, carries the events and the confirmation extents.

	Set of execution blocks	Blocks	Windows	Stars	Systems	Hours	Results it enters
1	Archival scope: blocks the query exposes	656	–	–	–	–	rows 1.1 and 1.2
1.1	. . . processed with the frozen pipeline	479	–	–	–	–	panel 2
1.2	. . . not processed at the freeze	177	–	–	–	–	panel 3
2	Processed with the frozen pipeline[c]	484	–	–	–	–	rows 2.1 and 2.2
2.1	**Primary census**	407	1663	90	83	1098	every limit, the crossing list, every chance expectation
2.1.1	Class A, fine channels[a]	–	405	66	61	228	the drifting-carrier experiment
2.1.2	Class B, coarse channels[a]	–	1258	89	82	870	the spectral-excess experiment
2.2	Reserved hold-out	77	315	36	–	214	the rank displacement, the false-alarm tail factor, the radial control profile
3	Not processed at the freeze: what became of each	177	–	–	–	–	rows 3.1 to 3.5, the five fates
3.1	Calibrated and searched	144	–	–	–	–	rows 3.1.1 to 3.1.4
3.1.1	**Archival extension**, reported	115	463	16	16	269	recurrence, and the null on data the frozen pipeline had not seen
3.1.2	Searched after the extension was frozen	8	32	–	–	–	nothing; later than the reported extension
3.1.3	ϵ Eri, both pointings[d]	8	32	–	–	–	reported limits only; no census statistic
3.1.4	No window reached the export	13	0	–	–	–	nothing
3.2	No calibration in the ALMA delivery	18	–	–	–	–	nothing; the archival gap
3.3	Calibration failed	11	–	–	–	–	nothing; the archival gap
3.4	Not attempted	3	–	–	–	–	nothing; the archival gap
3.5	Working disk insufficient	1	–	–	–	–	nothing; the archival gap
4	Beyond 40 pc: out-of-sample control	47	198	22	–	88	the external null; no science count
5	Rank-calibration sample[b]	326	1322	56	–	–	the rank calibration and the positive control; no science count

[a] The two classes partition the census windows, not its blocks: 255 blocks carry a Class A window and 392 a Class B one, and most carry both, so the hours of the two rows partition the windows and not the observing time.
[b] Not an out-of-sample set and not an addition to this ledger: every one of its 327 blocks is counted in a row above, 264 in the primary census and 63 in the hold-out, and it is listed apart only to keep it from being counted twice.
[c] Panels 1 and 2 overlap rather than add: 479 of these 484 blocks are row 1.1, and the other 5 carry no member observing unit in the archive metadata, so the scope query does not expose them; 4 of the 5 are in the primary census.
[d] Every one is an ϵ Eri Band 6 block. The archival pointings put the star either at the centre of the field or well off it (response 0.23–0.29, against a retention limit of 0.22), and all of them are searched: their conventions are in Appendix A.
The same band's 12 windows that fall in the primary census are counted in row 2.1 and carry every census statistic; the on-axis pointings reach a response of 1.00.

processed, 77 are reserved (15.9 per cent), giving 315 windows toward 36 stars with no star and no system lost from the headline sample. The rule hashes the execution block, and ALMA's identifier carries the observing session above it, so two executions of one session can fall on opposite sides of the split: 49 of the 77 reserved blocks share a session with a searched one, across 40 sessions. A session-level hash would have been the stronger choice; the rule stands as it was drawn, because it was drawn before the first reserved block was searched. Two quantities are calibrated on those blocks and on nothing else.

First, the rank displacement. Out of sample the stellar rank has median 0.405 where one half is expected, with a block-clustered 95 per cent interval of 0.341–0.458, and 14 per cent of ranks below 0.1 against 6 per cent above 0.9, so the distribution is displaced rather than growing a tail. The displacement runs the way that matters: a rank below one half places the star high, and the star outranks all 512 probes in 14 of the survey's windows against 3.2 if the positions were interchangeable — 12 of them reaching the trigger as well, which is the conditioned count the chance comparison uses (Appendix D) — so the direction inflates star-first outcomes rather than suppressing them. That median is pooled over windows, and the reserved windows are not spread evenly over stars, so the comparison with the searched sample is made star by star in Appendix D: the two samples agree once each star carries equal weight, and the gap between the pooled medians is a property of how the windows are distributed over stars rather than of the reserved blocks themselves.

A median, however, constrains the centre of a distribution, and the crossings are drawn from its extreme tail. The figure that bears on the chance comparison is therefore the rate at the trigger itself, and the same reserved blocks measure it: stars reach the trigger in 8 of 62 fine-channel windows where those windows' own control positions predict 7.5, a ratio of 1.07 with 0.48–1.79 allowed by resampling the blocks, and carrying it through moves the expectation of Appendix D from 25.8 to 27.6 against the 32 found. That interval is wide — it permits a stellar exceedance rate anywhere from about half the control rate to nearly twice it — so what 8 events on 62 windows establish is compatibility with the empirical control expectation, not the shape of the tail, and a factor of two either way would not show.

Second, the radial behaviour of the control ensemble: the standardised level runs +0.12, +0.10, +0.12, +0.16 across the annulus, a mean per-window slope of $+0.067 \pm 0.033$, small and of the wrong sign to account for the displacement.

TABLE A2
The survey ledger, part two: events, dispositions and confirmation. Every headline count in this paper is read from here. Three counts are 52 over three different sets — the Class A crossings of the adopted ledger, those of the extraction catalogue, and the quality-passing crossings. The first two hold different rows, since re-extraction moved four crossings across the trigger each way; the last two hold the same rows, because the one block failing the quality criterion is also the survey's only block with a coarse-channel crossing. Row 3.1.1 is what this paper means by confirmation and the only such count it quotes: a repeat epoch at a frequency the first did not cover confirms nothing, so rows 3.1 and 3.2 count any second epoch and are not confirmation counts. Row 2.2.1 was also decomposed system by system, each expectation formed from that system's own control positions: 2 of the 61 systems fall below one chance in twenty, against 3 expected, and HD 14055, which holds 8 of the 56 crossings and 8 of the unattributed ones, gives 6 in census windows against 3.6 expected over its 38 Class A windows. No system departs from its own controls by more than chance allows.

	Count	Value	What it counts
1	Threshold crossings at $T_\star \geq 5$	56	census and archival tail
1.1	Class A, fine channels	52	the carrier experiment
1.1.1	… in the primary census	50	catalogued window by window
1.1.2	… in archival-tail blocks	2	both toward HD 14055; in no expectation
1.2	Class B, coarse channels	4	all in the one block failing §5.3.2
2	Quality-passing crossings	52	the same rows as 1.1
2.1	… attributed to a transition	18	inside the mask in the star's own frame
2.2	… unattributed	34	rows 2.2.1 and 2.2.2
2.2.1	… in the primary census	32	against 25.8 expected by chance, measured at the 405 windows' own control positions (11–43 allowed)
2.2.2	… in archival-tail blocks	2	outside that comparison
2.2.3	… above all 512 controls	2	the two events of Table 6
2.2.4	… recovered in a later block	0	no confirmed signal; 13 attributed ones do
3	Class A systems	61	of 83 surveyed
3.1	… observed again more than 1 day later	40	at any frequency, so not a confirmation count
3.1.1	… at a frequency the first block covered	**38**	**the confirmable set, and the only one quoted**; 12 of them over a year later
3.2	… observed once, or all within one day	21	18 in a single block, 3 within a day
3.3	… with no second block at a covered frequency	23	confirmation impossible: row 3.2 and those whose later blocks share no frequency; 36 of all 83
4	Class A union bandwidth	47.7 GHz	26 disjoint islands
4.1	… removed by the line mask	6.4 %	3.05 GHz: the figure the text quotes, against 4.3 per cent of the 118.1 GHz archival union

The *rank-calibration sample* is not an out-of-sample set and must not be confused with the hold-out. Its 326 execution blocks are re-used to calibrate the rank and to supply repeats the hold-out does not have; 264 are primary-census blocks, 63 hold-out blocks and 0 outside both, so it enters no count of the science sample because its blocks are counted there already. Over its 1322 windows the frozen pipeline returns 7 crossings outranking all their controls, 4 of them β Pictoris CO, and 3 unattributed against 3.7 expected.

TABLE A3
The 67 ALMA proposals the searched execution blocks belong to, in the standard ADS/JAO.ALMA#*code* form. Every block of the primary census belongs to one of these, and the blocks themselves are retrieved from the ALMA Science Archive by the execution-block identifiers Tables D1 and D2 give.

2012.1.00142.S, 2012.1.00198.S, 2012.1.00268.S, 2012.1.00385.S, 2012.1.00437.S, 2012.1.00993.S, 2013.1.00359.S, 2013.1.00588.S, 2013.1.00645.S, 2013.1.01147.S, 2013.1.01214.S, 2015.1.00307.S, 2015.1.00633.S, 2015.1.00783.S, 2015.1.01015.S, 2015.1.01260.S, 2016.1.00104.S, 2016.1.00358.S, 2016.1.00515.S, 2016.1.00637.S, 2016.1.00803.S, 2016.1.00817.S, 2016.1.00878.S, 2016.1.00880.S, 2016.2.00015.S, 2016.2.00200.S, 2016.A.00013.S, 2017.1.00167.S, 2017.1.00200.S, 2017.1.00215.S, 2017.1.00561.S, 2017.1.00698.S, 2017.1.00704.S, 2017.1.00786.S, 2017.1.00986.S, 2017.1.01583.S, 2017.1.01644.S, 2018.1.00045.S, 2018.1.00072.S, 2018.1.00126.S, 2018.1.00461.S, 2018.1.00470.S, 2018.1.00947.S, 2018.1.01149.S, 2018.1.01545.S, 2018.1.01833.S, 2019.1.00189.S, 2019.1.00696.S, 2019.1.01443.T, 2019.1.01517.S, 2019.1.01603.S, 2019.1.01681.S, 2019.2.00208.S, 2021.1.00485.S, 2021.1.01209.S, 2021.2.00097.S, 2022.1.00134.S, 2022.1.00338.L, 2022.1.00793.S, 2022.1.01163.S, 2023.1.00390.S, 2023.1.00487.S, 2024.1.00165.S, 2024.1.00595.S, 2024.1.00722.S, 2024.1.01095.S, 2024.A.00040.S

B.2. *What the recurrence statistic reads when nothing is there*

The exclusion in Table D1 is one formula. Writing S and $\sigma_{\rm d}$ for the discovery amplitude and its error, and $\bar{a}$ and $\bar{\sigma} = (\sum_i \sigma_i^{-2})^{-1/2}$ for the inverse-variance combination of the repeat readings at the predicted channel,

$$\sigma_{\rm excl} = \frac{S - \bar{a}}{\sqrt{\sigma_{\rm d}^2 + \bar{\sigma}^2}} = \frac{T_{\rm pers} - T_{\rm rep}}{\sqrt{1 + (T_{\rm pers}/T_\star)^2}}, \qquad \text{(B1)}$$

with $T_{\rm pers} = S/\bar{\sigma}$ and $T_{\rm rep} = \bar{a}/\bar{\sigma}$, so every exclusion follows from three columns of that table and nothing else. $T_{\rm rep}$ is read at the carried drift and at one cell; the largest value over every drift trial at the same cell is tabulated beside it but is never compared with $T_{\rm pers}$, since a maximum is positive on average whatever is there. No exclusion can exceed the strength of the discovery it is built from, nor is it the strongest statement available: an amplitude selected as the largest of 10^2 to 10^6 cells at $T_\star \simeq 5$ is biased high, so a real carrier would be fainter than measured, and what survives the bias is the limit $(T_{\rm rep}+5)/T_{\rm pers}$ — the fraction of the discovery amplitude still ruled out at the trigger.

The same three numbers say where the test has power and where it has none. The criterion asks the repeat to reach the trigger, so a carrier radiating through both observations at the discovery amplitude satisfies it with probability $\Phi(T_{\rm pers} - 5)$, and at the trigger amplitude $5S/T_\star$ — the faintest carrier that could have produced the crossing at all — with $\Phi(5T_{\rm pers}/T_\star - 5)$. Where they are small the crossing is untested rather than excluded, and is reported that way.

The reading at the predicted channel is centred on zero and scatters by one, measured and not merely designed. Over 511 single-cell readings of every repeat block in the census and the hold-out its mean is $+0.014$ and its scatter 1.046, and at the control positions of the discovery windows -0.036 and 1.034; over the 2 crossings whose per-integration repeat spectra survive, the largest value anywhere within $\pm 15\,{\rm km\,s^{-1}}$ of a predicted channel is 3.19 at

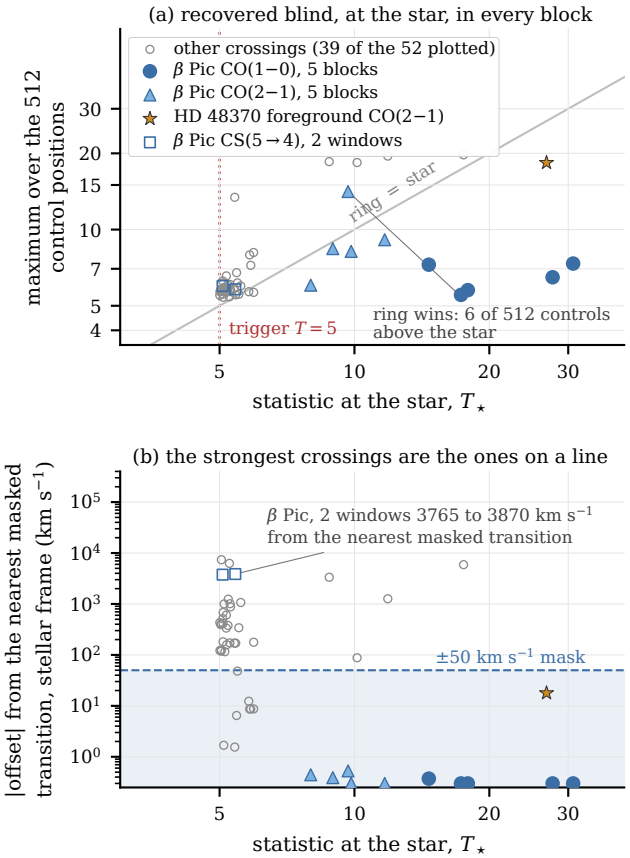


Fig. B1.— The β Pictoris positive control, in both directions. Both panels draw the same 52 of the 56 crossings, the remainder having no retained control ring. *(a)* The spatial screen: the stellar statistic against the maximum over that window's 512 control positions. Filled symbols are the blocks in which the blind search recovers circumstellar carbon monoxide at β Pictoris, 9 of them with the star above every control and one with the ring above it, block `X7116f1_X1785`, Band 6, where the ring reaches 14.13 against 9.68 at the star (§5.4). *(b)* The same crossings by velocity offset from the nearest masked transition *in their own star's rest frame*, with the adopted $\pm 50\,\mathrm{km\,s^{-1}}$ half-width marked: the 10 β Pictoris carbon-monoxide crossings land 0.1–0.5 $\mathrm{km\,s^{-1}}$ from CO, and its 2 CS(5→4) crossings 3765–3870 $\mathrm{km\,s^{-1}}$ from the nearest transition of the mask. 39 further crossings are grey, HD 48370 marked.

the carried drift. That is a scan across frequency at one drift, and it is the one reading here that needs those spectra; the 4.71 of Table 6 is its converse, one channel read at every trial of the grid, and that was recorded for each of the 444 repeat measurements at the time it was made. The search of the repeat epochs over whole windows at every drift is the survey's own search, one catalogue row per window, which is where the 28 exceedances of §5.6 are counted. A maximum behaves as a maximum should: the largest of the 8 retained control probes of a repeat block averages 1.449 against 1.424 expected, the largest of the n_{rep} per-block readings +1.102 against +1.130, and the combined T_{rep} of Eq. B1 −0.119. Widening the window costs trials and not calibration: the exceedances of §5.6 match the same accounting, and HD 69830's windows are the only ones too narrow for the excursion their crossing's drift implies.

C. THE LINE MASK

A crossing that falls on a known molecular transition has an astrophysical explanation to hand, and the mask exists to say which crossings those are. It reports the frequency coincidence and the judgement about the coinciding species separately, so the reproducible part can be recomputed and the arguable part argued with.

The list, and when it was made. The mask holds every transition of the 12 species these spectral setups cover — the three CO isotopologues, HCN, HCO^+, CS, CN, SiO, H_2CO, SO, [C I] and hydrogen recombination — that falls inside one of the 46 searched frequency intervals with an upper-state energy below 150 K. That rule is the whole of it, and it admits 111 transitions (Table C1). No strength cut is applied across species: any threshold admitting HCN's strong lines would discard the brightest lines in these fields. Within a species the main hyperfine components are kept and the satellites dropped. The list is a single Splatalogue query of CDMS and JPL (Pickett et al. 1998; Müller et al. 2005) made on 2026 October 6, after the disposition criteria were fixed on 2026 September 11, and frozen before any crossing was compared with it.

The species are the ones the observers went to the telescope for, and that is a limitation. The spectral setups are tuned to them: 39 per cent of the 405 Class A windows are centred within $\pm 50\,\mathrm{km\,s^{-1}}$ of a transition of the list and 88 per cent contain one somewhere, against 1–8 per cent centred when the same windows are displaced bodily by 1 to 7 GHz. The list is therefore set by the science goals of the programmes that took the data and not by anything in this search, which is why it works as well as it does on these fields and also where it is blind: a setup tuned to a line the list does not hold is screened only where that line and a masked transition happen to coincide in frequency.

Attribution is decided in the star's own rest frame, and the sign is not the usual one. A crossing's frequency is carried from the topocentric value to the barycentre using its own block's pointing and mid-time, and then to the stellar rest frame using the star's systemic velocity. That frame, and no other, is where a coincidence is judged. The barycentric term alone spans $58\,\mathrm{km\,s^{-1}}$ across the survey, so a tolerance applied in the observed frame would spend most of its budget on the Earth's motion. The 3 HD 285968 epochs show the transform working in the only way that can be checked: $12\,\mathrm{km\,s^{-1}}$ apart as observed, they agree to $0.03\,\mathrm{km\,s^{-1}}$ in the star's frame. Throughout, $\Delta v_\star = c\,(\nu_\star - \nu_0)/\nu_0$, so a positive offset means the crossing lies *above* the transition in frequency — the opposite sign to the velocity convention of a spectral line.

The local standard of rest is read as well, as a check. Gas along the line of sight is at rest in that frame rather than in the star's, so for a star with a large LSR velocity a foreground line could fall outside a stellar-frame mask by construction. Every crossing is therefore carried one further step, the projection of the standard solar motion, and read there too. The two offsets differ by the star's own LSR velocity and by nothing else, which here runs 0.3–18.7 $\mathrm{km\,s^{-1}}$ about a median of 3.0. The check moves nothing: no crossing is attributed there that the star's frame does not attribute already, and nor could one have been, since the unattributed crossing nearest a transition lies $88\,\mathrm{km\,s^{-1}}$ from one, $38\,\mathrm{km\,s^{-1}}$ beyond the mask and twice the largest of those velocities. One coincidence runs the other way and exists only in the star's frame, HD 161868's $-48.0\,\mathrm{km\,s^{-1}}$ from $SO(2_3–2_1)$ against −53.4

TABLE C1
THE LINE MASK, BY SPECIES: THE NUMBER OF TRANSITIONS N IT HOLDS, THE FREQUENCY RANGE THEY SPAN INSIDE THE SEARCHED INTERVALS AND THEIR UPPER-STATE ENERGIES. HYDROGEN RECOMBINATION LINES CARRY NO TABULATED E_u. THE CRITERION AND THE DATED QUERY ABOVE DEFINE THE LIST EXACTLY, SO THE INDIVIDUAL TRANSITIONS CAN BE RECOVERED FROM THE LINE CATALOGUES THEMSELVES.

Species	N	ν_0 (GHz)	E_u (K)
H	5	92.03 – 231.90	–
SO	15	100.03 – 682.68	16–143
H_2CO	13	101.33 – 491.97	10–106
$C^{18}O$	3	109.78 – 329.33	5–32
CN	50	113.12 – 680.30	5–114
CO	3	115.27 – 345.80	6–33
CS	3	195.95 – 342.88	24–66
SiO	2	217.10 – 347.33	31–75
^{13}CO	8	220.40 – 330.59	16–32
HCN	7	354.50 – 354.51	43–43
HCO^+	1	356.73 – 356.73	43
[C I]	1	492.16 – 492.16	24

in the local standard of rest, which is a further reason to decide in the frame circumstellar gas shares.

What the second reading establishes is which coincidences are foreground. Of the 24 stars carrying crossings, the composite CO(1 → 0) survey of Dame & Thaddeus (2022), which contains Dame et al. (2001), covers 11 and holds emission toward 5 of them, between −18 and +22 km s^{-1}; the all-sky Planck CO map (Planck Collaboration et al. 2016) stays below 0.178 K km s^{-1} toward every other, including the 13 outside the CfA footprint. Toward ALMA J153702653−33192492, HD 139084 B, HD 285968 and HD 48370 the CO spectrum is non-zero at the crossing's own LSR velocity, within 3 km s^{-1} of its own peak and at 0.23 K or brighter against a 0.18 K channel: 6 of the 21 coincidences are foreground gas and not circumstellar. Two are worth naming. ALMA J153702653−33192492 is a serendipitous field star toward the Lupus clouds, at 35.0 pc with the clouds far behind it, and its CO(2 → 1) crossing lies 0.5 km s^{-1} from a 6.6 K km s^{-1} cloud line at +4.9 km s^{-1}: the field is a molecular sightline and not a disc. The velocity says so and the field does not: its brightest control position stands above the star in all six of its windows, but so it does in 393 of the 405 Class A windows of the survey, so that comparison distinguishes nothing. Its other 5 crossings lie at least 122 km s^{-1} from every transition of the list, and the mask disposes of none of them. HD 48370 is the case where the two readings part: its coincidence lies −17.8 km s^{-1} from the star's own CO(2 → 1), while in the local standard of rest it sits at +23.9 km s^{-1}, on a catalogued cloud line peaking at +21.8. The offset follows the foreground gas and not the star, so that window's extended emission is foreground cloud and not a resolved disc.

The coincidences, and the one species we do not read as emission. Of the 56 crossings, 21 fall within ±50 km s^{-1} of a transition of the list in their own star's frame; that count is the first layer and all the velocity criterion decides. Coincidences with sulphur monoxide account for 3 — CD−57 1054, CP−72 2713 and HD 161868, at −1.7, −12.3 and −48.0 km s^{-1} from transitions of upper-state energy 21–87 K, each inside a Band 7 window that also covers CO(3 → 2) — and two things argue against reading those as emission. Sulphur monoxide has not been reported in a debris disc, where the second-generation gas is CO, C and O; and toward CP−72 2713 the CO(3 → 2) line falls in the same window and is absent to 10.5 mJy in a 0.42 km s^{-1} channel while the crossing's fitted amplitude is 11.3 ± 2.0 mJy, so an SO line there would have to be at least as bright as an undetected CO(3 → 2) line in the same data — but by only 0.43 of the fitted error, so it is the species and not this margin that carries the weight. Both arguments are about the species and not about any one crossing, so all 3 are treated alike: Table D1 prints the offset beside each with a dagger and disposes of none of them, giving 18 attributed against 38 unattributed, of which two outrank every control position in their own windows. Reading them as attributions instead gives 21 and 35, and 1 such crossing; not one of the 3 is present again in any block covering it, and none is confirmed, so the choice moves counts and no disposition. Table D1 forms the exclusion of Eq. B1 for each of them, at 4.1–5.5σ, but that significance is built on the discovery amplitude; read at the trigger amplitude instead, a crossing whose repeat leaves the criterion without power is reported in §5.6 as untested rather than excluded, and that applies to these as to any other. The reserved hold-out is treated identically, leaving 1 attributed and 7 unattributed there (§5.6.2).

The half-width, and what it costs. A one-channel test is too tight for circumstellar gas, whose emission is shifted by many channels: the width follows from Keplerian motion where these molecules survive, $\lesssim$13 km s^{-1} beyond 10 au, and is ±50 km s^{-1}. At that width the mask removes 3.05 GHz of the 47.7 GHz Class A union, 6.4 per cent of it, against 5.04 GHz of the 118.1 GHz both classes cover, 4.3 per cent: the cost falls on the primary experiment, because its windows are the line-tuned ones. 6.4 per cent is the cost of the mask this paper applies, and is the figure quoted elsewhere. Attributing on either frame instead would mask the union of two tubes separated by the star's LSR velocity, 3.72 GHz or 7.8 per cent of the Class A union — 22 per cent more bandwidth for no further attribution, which is the measurement behind deciding in one frame. Between ±20 and ±100 km s^{-1} the mask attributes 17 to 19 of the crossings, and the 2 that outrank all 512 control positions of their own windows are the same two at every width.

What a bright line costs is its channels, and nothing beyond them. Because every above-trigger cell of a window is recorded (§5.3), a weaker carrier beside a molecular line is recorded too, and the price of line confusion is the masked channels and not the windows holding them. Only the 3 fine windows narrower than the mask itself are lost outright, 0.108 GHz, 0.23 per cent of the Class A union; the one masked window that still holds a trigger-level cell outside the mask, β Pic at 115.6200 GHz, does not recur in the 4 other blocks covering that cell.

The labelled crossings are tested for recurrence too. 15 of the 18 attributed crossings have a later block covering the predicted cell, and the 13 that are present again, toward β Pic and HD 285968, are the positive control of that stage. For the 2 not recovered a repeat window's own largest statistic bounds the predicted cell from above and Table D1 prints the row as a bound; 3 have no covering block at all (HD 139084 B, HD 48370 and HIP 10679).

Image-sideband leakage accounts for no crossing, and a longer list would not be a better one. The mask holds signal-sideband frequencies, and the 2 unattributed

crossings that share a block with a CO($2 \rightarrow 1$) crossing in the opposite sideband would need a first local oscillator outside the range their own block's spectral windows permit, 235.02–235.68 GHz. In the other direction, every Splatalogue transition over these intervals with an Einstein coefficient above $10^{-5}\,\mathrm{s}^{-1}$ — 5 114 transitions of 235 species — would mask 75.1 per cent of the union and move 27 crossings across the line, one of them rank-leading. At these frequencies almost anything lies within $\pm 50\,\mathrm{km\,s}^{-1}$ of something in such a list — 48 of the 56 crossings do, at a median $8.7\,\mathrm{km\,s}^{-1}$ from the nearest of them — which is why the species are restricted to what these setups cover and why the outcome is a label rather than a verdict.

D. THE FALSE-ALARM EXPECTATION AND THE CONTROL ENSEMBLE

This appendix gives the chance expectation for the unattributed crossings, what it is conditioned on, and what the control ensemble calibrating it can support.

D.1. *What the expectation is conditioned on*

One attribution set serves both sides of the comparison, the set of §5.3: the mask of Appendix C within $\pm 50\,\mathrm{km\,s}^{-1}$ in the star's own rest frame, sulphur monoxide excluded. So the 3 sulphur monoxide coincidences are unattributed crossings, and a control maximum on one of those transitions is an unattributed exceedance. A systemic velocity is measured for the stars of 277 of the census windows and the others are taken at rest, which moves the masked share by 3.0 points.

The expectation is built from the control positions, the only empirical null the data provide, and each Class A window contributes two quantities measured on itself. The fraction of its own 512 control positions that reaches the trigger is the probability that a position in that field does so — no cell count enters it and no form is assumed for the tail — and summed over the windows 36.2 are expected to trigger by chance. How many of those would be attributed is measured on the channels the control maxima land on, which the retained spectra record: 88 of 302 exceedances fall inside the mask, and the attribution therefore takes 29 per cent of the expectation. A maximum free to land anywhere on its own grid would take 20 per cent, the median window offering 6.0 per cent of its channels; the difference is that 15 narrow line-tuned windows lie wholly inside the mask and their fields, which hold real emission, are where a control position most often triggers, so a uniform landing would under-attribute them and leave the unattributed expectation too high. What is left is 25.8 expected unattributed Class A crossings against 32 observed.

Both sides run over the 405 Class A windows of the primary census, so the expectation is a sum over rows and nothing is carried at an assumed rate. A further 15 Class A windows arrived with the archival tail and lie outside it, together with the two unattributed crossings toward HD 14055 they hold, which are reported beside the comparison and enter neither side of it. No window holds more than one primary crossing, so both sides count windows; the weaker cells behind those maxima are counted at their own grain in §5.3.

The one assumption the searched data cannot test is that a star behaves like a control position in its own field, and the reserved blocks measure it in the quantity used here. Their stars reach the trigger in 8 of 62 fine-channel windows where the control positions of those same windows predict 7.5: a ratio of 1.07, with 0.48–1.79 allowed by resampling the reserved blocks, and no two of the events in one block. In the 253 coarse-channel reserved windows no star reaches the trigger, against 0.04 predicted. The same blocks place the stellar rank below one half (Appendix B), which is the same departure read in the body of the distribution rather than in its tail and runs the same way. Carrying the measured ratio through gives 27.6 against the 25.8 above, and the observed 32 sits inside the range either way; the ratio is itself consistent with unity, so the uncorrected sum is the one quoted.

The 11–43 quoted beside the expectation is what 20 000 resamplings of the execution blocks allow, each window's rate redrawn from its own 512 probes and the attributed share from its own 302 exceedances: windows share blocks and are not independent trials, and both ensembles are finite. Those terms alone give 23–29, so what widens the range is the calibration above. That is a range on a rate, and a count drawn from it scatters as well, so the number of unattributed crossings chance alone would deliver runs from 9 to 47: an added population of as many as 23 events would be absorbed by that width, and that is the limit on what the comparison can exclude.

Every Class B crossing in the survey comes from the one execution block of §D.2, and over both classes together 25.9 are expected against the same 32 observed: a coarse single pointing holds too few cells for its own field to reach the trigger by chance, and only 459 of the 1663 windows hold a control position above $T_{\mathrm{trig}} = 5$ at all. The Class A comparison is the survey's.

The statistic is always the one measured over the window's whole field. 157 windows admit integrations inside the primary beam that a narrower field would discard, recovering a median 2.1 times the on-source time and moving the statistic by a median 0.07; the whole-field value is the one adopted, on 8 rows of Table D1, and each window's control ensemble is measured with it, so neither side of the comparison mixes two fields. Where the extraction catalogue's disposition column disagrees — on 4 crossings, and the species list moves none into attribution — the ledger is the authority.

D.2. *Execution-block quality*

The comparison with the control positions assumes each block's ensemble behaves like noise, and one block makes that false: every one of its 512 probes exceeds the trigger, the smallest at 6.6. A criterion for the condition is therefore applied to every block rather than to the one that prompted it.

Two criteria are built from that statistic and they are not the same test. The statistic is the median of a window's own 512-position control ensemble, which separates a field with a source in it from a field that is no null at all. A *window* is flagged when that median exceeds twice the median of the same quantity over the survey, 5.5 against 2.74; the flag is a property of the window and cuts across attribution by design, firing on 5 of the 1663 windows in 2 blocks toward 2 stars (BD+05 1668 and HD 48370), and every one of those windows carries a crossing, attributed and unattributed alike. A *block* is

judged on the same median divided by the median over the windows of its own class, since a maximum over a larger channel×drift grid sits higher and one threshold should mean the same thing in both; a block fails when the median of that ratio over its windows exceeds twice the scale of its class. The factor is the same in both. Neither was fixed in advance: the condition was noticed in one block and the criterion written afterwards, so it is post hoc, and what makes it trustworthy is that it does not depend on where the line was drawn — the failing block sits at 4.1 and the next worst at 1.12, so any threshold between them selects the same single block.

Applied to all 407 blocks of the primary census, where the median block sits at 1.00, the criterion fails 1: A002_Xc7fa6f_X28a8 toward BD+05 1668. Nothing else is close, and the unscaled form of the rule fails the same 1 block. Its 4 windows are all coarse-channel and hold the survey's entire Class B crossing population, so it carries 4 of the 56 threshold crossings — 0 of them outranking all 512 control positions of their own window — and no Class A statistic here has ever contained it. Those 4 appear in Table D1 with their block named and enter no count elsewhere.

D.3. *The control ensemble: its calibration, and its resolution*

The ring geometry is set in §4.1; its calibration and its resolution are the subject here. Over the 1657 census windows that retain a control vector the stellar ranks are consistent with the one half exchangeability implies — median 0.49 with each star carrying equal weight, 0.47 to 0.51 — in both array strata, the compact-array windows measured on each observation's own antenna table rather than on a 12 m beam. That is a weak statement and not a strong one, since a shift smaller than about 0.042 could not have been seen in a sample of this size. The reserved blocks lean further from one half when their windows are pooled, but those windows are concentrated on a few stars — 108 of 315 belong to five — and the two samples agree once each star carries equal weight, 0.44 (0.33–0.52) against the 0.49 above. Compared star by star over the 36 stars both samples hold the difference is −0.115 with −0.17 to +0.07 allowed, so the pooled gap is a weighting effect and not a property of the reserved blocks (Appendix B.1). The expectations above are summed from the exceedance rate measured at each window's control positions, so they do rest on that transfer and carry its measured accuracy.

The rank is not a significance test, and the reason is structural. Ranking the star against 512 control positions of the same field decides which windows deserve examination, but $1/(512+1) = 1.95\times10^{-3}$ is the *resolution* of the ensemble — the smallest rank it can express — and not the probability that noise produced the window. A survey of 1663 windows would need a rank 65 times finer, and correlation makes it coarser still: controls inside the synthesised beam are not independent, so an annulus resolves no better than 0.023–0.032 however many probes are drawn into it, with the primary beam bounding how many a single pointing contains. That floor is the design's central limitation: no single window here can be significant on the rank alone, which is why confirmation requires independent recurrence.

D.4. *The trials budget*

The searched cells counted in §4.2 are not independent trials: channel smoothing and a drift grid stepped twice per channel correlate neighbouring ones, which alone would leave 0.19 of them independent. The control ensembles say otherwise, their maxima sitting at 0.995 of the prediction for the maximum of the full searched count treated as independent and Gaussian — a budget predicting 480 windows with a control above $T_{\rm trig}$ against the 459 observed — while fitting each window's effective count against its own ensemble returns 0.81 of its searched cells. Nothing turns on which is right, because the expectation above is summed from measured rates and not from a cell count: by the cell-count route the number reaching the trigger is 36.6 rather than 36.2, 3 per cent away.

The on-star crossings do exceed that budget, and the attribution accounts for it. A Gaussian tail over every searched cell predicts 37.9 stellar exceedances against the 56 found, but 18 are attributed by the set above, 4 come from the block of §D.2 and 2 fall outside the census windows, leaving 32 against the 25.9 predicted. No excess survives the attribution.

One stratum departs from uniformity, and the departure is where the line emission is. Counting the windows whose stellar peak both reaches the trigger and exceeds all 512 control positions of its own window, the 405 fine-channel windows hold 12 against 0.79 expected, while the coarse stratum, three times larger and predicting 2.45, contains 0. A coarse channel dilutes a narrow circumstellar line, so the class that can resolve one is the class that finds it — and an unresolved carrier would appear in Class A first for the same reason. Reaching the trigger is holding a crossing, so each of those windows holds one and 10 hold a crossing the mask attributes: the departure is a statement about the windows carrying line emission and not about the fine stratum at large. Drop the trigger condition and the star leads its own ring in 2 further windows, at 4.20 and 4.36, giving 14 over all 1663 windows against 3.2 expected. One quantity, two conditions; the chance comparison uses the conditioned one.

What is left is bounded, and the bounds are null. The real stellar ranks agree at $p = 0.38$ with the 220 672 ranks each control takes against the other probes of its own window; radial suppression is below 10 per cent in the mean; and scrambling a window so nothing coherent survives along a drift track carries 4 past the rank floor and changes no disposition.

E. RADIO-FREQUENCY INTERFERENCE SCREENING

Interference is the first alternative explanation for an unattributed crossing, so the screening this survey has and the coverage it lacks are both set out here. Millimetre interferometry is an unusually poor environment for terrestrial interference, though not an empty one: ITU Radio Regulations Article 5 (International Telecommunication Union 2024) carries active allocations throughout 90–275 GHz and WRC-19 identified parts of 275–450 GHz for land-mobile and fixed use. The specific hazard is the 94.0–94.1 GHz Earth exploration-satellite allocation used by spaceborne cloud radar (Stephens et

block of §5.3.2. Every row is the primary crossing of its window; the 21 weaker cells behind those maxima are not listed individually, and the three of them the mask does not attribute are described in §5.3. The columns are the star, its execution block, the ALMA band, the crossing frequency as the correlator delivered it, or, where no peak frequency was measured, recovered from the line offset, and the same frequency in the star's rest frame, the search statistic $T_\star$, the largest statistic $T_{\rm ctrl}$ among the window's 512 control positions, the nearest masked transition and the stellar-frame offset $\Delta v_\star$ from it, then the recurrence test: $n_{\rm rep}$ covering repeat blocks spanning Δt days, the statistic $T_{\rm pers}$ a carrier persisting at the discovery flux would have produced at the predicted stellar-frame cell, the statistic $T_{\rm rep}$ measured there at the carried drift, and the exclusion of Eq. B1, which three of these columns reproduce, two of them also giving the limit defined in Appendix B. Where the emission is present again in a covering block the test returns a recovery and not an exclusion, so both $T_{\rm rep}$ and $\sigma_{\rm excl}$ are blank and the disposition says which it was. $T_\star$ and $T_{\rm ctrl}$ come from the same measurement of the same window; $T_{\rm ctrl}$ gates nothing and says only which crossing was examined first, and the drift-maximised reading of the cell is given in Table 6 for the two leading events and is not tabulated for the rest. Rows whose window is data-quality flagged (§D.2) are marked, and on those $T_{\rm pers}$ and the exclusion are withheld: both are built from the discovery amplitude, and in a window whose control ring stands as high as its star that amplitude cannot be assigned to a point source at the star. 1 of those rows lies outside the failing block — HD 48370, foreground cloud emission along its sightline — and is withheld for that reason and not for its block. A crossing is attributed by the set defined above, and a dagger marks the 3 rows whose coinciding species that set excludes. The transition named is the nearest in the star's frame, so where it is far from every transition it is the magnitude and not the name that carries information. Block identifiers omit the constant `A002_` prefix.

Star	block	B	$\nu_{\rm topo}$ (GHz)	$\nu_\star$ (GHz)	$T_\star$	$T_{\rm ctrl}$	line	$\Delta v_\star$	$n_{\rm rep}$	Δt (d)	$T_{\rm pers}$	$T_{\rm rep}$	$\sigma_{\rm excl}$	disposition
BD+05 1668	Xc7fa6f_X28a8	6	240.1016	240.1054	73.398	74.69	CS(5–4)	−5911.9	29	0.9–113	n/a	+0.15	n/a	unattributed; flagged
BD+05 1668	Xc7fa6f_X28a8	6	224.7422	224.7458	48.303	83.74	$H_2CO(3_{1,2}–2_{1,1})$	−1264.5	29	0.9–113	n/a	+0.37	n/a	unattributed; flagged
BD+05 1668	Xc7fa6f_X28a8	6	226.7422	226.7458	41.355	75.48	$CN(2_{3/2}–1_{1/2})$	+88.0	29	0.9–113	n/a	+0.66	n/a	unattributed; flagged
BD+05 1668	Xc7fa6f_X28a8	6	242.1953	242.1992	39.586	83.29	CS(5–4)	−3349.2	29	0.9–113	n/a	+0.77	n/a	unattributed; flagged
β Pic	Xf5d76d_Xeb8	3	115.2603	115.2711	30.765	7.34	CO(1–0)	−0.2	4	0.0–53	n/a	—	n/a	attributed; recurs
β Pic	Xf5d76d_Xe19	3	115.2603	115.2711	27.684	6.48	CO(1–0)	−0.2	4	0.0–53	n/a	—	n/a	attributed; recurs
HD 48370	Xc26103_X155a	6	230.5117	230.5243	26.834	18.36	CO(2–1)	−17.8	—	—	n/a	—	n/a	attributed; flagged
β Pic	Xf4df6f_Xc7	3	115.2620	115.2711	17.907	5.76	CO(1–0)	−0.2	4	53	n/a	—	n/a	attributed; recurs
β Pic	Xf5d76d_Xcc5	3	115.2604	115.2711	17.288	5.52	CO(1–0)	−0.2	4	0.0–53	n/a	—	n/a	attributed; recurs
β Pic	Xf5d76d_X32f1	3	115.2606	115.2713	14.654	7.27	CO(1–0)	+0.4	4	0.0–53	n/a	—	n/a	attributed; recurs
β Pic	Xd9668b_X3a90	6	230.5161	230.5378	11.681	9.12	CO(2–1)	−0.3	5	1.0–1983	n/a	—	n/a	attributed; recurs
β Pic	X6f1341_X1484	6	230.5284	230.5379	9.832	8.20	CO(2–1)	−0.1	5	25–1984	n/a	—	n/a	attributed; recurs
β Pic	X7116f1_X1785	6	230.5272	230.5384	9.677	14.13	CO(2–1)	+0.5	5	25–1959	n/a	—	n/a	attributed; recurs
β Pic	Xd9668b_Xa9df	6	230.5160	230.5377	8.948	8.41	CO(2–1)	−0.4	5	1.0–1984	n/a	—	n/a	attributed; recurs
β Pic	Xa7a216_X23e4	6	230.5273	230.5377	7.988	6.03	CO(2–1)	−0.4	5	601–1313	n/a	—	n/a	attributed; recurs
61 Vir	Xc079b5_X82f	7	346.3138	346.3232	5.958	5.65	$SO(9_8–8_7)$	−177.6	3	0.1–3.0	9.7	+0.39	4.9	unattributed; no recurrence
HD 285968	Xd98580_X3bf6	6	230.5021	230.5447	5.956	8.12	CO(2–1)	+8.7	3	13–44	n/a	—	n/a	attributed; recurs
HD 285968	Xdb7ab7_X5b4e	6	230.5116	230.5447	5.868	7.22	CO(2–1)	+8.7	3	5.0–44	n/a	—	n/a	attributed; recurs
HD 285968	Xdb4b9a_X108f	6	230.5100	230.5447	5.835	7.94	CO(2–1)	+8.7	3	5.0–39	n/a	—	n/a	attributed; recurs
CP−72 2713	Xff0235_X4a6d	7	344.2697	344.2965	5.809	5.68	$SO(8_8–7_7)$	$-12.3^\dagger$	5	0.1–2.1	6.3	−0.82	4.8	unattributed; no recurrence
HD 139084 B	Xcd8029_Xb6b0	7	345.8241	345.8241	5.763	17.70	CO(3–2)	+24.4	—	—	—	—	—	attributed
HD 14055	Xfff3d8_Xd46	7	331.7699	331.7664	5.575	5.99	$^{13}CO(3–2)$	+1068.7	30	0.9–650	170.4	−0.02	5.6	unattributed; no recurrence
HD 23484	X10b22d7_X2445	6	230.7904	230.8006	5.520	5.55	CO(2–1)	+341.5	4	0.1–2.1	10.4	+1.25	4.3	unattributed; no recurrence
HD 161868	Xff45a6_X10ca8	7	345.6338	345.6491	5.481	6.16	$SO(2_3–2_1)$	$-48.0^\dagger$	13	0.0–6.1	92.2	+0.24	5.5	unattributed; no recurrence
HIP 10679	Xa95c04_X14cd	6	220.4185	220.4035	5.456	6.76	$^{13}CO(2–1)$	+6.5	—	—	—	—	—	attributed
HD 14055	X11adad7_X25c5e	7	345.4564	345.4303	5.448	6.82	$SO(2_3–2_1)$	−237.8	30	0.1–653	14.0	+0.03	5.1	unattributed; no recurrence
HD 207129	Xc19d6f_X5706	6	230.4226	230.4070	5.433	5.52	CO(2–1)	−170.3	4	0.8–425	69.7	−0.93	5.5	unattributed; no recurrence
β Pic	Xd9668b_X3a90	6	241.7512	241.7739	5.420	5.81	CS(5–4)	−3869.7	1	1.0	5.7	+0.72	3.4	unattributed; no recurrence

TABLE D2

The crossing ledger, continued. The remaining 28 of the 56 threshold crossings, in the same order and with the same columns as Table D1, whose caption defines them. These too are primary crossings, one to a window.

Star	block	B	ν_{topo} (GHz)	$\nu_\star$ (GHz)	$T_\star$	T_{ctrl}	line	$\Delta v_\star$	n_{rep}	Δt (d)	T_{pers}	T_{rep}	σ_{excl}	disposition
HR 1010	Xc6fa08_X1053	6	230.5133	230.5287	5.410	5.65	CO(2–1)	−12.0	7	2.0–14	8.4	<+4.60	>2.1	attributed; not recovered
ALMA J153702653-33192492	Xff0235_X8392	6	230.5222	230.5392	5.405	13.41	CO(2–1)	+1.6	6	18–141	6.2	<+4.15	>1.4	attributed; not recovered
HD 110411	Xe45e29_Xb7d4	6	219.4538	219.4357	5.393	5.85	$C^{18}O$(2–1)	−170.2	1	10	6.7	+0.62	3.8	unattributed; no recurrence
HD 31392	X129754b_X6946	6	231.1998	231.2231	5.289	5.81	H(30α)	−876.3	8	0.1–41	15.7	−0.26	5.1	unattributed; no recurrence
HD 14055	Xff99e1_X24be0	7	331.7078	331.7035	5.277	5.69	^{13}CO(3–2)	+1011.6	30	0.9–652	252.2	+0.15	5.3	unattributed; no recurrence
AU Mic	Xa42f75_X319	6	230.6864	230.6705	5.274	6.07	CO(2–1)	+172.3	2	310–455	6.4	−0.16	4.2	unattributed; no recurrence
HD 69830	X12209f7_X16edb	7	322.4325	322.4444	5.259	5.42	$C^{18}O$(3–2)	−6268.5	5	0.1–29	8.9	−0.48	4.8	unattributed; no recurrence
ALMA J153702653-33192492	Xfe62c1_X398	6	216.1844	216.2036	5.228	6.02	SiO(5–4)	−1244.6	7	18–126	11.0	+1.03	4.3	unattributed; no recurrence
AU Mic	X899169_X1838	6	230.8252	230.8294	5.221	5.85	CO(2–1)	+379.0	2	145–310	10.0	+0.19	4.5	unattributed; no recurrence
HD 14055	Xff99e1_X2c3bb	7	345.5256	345.5216	5.186	5.66	$SO(2_3–2_1)$	−158.6	30	0.1–651	172.8	+2.00	5.1	unattributed; no recurrence
HD 14055	Xfffde1_X3835	7	345.0099	345.0070	5.177	6.56	$SO(2_3–2_1)$	−604.8	30	0.1–649	187.2	−1.78	5.2	unattributed; no recurrence
HD 14055	Xff99e1_X2b2ee	7	344.7007	344.6965	5.175	5.92	$SO(8_8–7_7)$	+336.0	30	0.1–651	210.7	−0.00	5.2	unattributed; no recurrence
LHS 1140	Xd9d7c7_X6fd2	6	230.6355	230.6268	5.136	5.68	CO(2–1)	+115.5	4	5.0–27	7.3	+0.33	4.0	unattributed; no recurrence
HD 10647	Xff294f_Xaf70	7	331.6570	331.6943	5.120	5.91	^{13}CO(3–2)	+1003.3	10	0.1–592	125.8	+0.39	5.1	unattributed; no recurrence
CD−57 1054	Xcf525f_X5aa4	7	346.5099	346.5265	5.108	5.71	$SO(9_8–8_7)$	$-1.7^{\dagger}$	1	39	6.0	−0.27	4.1	unattributed; no recurrence
ALMA J153702653-33192492	Xf8f6a9_X10359	6	217.7292	217.7269	5.091	5.41	$H_2CO(3_{0,3}–2_{0,2})$	−680.5	7	3.8–242	11.6	+0.07	4.6	unattributed; no recurrence
GJ 849	Xd9668b_X841a	6	230.8769	230.8581	5.090	5.83	CO(2–1)	+416.3	5	0.1–51	9.3	−1.69	5.3	unattributed; no recurrence
ALMA J153702653-33192492	Xfe62c1_X398	6	218.8729	218.8923	5.090	5.56	$H_2CO(3_{2,1}–2_{2,0})$	+181.2	7	18–126	10.5	−1.04	5.0	unattributed; no recurrence
HD 14055	X1001fdc_X29ae	7	331.1559	331.1546	5.082	6.26	^{13}CO(3–2)	+513.8	—	—	—	—	—	unattributed; no spectrum
β Pic	Xd9668b_Xa9df	6	241.8367	241.8594	5.079	5.99	CS(5–4)	−3765.1	1	1.0	4.8	−0.63	3.9	unattributed; no recurrence
ALMA J153702653-33192492	Xf8f6a9_X10359	6	218.6711	218.6688	5.048	5.77	$H_2CO(3_{2,1}–2_{2,0})$	−125.1	7	3.8–242	11.6	+0.00	4.6	unattributed; no recurrence
HD 69830	X12277ed_X178c6	7	321.2005	321.2149	5.046	6.13	$C^{18}O$(3–2)	−7387.7	5	6.0–36	10.2	−0.46	4.7	unattributed; no recurrence
HR 1010	Xc66ce1_Xaa9	6	230.8348	230.8491	5.044	5.72	CO(2–1)	+404.6	7	1.1–14	10.3	−1.53	5.2	unattributed; no recurrence
61 Vir	Xc067f7_X822	7	345.5599	345.5678	5.038	5.79	$SO(2_3–2_1)$	−118.5	3	2.0–3.0	9.3	+0.45	4.2	unattributed; no recurrence
HD 170773	Xff0235_X8c42	7	331.0101	331.0234	5.029	5.75	^{13}CO(3–2)	+394.8	4	0.9–113	74.8	+1.56	4.9	unattributed; no recurrence
ALMA J153702653-33192492	Xd9d7c7_Xbb64	7	347.2179	347.1890	5.009	5.68	SiO(8–7)	−122.2	1	0.1	5.0	−2.06	5.0	unattributed; no recurrence
HD 14055	X1190de8_Xd526	7	344.4676	344.4515	5.006	5.96	$SO(8_8–7_7)$	+122.7	30	20–611	11.6	−0.60	4.8	unattributed; no recurrence
η Crv	X122b6ff_X1041e	7	357.2815	357.2475	5.006	5.50	HCO^+(4–3)	+431.3	1	87	11.0	−0.62	4.8	unattributed; no recurrence

al. 2002; Wehr et al. 2023), a documented contaminant of ALMA Band 3 (Emerson 2004); 1 of the 1663 census windows overlaps it, carrying 0 threshold crossings and 0 rank-passing events, so no result here rests on it. What favours this band is that terrestrial transmitters are sparse compared with centimetre wavelengths and, decisively, that an interferometer is not a total-power detector: a signal not in the far field at the phase centre acquires a rapidly varying geometric phase across the array, and is suppressed in the correlation rather than imaged.

The coverage of the screening is uneven. The allocation test above and the sky- and intermediate-frequency tests below are computed for every searched window of the census, as is the 512-position rank screen, which asks whether an excess is at the star or everywhere in the field. Recurrence reaches only the crossings with a covering repeat block, and the visibility fit is a consistency check rather than a filter. The correlator flagging is the observatory's own, replayed from each block's delivered calibration script, and was not retained window by window. *No per-channel interference flag of our own exists, and none can be recovered.* Four measurements follow, and each could have failed.

First, clustering in sky frequency. If interference dominated the crossing population, the crossings would pile up at particular sky frequencies. They do cluster — but on the survey's own molecular lines. The test needs a measured peak frequency in the frame an interfering carrier would stand still in, so it runs topocentrically over the 51 of the 56 crossings for which one was measured, against the transitions the search was tuned to. The other 5 are carbon monoxide toward β Pic, four in CO(1–0) and one in CO(2–1) of its ten such crossings; for those the catalogue carries only the velocity offset from the line, so the frequency Table D1 prints is recovered from that offset rather than measured, and a frequency reconstructed from a line offset cannot be used to ask whether frequencies crowd onto lines. Of the 51, 11 fall within $\pm 50\,\mathrm{km\,s^{-1}}$ of a searched transition against 4.8 expected if each window's peak fell anywhere on that window's own channel grid, $p = 0.0010$. The densest band, 230.422–230.723 GHz, holds 12 of them, 9 within that velocity window of CO(2→1); and once the line-attributed crossings are set aside, the 40 that remain show no clustering at all ($p = 0.107$). *The concentration is the attribution, not a residue left over after it.* Table D1 counts the same band as 14 crossings with 11 attributed, over a different set — all 56 of them, each attributed in its own star's rest frame — and the two rows it adds are the ones this test cannot hold: one whose peak frequency the release records only as an offset from the line and one whose window the release does not record as a crossing at all.

Second, the same test against the atmosphere. A residual of the atmospheric calibration stands still in topocentric frequency exactly as a transmitter does, and it is narrow where the atmosphere has lines. 570 transitions of ozone, water and oxygen lie within 89.6–873.1 GHz (Pickett et al. 1998), a median 0.9 GHz apart. No crossing of either sample falls within 25 MHz of one; the median separation is 0.81 GHz; and 1 crossing lies inside the same $\pm 50\,\mathrm{km\,s^{-1}}$ window used above, against 2.1 expected from the windows' own channel grids. The one concentration a reader will notice — 15 crossings of the census and the hold-out between 344.2 and 346.4 GHz — is not atmospheric either, because that interval holds 0 telluric transitions; 12 of the 43 unattributed crossings whose window the catalogue tabulates fall in it against 11.8 ± 0.9 expected from where the spectral windows were tuned: the carbon monoxide tuning, seen from the other side.

Third, fixed intermediate frequency. An artefact of the signal chain would sit at a fixed position within the spectral window rather than at a fixed sky frequency. This test is run over every window for which the catalogue releases a peak frequency — 1401 of the 1663, of which only 51 carry a threshold crossing — because such an artefact would occupy the same intermediate frequency in all of them and not only where the trigger fires. It does not: the peaks' fractional positions within their own windows have median 0.50 and are uniform.

Fourth, a cross-target occupancy screen over every window the pipeline extracted, not only the crossings. That set is wider than the census: 3027 windows toward 133 stars, being the census together with the rows its audit removed, the reserved hold-out, the archival extension, the out-of-sample control, and 52 windows in 13 blocks that entered none of them. Interference belongs to the observatory and not to the star, so a terrestrial carrier should recur at the same sky frequency toward *unrelated* stars. It does not, and the sharpest place to look is the two unattributed crossings that outrank every control position in their own window, 61 Vir and CP−72 2713, both of which carry a measured frequency. Taking them in that order, 346.314 and 344.270 GHz are covered by 36 and 42 further windows, toward 24 and 8 other stars in 32 and 42 other blocks, and none of them carries a crossing in the same channel. With each window's peak randomised on its own channel grid, the number of cross-star coincidences within 5 MHz is 569 against 554 ± 13 expected ($p = 0.15$); among the 109 windows whose peak exceeds the trigger it is 4 against 5.2 ± 1.9, i.e. *below* the null, and with the molecular-line mask applied 1 against 2.1 ± 1.3. There is no excess to explain. The mask has to be applied first, because 35 of those 109 windows sit on a masked transition (CO(2–1) ×17, CO(1–0) ×10, CO(3–2) ×4, ^{13}CO(2–1) ×3, SiO(5–4) ×1), and a carbon monoxide line falls at nearly the same sky frequency toward every nearby star — exactly the coincidence the screen is looking for.

One execution block fails these tests, and it fails them as a block rather than as a carrier. Its 4 crossings toward BD+05 1668 are the highest statistics in the survey outside β Pictoris and HD 48370, and §5.3.2 identifies the failure. Whether interference accounts for it is what belongs here, and three readings say not: the visibility fit returns a large imaginary part, placing the emission away from the phase centre; the block's 24 sibling executions of the same field, array and spectral setup produce no crossing at all; and each of the 4 frequencies is covered by 127–137 further windows toward 17–22 other stars, of which 0 carries a crossing in the same channel.

One reading does point at a mechanism, and it can be tested across the survey. 2 of the 4 sit at the identical channel of two different spectral windows, which is the signature of a fixed intermediate frequency; and a feature

with no geometric fringe phase — a spur common to every antenna — does not fringe-track, so it gathers at the phase centre rather than at the star. Here the two are a twentieth of an arcsecond apart, so the block cannot test its own hypothesis; the survey can, because the star lies 5.0–18.0 arcsec from the phase centre in 83 windows whose control annulus brackets it, putting a median of 15 probes within two synthesised beams of the phase centre and the star well outside. Against probes at the same distance from the star but elsewhere in azimuth — the radius has to be matched, or a centrally concentrated disc would imitate the effect — those probes are higher by $+0.08 \pm 0.07$ in the statistic, consistent with zero and bounded by 0.23, smaller by a factor of about 37 than the 8.4 by which this block's ring is raised. *The phase centre is not where the excess lives.* The evidence supports a data-quality failure of one execution block, and the block criterion of Appendix D.2 — which fails 1 block of the 407 searched — is what disposes of these 4 crossings (§5.3.2).

None of this substitutes for the ON/OFF cadence a dedicated search uses. The claim is only that *no identified interference mechanism accounts for the unattributed events*: allocation tables cannot exclude harmonics, intermodulation products, unconventional emitters or near-field sources, and archival data without a simultaneous off-source strategy cannot do better. The four tests establish that the crossing population carries none of the signatures interference leaves — a statement about the population, not a clearance of any one crossing.

F. PER-SYSTEM SENSITIVITY

Table F1 gives the sensitivity reached toward each system of the carrier search together with the three things that decide what that number means: where in frequency it was obtained, over how wide a channel, and whether the archive holds a second epoch capable of confirming a crossing there. The limit quoted for a system is that of its best window, so a system whose coverage is wide is not penalised for the shallowest window in it; the whole distribution, window by window, is the per-window result the sensitivity of §5.1 is drawn from.

Three features of the table bear on how it should be read. The limits span more than two orders of magnitude, so a single survey depth would describe no part of the sample. The frequency at which a system's best limit falls is set by what the archive happens to hold toward it and not by any property of the star. And a limit in the first column says nothing about whether a crossing at that system could have been followed up: the last column does, and for 23 of the 61 systems no sensitivity would have been enough, because no second block covers a frequency the first one did.

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TABLE F1

Per-system sensitivity of the carrier search. For each of the 61 systems the fine-channel search covers, the $EIRP_{90}$ of its best window — the equivalent isotropic radiated power at which nine injected carriers in ten are recovered, $3.03\,P_{\rm trig}$ with that window's own annual-parallax and smearing retention applied — with that window's centre frequency ν and native channel width $\Delta\nu$, and the system's merged frequency coverage $\Delta\nu_{\rm cov}$. Rows are ordered by the limit, which spans 8.7×10^{13}–3.7×10^{16} W; the median system is covered over 1.84 GHz, and channel widths across the search run from 15.3 to 1953 kHz. The last column records whether two execution blocks cover a common frequency more than one day apart (d) or more than a year apart (yr): 38 systems reach the first and 12 the second, and these are lower bounds, since two blocks can be shown to be separated but never shown to be simultaneous. For the remaining 23 systems a crossing could not have been confirmed at any sensitivity. A gravitationally bound pair is one system and takes its primary's name.

System	$EIRP_{90}$ (10^{14} W)	ν (GHz)	$\Delta\nu$ (kHz)	$\Delta\nu_{\rm cov}$ (GHz)	conf.	System	$EIRP_{90}$ (10^{14} W)	ν (GHz)	$\Delta\nu$ (kHz)	$\Delta\nu_{\rm cov}$ (GHz)	conf.
LP 876-10	0.87	345.83	488	1.7	d	HD 170773	17	230.13	488	7.4	d
β Pic	1.0	230.52	15.3	4.4	yr	HD 61005	19	230.52	244	19	d
AU Mic	1.2	230.56	488	3.5	yr	HR 1010	19	230.53	977	2.8	d
τ Cet	1.2	230.55	488	1.8	d	LHS 1140	19	230.00	488	1.8	d
HD 69830	1.2	230.06	488	3.8	d	κ CrB	19	230.57	977	1.7	–
61 Vir	1.3	345.79	488	1.8	yr	HD 155555C	21	346.01	488	1.7	–
ϵ Eri	1.4	229.98	977	2.3	yr	L 836-122	22	346.00	488	1.7	–
η Crv	1.4	345.80	61.0	7.2	yr	CD−57 1054	23	345.99	488	1.7	–
TRAPPIST-1	1.6	346.58	488	1.7	d	CP−72 2713	24	345.12	488	5.8	–
AT Mic AB	1.7	346.04	488	1.7	–	HD 14055	24	230.11	488	7.2	d
GJ 3379	2.2	229.96	488	1.9	d	TWA 7	24	345.99	244	0.87	d
GJ 14	3.1	227.22	488	3.5	d	GJ 2006A	24	346.02	488	1.7	–
HN Lib	3.9	229.99	488	1.8	d	Wolf 219	26	230.13	488	1.7	d
HD 10647	4.4	345.79	488	6.1	yr	LP 736-15	28	345.82	977	1.7	–
HD 202628	5.4	230.51	488	1.8	yr	TYC 2211-1309-1	28	346.00	488	1.7	–
HD 53143	5.4	230.53	977	1.8	yr	UCAC3 176-23654	29	346.02	488	1.7	–
HD 172555	5.8	230.53	15.3	5.3	–	PM J03433+1958	31	230.13	488	1.7	–
HD 207129	6.4	230.57	488	1.8	yr	UCAC2 20312880	32	345.96	488	1.7	–
GJ 849	6.8	230.02	488	1.9	d	LP 353-51	33	346.02	488	1.7	–
TWA 3A	7.6	230.54	122	0.44	d	Barta 161 12	34	346.02	488	1.7	–
HD 139664	7.8	230.55	488	1.7	–	51 Eri	37	230.53	1953	1.7	–
HD 92945	8.3	355.11	488	3.7	d	HD 23484	38	230.53	977	1.8	d
TX PsA	8.5	346.03	488	1.7	–	HD 139084	54	346.01	488	1.7	–
HD 285968	9.1	229.97	488	1.8	d	WD 0407+179	84	230.13	488	1.7	–
HD 161868	9.9	227.22	488	7.4	d	HD 48370	89	230.73	122	3.3	yr
HD 38858	13	230.50	488	1.8	yr	HD 45184	93	230.53	977	1.7	d
HD 110411	13	219.58	244	4.7	yr	ALMA J153702653-33192492	99	345.98	488	14	d
HIP 490	13	230.53	244	0.86	–	HD 31392	130	230.52	977	1.7	d
HIP 10679	14	230.55	15.3	0.11	–	HD 89452	260	230.53	977	1.7	d
HD 107146	14	355.14	488	3.7	d	HD 377	370	230.54	977	1.7	d
LP 476-207	15	346.02	488	1.7	–						